Math 521, Section 2
Homework 1

Problem 1. Using induction, prove that for all $n \in \mathbb{N}$ we have

$$3 + 11 + \cdots + (8n - 5) = 4n^2 - n.$$

Problem 2. Compute the following sum:

$$\frac{1}{1 \cdot 4} + \frac{1}{4 \cdot 7} + \cdots + \frac{1}{(3n-2)(3n+1)}.$$

Prove that your answer is true for all $n \in \mathbb{N}$ using induction.

Problem 3. Let $f : A \to B$ be a function.

(a) Prove that f is injective if and only if $f^{-1}(f(X)) = X$ for all $X \subseteq A$.
(b) Prove that f is surjective if and only if $f(f^{-1}(Y)) = Y$ for all $Y \subseteq B$.

Problem 4. Let $f : A \to B$ be a function. Prove that the following two statements are equivalent:

(a) The function f is surjective.
(b) For any set C and for any functions $g : B \to C$ and $h : B \to C$ such that $g \circ f = h \circ f$, we have $g = h$.

Problem 5. Let $f : B \to C$ be a function. Prove that the following two statements are equivalent:

(a) The function f is injective.
(b) For any set A and for any functions $g : A \to B$ and $h : A \to B$ such that $f \circ g = f \circ h$, we have $g = h$.

Problem 6. Let $f : A \to B$ be a function. Suppose that there exists a function $g : B \to A$ such that $(g \circ f)(a) = a$ for all $a \in A$ and $(f \circ g)(b) = b$ for all $b \in B$. Prove that f is bijective.

Homework 1

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MATH 521 — Analysis I
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Contents

Problem 1

Using induction, prove that for all $n \in \mathbb{N}$ we have

$$3 + 11 + \cdots + (8n - 5) = 4n^2 - n.$$

Proof.
For $n \in \mathbb{N}$, let $P(n)$ denote the statement

$$3 + 11 + \cdots + (8n - 5) = 4n^2 - n.$$

We argue by induction.

Base case. For $n = 1$, the left-hand side is 3, while the right-hand side is $4 \cdot 1^2 - 1 = 3$. Hence $P(1)$ holds.

Inductive step. Assume $P(n)$ holds for some $n \in \mathbb{N}$, i.e.

$$3 + 11 + \cdots + (8n - 5) = 4n^2 - n.$$

We must show that $P(n + 1)$ holds. Observe that the next term in the sum is

$$8(n + 1) - 5 = 8n + 3.$$

Adding this term to both sides of the inductive hypothesis gives

$$3 + 11 + \cdots + (8n - 5) + (8n + 3) = (4n^2 - n) + (8n + 3).$$

The left-hand side is exactly the sum $3 + 11 + \cdots + (8(n + 1) - 5)$, and the right-hand side simplifies to

$$(4n^2 - n) + (8n + 3) = 4n^2 + 7n + 3.$$

On the other hand,

$$4(n + 1)^2 - (n + 1) = 4(n^2 + 2n + 1) - n - 1 = 4n^2 + 7n + 3.$$

Therefore

$$3 + 11 + \cdots + (8(n + 1) - 5) = 4(n + 1)^2 - (n + 1),$$

so $P(n+1)$ holds. By the Principle of Mathematical Induction, $P(n)$ is true for all $n \in \mathbb{N}$. $\square$

Problem 2

Compute the following sum:

$$\frac{1}{1\cdot 4}+\frac{1}{4\cdot 7}+\cdots+\frac{1}{(3n-2)(3n+1)}.$$

Prove that your answer is true for all $n\in\mathbb{N}$ using induction.

Discussion & Scratch Work

We first compute the sum in closed form. For $k\in\mathbb{N}$,

$$\frac{1}{(3k-2)(3k+1)}=\frac{A}{3k-2}+\frac{B}{3k+1}.$$

Solving

$$1=A(3k+1)+B(3k-2)$$

gives $A=\frac{1}{3}$ and $B=-\frac{1}{3}$. Hence

$$\frac{1}{(3k-2)(3k+1)}=\frac{1}{3}\left(\frac{1}{3k-2}-\frac{1}{3k+1}\right).$$

Therefore the partial sums telescope:

$$\sum_{k=1}^{n}\frac{1}{(3k-2)(3k+1)}=\frac{1}{3}\left(1-\frac{1}{3n+1}\right)=\frac{n}{3n+1}.$$

We now prove this identity by induction.

Proof.
For $n\in\mathbb{N}$, let $P(n)$ denote the statement

$$\frac{1}{1\cdot 4}+\frac{1}{4\cdot 7}+\cdots+\frac{1}{(3n-2)(3n+1)}=\frac{n}{3n+1}.$$

We will prove that $P(n)$ is true for all $n\in\mathbb{N}$ using induction.

Base case. $P(1)$ is the statement

$$\frac{1}{1\cdot 4}=\frac{1}{3\cdot 1+1},$$

which is true.

Inductive step. Assume $P(n)$ is true for some $n\in\mathbb{N}$, that is,

$$\frac{1}{1\cdot 4}+\frac{1}{4\cdot 7}+\cdots+\frac{1}{(3n-2)(3n+1)}=\frac{n}{3n+1}.$$

Adding the next term $\dfrac{1}{(3n+1)(3n+4)}$ to both sides, we obtain

$$\frac{1}{1\cdot 4}+\cdots+\frac{1}{(3n-2)(3n+1)}+\frac{1}{(3n+1)(3n+4)}=\frac{n}{3n+1}+\frac{1}{(3n+1)(3n+4)}.$$

The left-hand side is exactly the sum in $P(n+1)$. On the right-hand side,

$$\frac{n}{3n+1}+\frac{1}{(3n+1)(3n+4)}=\frac{1}{3n+1}\left(n+\frac{1}{3n+4}\right)=\frac{1}{3n+1}\cdot\frac{n(3n+4)+1}{3n+4}.$$

Since

$$n(3n+4)+1=3n^2+4n+1=(3n+1)(n+1),$$

we obtain

$$\frac{n}{3n+1}+\frac{1}{(3n+1)(3n+4)}=\frac{n+1}{3n+4}=\frac{n+1}{3(n+1)+1}.$$

Thus $P(n+1)$ is true. By the Principle of Mathematical Induction, $P(n)$ is true for all $n\in\mathbb{N}$. $\square$

Problem 3

Let $f : A \to B$ be a function.

(a) Prove that f is injective if and only if $f^{-1}(f(X)) = X$ for all $X \subseteq A$.

(b) Prove that f is surjective if and only if $f\big(f^{-1}(Y)\big) = Y$ for all $Y \subseteq B$.

Discussion & Scratch Work

Key facts (true for any function $f : A \to B$):

$$X \subseteq f^{-1}(f(X)) \quad \text{for all } X \subseteq A, \qquad f\big(f^{-1}(Y)\big) \subseteq Y \quad \text{for all } Y \subseteq B.$$

The nontrivial directions use injectivity/surjectivity.

Proof.
Let $f : A \to B$ be a function.

(a) ($\Rightarrow$) Assume f is injective. Let $X \subseteq A$. We prove $f^{-1}(f(X)) = X$ by double inclusion.

First, if $x \in X$, then $f(x) \in f(X)$, so by definition of preimage, $x \in f^{-1}(f(X))$. Hence $X \subseteq f^{-1}(f(X))$.

Conversely, let $a \in f^{-1}(f(X))$. Then $f(a) \in f(X)$, so there exists $x \in X$ with $f(a) = f(x)$. Since f is injective, $a = x \in X$. Thus $f^{-1}(f(X)) \subseteq X$.

Therefore $f^{-1}(f(X)) = X$.

($\Leftarrow$) Assume that for all $X \subseteq A$ we have $f^{-1}(f(X)) = X$. To show f is injective, let $a_1, a_2 \in A$ and suppose $f(a_1) = f(a_2)$. Set $X = \{a_1\}$. Then $f(a_2) = f(a_1) \in f(X)$, so $a_2 \in f^{-1}(f(X)) = X = \{a_1\}$. Hence $a_2 = a_1$. Therefore f is injective.

(b) ($\Rightarrow$) Assume f is surjective. Let $Y \subseteq B$. We prove $f(f^{-1}(Y)) = Y$ by double inclusion.

First, if $b \in f(f^{-1}(Y))$, then by definition there exists $a \in f^{-1}(Y)$ with $f(a) = b$. But $a \in f^{-1}(Y)$ means $f(a) \in Y$, so $b \in Y$. Hence $f(f^{-1}(Y)) \subseteq Y$.

Conversely, let $y \in Y$. Since f is surjective, there exists $a \in A$ such that $f(a) = y$. Then $f(a) = y \in Y$, so $a \in f^{-1}(Y)$, and therefore $y = f(a) \in f(f^{-1}(Y))$. Thus $Y \subseteq f(f^{-1}(Y))$.

Therefore $f(f^{-1}(Y)) = Y$.

($\Leftarrow$) Assume that for all $Y \subseteq B$ we have $f(f^{-1}(Y)) = Y$. Take $Y = B$. Then

$$f(f^{-1}(B)) = B.$$

But $f^{-1}(B) = A$ (every $a \in A$ satisfies $f(a) \in B$), so this reads $f(A) = B$. Hence f is surjective.

$\square$

Problem 4

Let $f : A \to B$ be a function. Prove that the following two statements are equivalent:

(a) The function f is surjective.

(b) For any set C and for any functions $g : B \to C$ and $h : B \to C$ such that $g \circ f = h \circ f$, we have $g = h$.

Discussion & Scratch Work

This is the "right-cancellation" property: f is surjective iff precomposition with f detects equality of functions $B \to C$. The forward direction uses surjectivity to pull back an arbitrary $b \in B$ to some $a \in A$. The reverse direction is proved by contradiction: if f is not surjective, choose $b_0 \in B \setminus f(A)$ and build two functions $g, h : B \to C$ that agree on $f(A)$ but differ at b_0.

Proof.
Let $f : A \to B$ be a function.

($\Rightarrow$) Assume that f is surjective. Let C be any set, and let $g, h : B \to C$ satisfy $g \circ f = h \circ f$. We show that $g = h$.

Fix $b \in B$. Since f is surjective, there exists $a \in A$ such that $f(a) = b$. Then

$$g(b) = g(f(a)) = (g \circ f)(a) = (h \circ f)(a) = h(f(a)) = h(b).$$

Since $b \in B$ was arbitrary, we conclude $g = h$.

($\Leftarrow$) Assume that for every set C and all functions $g, h : B \to C$,

$$g \circ f = h \circ f \quad \Longrightarrow \quad g = h.$$

We prove that f is surjective. Suppose, for contradiction, that f is not surjective. Then there exists $b_0 \in B \setminus f(A)$.

Let $C = \{0, 1\}$, and define $g, h : B \to C$ by

$$g(b) = 0 \text{ for all } b \in B, \qquad h(b) = \begin{cases} 1, & b = b_0, \\ 0, & b \neq b_0. \end{cases}$$

Then $g \neq h$ (since $g(b_0) = 0$ but $h(b_0) = 1$). However, for every $a \in A$ we have $f(a) \neq b_0$ (because $b_0 \notin f(A)$), so $h(f(a)) = 0 = g(f(a))$. Hence

$$(g \circ f)(a) = g(f(a)) = 0 = h(f(a)) = (h \circ f)(a) \quad \text{for all } a \in A,$$

which implies $g \circ f = h \circ f$. By the assumed property, this would force $g = h$, a contradiction. Therefore f must be surjective. $\square$

Problem 5

Let $f : B \to C$ be a function. Prove that the following two statements are equivalent:

(a) The function f is injective.

(b) For any set A and for any functions $g : A \to B$ and $h : A \to B$ such that $f \circ g = f \circ h$, we have $g = h$.

Discussion & Scratch Work

This is the "left-cancellation" property: f is injective iff postcomposition with f detects equality of functions $A \to B$. The forward direction uses injectivity to cancel f from the equation $f \circ g = f \circ h$ pointwise. The reverse direction is proved by contradiction: if f is not injective, pick $b_1 \neq b_2$ with $f(b_1) = f(b_2)$ and build two functions $g, h : A \to B$ that differ at one point but become equal after applying f.

Proof.
Let $f : B \to C$ be a function.

($\Rightarrow$) Assume that f is injective. Let A be any set, and let $g, h : A \to B$ satisfy $f \circ g = f \circ h$. We show that $g = h$.

Fix $a \in A$. Then
$$f(g(a)) = (f \circ g)(a) = (f \circ h)(a) = f(h(a)).$$
Since f is injective, this implies $g(a) = h(a)$. Because $a \in A$ was arbitrary, we conclude $g = h$.

($\Leftarrow$) Assume that for every set A and all functions $g, h : A \to B$,
$$f \circ g = f \circ h \implies g = h.$$
We prove that f is injective. Suppose, for contradiction, that f is not injective. Then there exist $b_1, b_2 \in B$ with $b_1 \neq b_2$ and $f(b_1) = f(b_2)$.

Let $A = \{0, 1\}$ and define $g, h : A \to B$ by
$$g(0) = b_1, \quad g(1) = b_1, \qquad h(0) = b_1, \quad h(1) = b_2.$$
Then $g \neq h$ (since $g(1) = b_1 \neq b_2 = h(1)$). However, for each $a \in A$ we have
$$(f \circ g)(a) = f(g(a)) = \begin{cases} f(b_1), & a = 0, \\ f(b_1), & a = 1, \end{cases} \qquad (f \circ h)(a) = f(h(a)) = \begin{cases} f(b_1), & a = 0, \\ f(b_2), & a = 1, \end{cases}$$
and since $f(b_1) = f(b_2)$, it follows that $(f \circ g)(a) = (f \circ h)(a)$ for all $a \in A$. Thus $f \circ g = f \circ h$. By the assumed property, this would force $g = h$, a contradiction. Therefore f must be injective. $\square$

Problem 6

Let $f : A \to B$ be a function. Suppose that there exists a function $g : B \to A$ such that $(g \circ f)(a) = a$ for all $a \in A$ and $(f \circ g)(b) = b$ for all $b \in B$. Prove that f is bijective.

Discussion & Scratch Work

> The hypotheses say that g undoes the action of f on both sides:
>
> $$(g \circ f)(a) = a \quad \text{for all } a \in A, \qquad (f \circ g)(b) = b \quad \text{for all } b \in B.$$
>
> From the first identity, applying g to an equality $f(a_1) = f(a_2)$ forces $a_1 = a_2$, which implies injectivity. From the second identity, every $b \in B$ can be written as $b = f(g(b))$, which implies surjectivity.

Proof.
Let $f : A \to B$ and suppose there exists $g : B \to A$ such that

$$(g \circ f)(a) = a \quad \text{for all } a \in A, \qquad (f \circ g)(b) = b \quad \text{for all } b \in B.$$

We show that f is injective and surjective.

Injective. Let $a_1, a_2 \in A$ and assume $f(a_1) = f(a_2)$. Applying g to both sides gives

$$g(f(a_1)) = g(f(a_2)).$$

But by the first hypothesis, $g(f(a)) = a$ for any $a \in A$, so $a_1 = a_2$. Hence f is injective.

Surjective. Let $b \in B$. Set $a = g(b) \in A$. Then

$$f(a) = f(g(b)) = (f \circ g)(b) = b.$$

Thus every $b \in B$ is hit by f, so f is surjective.

Since f is both injective and surjective, f is bijective. $\square$

Appendix: Toolbox

Definition 1 (Natural Numbers). *The set of natural numbers* $\mathbb{N} = \{1, 2, 3, \dots\}$ *is defined axiomatically by the Peano axioms:*

(a) $1 \in \mathbb{N}$.

(b) *If* $n \in \mathbb{N}$*, then* $n + 1 \in \mathbb{N}$.

(c) 1 *is not the successor of any natural number.*

(d) *If* $n + 1 = m + 1$*, then* $n = m$.

(e) *(*Induction Axiom*) If* $S \subseteq \mathbb{N}$ *satisfies* $1 \in S$ *and* $n \in S \Rightarrow n + 1 \in S$*, then* $S = \mathbb{N}$.

Theorem 1 (Principle of Mathematical Induction). *Let* $P(n)$ *be a statement defined for all* $n \in \mathbb{N}$*. If:*

(a) $P(1)$ *is true (base case), and*

(b) *for all* $n \in \mathbb{N}$, $P(n) \Rightarrow P(n + 1)$ *(inductive step),*

then $P(n)$ *is true for all* $n \in \mathbb{N}$.

Remark. *The Principle of Mathematical Induction follows from the Peano axioms. Given a statement* $P(n)$*, define*

$$S = \{n \in \mathbb{N} : P(n) \text{ is true}\}.$$

If $1 \in S$ *and* $n \in S \Rightarrow n + 1 \in S$*, then* $S = \mathbb{N}$ *by the Induction Axiom.*

Definition 2 (Function). *Let* A *and* B *be nonempty sets. A* function $f : A \to B$ *assigns to each element* $a \in A$ *exactly one element* $f(a) \in B$*. The set* A *is called the* domain *and the set* B *the* codomain.

Definition 3 (Composition of Functions). *Let* $f : A \to B$ *and* $g : B \to C$ *be functions. The* composition *of* g *with* f *is the function* $g \circ f : A \to C$ *defined by*

$$(g \circ f)(a) = g(f(a)) \quad \text{for all } a \in A.$$

Definition 4 (Image and Preimage). *Let* $f : A \to B$ *be a function.*

- *For* $X \subseteq A$*, the* image *of* X *under* f *is*

$$f(X) = \{f(x) : x \in X\}.$$

- *For* $Y \subseteq B$*, the* preimage *of* Y *under* f *is*

$$f^{-1}(Y) = \{a \in A : f(a) \in Y\}.$$

Definition 5 (Injective, Surjective, Bijective). *Let* $f : A \to B$ *be a function.*

- *f is* injective *if for all $a_1, a_2 \in A$,*

$$f(a_1) = f(a_2) \Rightarrow a_1 = a_2.$$

- *f is* surjective *if $f(A) = B$, i.e. for every $b \in B$ there exists $a \in A$ such that $f(a) = b$.*
- *f is* bijective *if it is both injective and surjective.*

Remark. *For a general function $f : A \to B$, the notation $f^{-1}(Y)$ denotes a* preimage *and does not require f to be invertible. Moreover, the properties of being injective, surjective, or bijective depend not only on the rule defining f, but also on the choice of domain A and codomain B.*

Math 521, Section 2
Homework 2

Problem 1. Prove that if A and B are both countable sets and $A \cap B = \emptyset$, then $A \cup B$ is countable. (Hint: We proved that $\mathbb{Z}$ is countable in class. Try using a similar idea here.)

Problem 2. Let A_n be a countable set for each $n \in \mathbb{N}$, and suppose that $A_m \cap A_n = \emptyset$ whenever $m \neq n$. Prove that the union $A = A_1 \cup A_2 \cup A_3 \cup \ldots$ is countable. (Hint: We proved that $\mathbb{N} \times \mathbb{N}$ is countable in class. Try using a similar idea here.)

Problem 3. Recall that given two sets A and B, we write $A \sim B$ if and only if there exists a bijective function $f : A \to B$. Prove that $\sim$ is an equivalence relation.

Problem 4. For $a, b \in \mathbb{Z}$, define $a \sim b$ if and only if $b - a$ is divisible by 2.

(a) Prove that $\sim$ is an equivalence relation on $\mathbb{Z}$.
(b) Find the equivalence classes.

Problem 5. Let $(F, +, \cdot)$ be a field.

(a) Prove that for any $x \in F$, the function $f : F \to F$, $f(y) = x + y$ is bijective.
(b) Prove that for any $x \in F \setminus \{0\}$, the function $g : F \to F$, $g(y) = x \cdot y$ is bijective.

Problem 6. Let $(F, +, \cdot)$ be a field. Suppose that $F = \{0, 1, a, b\}$ is a set with exactly four elements, where 0 and 1 denote the identities for $+$ and $\cdot$ respectively, and a, b denote the remaining two elements of F. Fill in the addition and multiplication tables below, and justify your answer.

$+$	0	1	a	b
0				
1				
a				
b				

$\cdot$	0	1	a	b
0				
1				
a				
b				

(Hint: Using the previous problem, show that each row and column in the addition table contains every element of F exactly once, like a Sudoku puzzle. Show that the same is true for the *nonzero* rows and columns of the multiplication table. Note that for each entry, there is only one correct answer.)

Homework 2

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Contents

Problem 1

Prove that if A and B are both countable sets and $A \cap B = \varnothing$, then $A \cup B$ is countable. (Hint: We proved that $\mathbb{Z}$ is countable in class. Try using a similar idea here.)

Discussion & Scratch Work

Because A and B are countable, there exist enumerations

$$A = \{a_1, a_2, a_3, \dots\}, \qquad B = \{b_1, b_2, b_3, \dots\}.$$

Intuitively, this means we can list the elements of A and B in a sequence. To show that $A \cup B$ is countable, we want to construct a single list that contains every element of A and B exactly once. The idea is to interleave the two lists, alternating between an element of A and an element of B. The assumption $A \cap B = \varnothing$ ensures that no element appears in both lists, so this alternating procedure never produces duplicates.

Proof.
Since A and B are countable, there exist bijections

$$f : \mathbb{N} \to A \quad \text{and} \quad g : \mathbb{N} \to B.$$

Write $a_n = f(n)$ and $b_n = g(n)$, so that

$$A = \{a_1, a_2, a_3, \dots\} \quad \text{and} \quad B = \{b_1, b_2, b_3, \dots\}.$$

Define a function $h : \mathbb{N} \to A \cup B$ by

$$h(2n-1) = a_n \quad \text{and} \quad h(2n) = b_n \qquad (n \in \mathbb{N}).$$

We show that h is bijective.

First, h is surjective. Let $x \in A \cup B$. If $x \in A$, then $x = a_n$ for some n, and hence $x = h(2n-1)$. If $x \in B$, then $x = b_n$ for some n, and hence $x = h(2n)$. Thus every element of $A \cup B$ is in the image of h.

Next, h is injective. Suppose that $h(k) = h(\ell)$. If k and ℓ have different parity, then one of $h(k)$ lies in A and the other lies in B, which is impossible because $A \cap B = \varnothing$. Hence k and ℓ must have the same parity. If $k = 2m-1$ and $\ell = 2n-1$, then $a_m = a_n$, and since f is injective, we have $m = n$, so $k = \ell$. If $k = 2m$ and $\ell = 2n$, then $b_m = b_n$, and since g is injective, we again obtain $m = n$, so $k = \ell$.

Therefore h is a bijection from $\mathbb{N}$ onto $A \cup B$. It follows that $A \cup B$ is countable. $\square$

Problem 2

Let A_n be a countable set for each $n \in \mathbb{N}$, and suppose that $A_m \cap A_n = \varnothing$ whenever $m \neq n$. Prove that the union $A = A_1 \cup A_2 \cup A_3 \cup \cdots$ is countable. (Hint: We proved that $\mathbb{N} \times \mathbb{N}$ is countable in class. Try using a similar idea here.)

Discussion & Scratch Work

Because each A_n is countable, we can "list" its elements as

$$A_n = \{a_{n,1}, a_{n,2}, a_{n,3}, \dots\}.$$

So every element of the big union $A = \bigcup_{n\in\mathbb{N}} A_n$ comes with two pieces of information: (1) which set A_n it belongs to, and (2) its position in the list of A_n.

That suggests encoding an element of A by a pair $(n, k) \in \mathbb{N} \times \mathbb{N}$, where n tells us which A_n we are in, and k tells us which element of A_n we chose. Since we proved in class that $\mathbb{N} \times \mathbb{N}$ is countable, we can list all pairs (n, k) in a single sequence. If we map (n, k) to the k-th element of A_n, we get a single list that hits every element of A.

The disjointness assumption $A_m \cap A_n = \varnothing$ for $m \neq n$ is what prevents duplicates from coming from different A_n's.

Proof.
For each $n \in \mathbb{N}$, the set A_n is countable, so there exists a bijection

$$f_n : \mathbb{N} \to A_n.$$

Define a function $h : \mathbb{N} \times \mathbb{N} \to A$ by

$$h(n, k) := f_n(k) \qquad ((n, k) \in \mathbb{N} \times \mathbb{N}).$$

We claim that h is bijective.

Surjectivity. Let $x \in A$. By definition of $A = \bigcup_{n\in\mathbb{N}} A_n$, there exists some $n \in \mathbb{N}$ such that $x \in A_n$. Since f_n is surjective onto A_n, there exists $k \in \mathbb{N}$ with $f_n(k) = x$. Thus $h(n, k) = x$, so h is surjective.

Injectivity. Suppose $h(n, k) = h(m, \ell)$. Then

$$f_n(k) = f_m(\ell).$$

The left-hand side lies in A_n and the right-hand side lies in A_m. If $n \neq m$, then this would imply $A_n \cap A_m \neq \varnothing$, contradicting the hypothesis that the A_n are pairwise disjoint. Hence $n = m$. With $n = m$, we have $f_n(k) = f_n(\ell)$, and since f_n is injective, it follows that $k = \ell$. Therefore $(n, k) = (m, \ell)$, and h is injective.

So h is a bijection $\mathbb{N} \times \mathbb{N} \to A$.

Finally, we proved in class that $\mathbb{N} \times \mathbb{N}$ is countable, i.e. there exists a bijection

$$\varphi : \mathbb{N} \to \mathbb{N} \times \mathbb{N}.$$

Then the composition $h \circ \varphi : \mathbb{N} \to A$ is a bijection. Therefore A is countable. $\square$

Problem 3

Recall that given two sets A and B, we write $A \sim B$ if and only if there exists a bijective function $f : A \to B$. Prove that $\sim$ is an equivalence relation.

Discussion & Scratch Work

We want to prove that $\sim$ is an equivalence relation, so we must check three properties: reflexivity, symmetry, and transitivity.

The definition says $A \sim B$ means "there exists a bijection $f : A \to B$."

- **Reflexive:** any set A is in bijection with itself via the identity function defined by $f : A \to A$ with $f(a) = a$ for all $a \in A$.
- **Symmetric:** if $A \sim B$, there is a bijection $f : A \to B$. A bijection has an inverse function $f^{-1} : B \to A$, and f^{-1} is also a bijection, so $B \sim A$.
- **Transitive:** if $A \sim B$ and $B \sim C$, then there are bijections $f : A \to B$ and $g : B \to C$. Their composition $g \circ f : A \to C$ is again a bijection, so $A \sim C$.

So the whole proof is basically: identity gives reflexive, inverse gives symmetric, and composition gives transitive.

Proof.

We verify the three defining properties.

(Reflexivity). Let A be a set. Define a function $f : A \to A$ by $f(a) = a$ for all $a \in A$. Then f is bijective. Hence $A \sim A$.

(Symmetry). Suppose $A \sim B$. Then there exists a bijection $f : A \to B$. Since f is bijective, it has an inverse function $f^{-1} : B \to A$, and f^{-1} is bijective. Therefore $B \sim A$.

(Transitivity). Suppose $A \sim B$ and $B \sim C$. Then there exist bijections $f : A \to B$ and $g : B \to C$. The composition $g \circ f : A \to C$ is bijective, so $A \sim C$.

Thus $\sim$ is reflexive, symmetric, and transitive; hence it is an equivalence relation. $\square$

Problem 4

For $a, b \in \mathbb{Z}$, define $a \sim b$ if and only if $b - a$ is divisible by 2.

(a) Prove that $\sim$ is an equivalence relation on $\mathbb{Z}$.

(b) Find the equivalence classes.

Discussion & Scratch Work

Big picture. The relation is saying that two integers are equivalent when they have the *same parity* (both even or both odd). Another way to say this is that $a \sim b$ exactly when a and b differ by an *even* integer.

To prove that $\sim$ is an equivalence relation, we check the three properties from the notes:

- **Reflexive:** always true because $a - a = 0$ is divisible by 2.
- **Symmetric:** if $b - a$ is divisible by 2, then $a - b = -(b - a)$ is also divisible by 2.
- **Transitive:** if $b - a$ and $c - b$ are divisible by 2, then adding them gives $c - a = (c - b) + (b - a)$, also divisible by 2.

For part (b), once you realize this is "same parity," there should be exactly two equivalence classes: the evens and the odds. Concretely:

$$[0] = \{\text{even integers}\} = \{2k : k \in \mathbb{Z}\} = 2\mathbb{Z}, \qquad [1] = \{\text{odd integers}\} = \{2k+1 : k \in \mathbb{Z}\} = 2\mathbb{Z}+1.$$

Any even integer is equivalent to 0, and any odd integer is equivalent to 1.

Proof.
(a) We show that $\sim$ is reflexive, symmetric, and transitive on $\mathbb{Z}$.

Reflexive. Let $a \in \mathbb{Z}$. Then $a - a = 0$, and 0 is divisible by 2. Hence $a \sim a$.

Symmetric. Suppose $a \sim b$. By definition, $b - a$ is divisible by 2, so $b - a = 2k$ for some $k \in \mathbb{Z}$. Then

$$a - b = -(b - a) = -2k = 2(-k),$$

so $a - b$ is divisible by 2. Thus $b \sim a$.

Transitive. Suppose $a \sim b$ and $b \sim c$. Then $b - a = 2k$ and $c - b = 2\ell$ for some $k, \ell \in \mathbb{Z}$. Adding,

$$c - a = (c - b) + (b - a) = 2\ell + 2k = 2(k + \ell),$$

which is divisible by 2. Hence $a \sim c$.

Therefore $\sim$ is an equivalence relation on $\mathbb{Z}$.

(b) The equivalence classes are exactly the parity classes. First, if $n \in \mathbb{Z}$ is even, then $n - 0 = n$ is divisible by 2, so $n \sim 0$; hence every even integer lies in $[0]$. Conversely, if $n \in [0]$, then $n \sim 0$, so $n - 0 = n$ is divisible by 2, which means n is even. Thus

$$[0] = \{2k : k \in \mathbb{Z}\} = 2\mathbb{Z}.$$

Similarly, if $n \in \mathbb{Z}$ is odd then $n-1$ is even (hence divisible by 2), so $n \sim 1$ and $n \in [1]$; and if $n \in [1]$, then $n \sim 1$ so $n-1$ is divisible by 2, meaning n is odd. Hence

$$[1] = \{2k+1 : k \in \mathbb{Z}\} = 2\mathbb{Z}+1.$$

These are the two equivalence classes. $\square$

Problem 5

Let $(F, +, \cdot)$ be a field.

(a) Prove that for any $x \in F$, the function $f : F \to F$, $f(y) = x + y$ is bijective.

(b) Prove that for any $x \in F \setminus \{0\}$, the function $g : F \to F$, $g(y) = x \cdot y$ is bijective.

Discussion & Scratch Work

Big picture. In a field, "adding a fixed element" and "multiplying by a fixed nonzero element" are reversible operations.

- For (a), if we define $f(y) = x + y$, then we can undo it by adding the additive inverse $-x$:
$$f(y) = x + y \quad \Longrightarrow \quad y = (-x) + f(y).$$
So the inverse candidate is $F \to F$, $y \mapsto (-x) + y$.
- For (b), if $x \neq 0$ and we define $g(y) = x \cdot y$, then we can undo it by multiplying by x^{-1}:
$$g(y) = x \cdot y \quad \Longrightarrow \quad y = x^{-1} \cdot g(y).$$
So the inverse candidate is $F \to F$, $y \mapsto x^{-1} \cdot y$.

Once we exhibit an inverse function (and verify the two composition identities), bijectivity follows.

Proof.
(a) Fix $x \in F$ and define $f : F \to F$ by $f(y) = x + y$. Define $h : F \to F$ by
$$h(y) = (-x) + y.$$
We show that h is the inverse of f by checking both compositions.
For any $y \in F$,
$$(h \circ f)(y) = h(f(y)) = h(x + y) = (-x) + (x + y) = \big((-x) + x\big) + y = 0 + y = y,$$
using associativity of $+$, the definition of $-x$, and the additive identity.
For any $y \in F$,
$$(f \circ h)(y) = f(h(y)) = f((-x) + y) = x + ((-x) + y) = (x + (-x)) + y = 0 + y = y,$$
again by associativity and the definition of additive inverse.
Thus $h = f^{-1}$, so f is bijective.

(b) Fix $x \in F \setminus \{0\}$ and define $g : F \to F$ by $g(y) = x \cdot y$. Since $x \neq 0$, there exists $x^{-1} \in F$ such that $x \cdot x^{-1} = x^{-1} \cdot x = 1$. Define $k : F \to F$ by
$$k(y) = x^{-1} \cdot y.$$

We again check both compositions. For any $y \in F$,

$$(k \circ g)(y) = k(g(y)) = k(x \cdot y) = x^{-1} \cdot (x \cdot y) = (x^{-1} \cdot x) \cdot y = 1 \cdot y = y,$$

using associativity of $\cdot$ and the multiplicative identity.
For any $y \in F$,

$$(g \circ k)(y) = g(k(y)) = g(x^{-1} \cdot y) = x \cdot (x^{-1} \cdot y) = (x \cdot x^{-1}) \cdot y = 1 \cdot y = y.$$

Thus $k = g^{-1}$, so g is bijective. $\square$

Problem 6

Let $(F, +, \cdot)$ be a field. Suppose that $F = \{0, 1, a, b\}$ is a set with exactly four elements, where 0 and 1 denote the identities for $+$ and $\cdot$ respectively, and a, b denote the remaining two elements of F. Fill in the addition and multiplication tables below, and justify your answer.

$+$	0	1	a	b
0				
1				
a				
b				

$\cdot$	0	1	a	b
0				
1				
a				
b				

(Hint: Using the previous problem, show that each row and column in the addition table contains every element of F exactly once, like a Sudoku puzzle. Show that the same is true for the nonzero rows and columns of the multiplication table. Note that for each entry, there is only one correct answer.)

Discussion & Scratch Work

Strategy. We know $F = \{0, 1, a, b\}$ is a field, so $(F, +)$ is an abelian group of order 4 and $F \setminus \{0\} = \{1, a, b\}$ is an abelian group under $\cdot$ of order 3.

Addition table. Since $|F| = 4 = 2^2$, the characteristic of F must be 2 (the characteristic of a field is a prime p and $|F|$ is a power of p). Hence

$$1 + 1 = 0.$$

In fact, in an additive group of order 4 with characteristic 2, every element satisfies $x + x = 0$. So

$$1 + 1 = a + a = b + b = 0.$$

Now use the "Sudoku" constraint: for each fixed $x \in F$, the map $y \mapsto x + y$ is a bijection (Problem 5 style), so each row/column of the addition table contains each element of F exactly once. For example, in the row for 1, the entries are $1 + 0 = 1$ and $1 + 1 = 0$, so the remaining two entries must be a and b. Also $1 + a$ cannot be 0 (else $a = -1 = 1$) and cannot be 1 (else $a = 0$) and cannot be a (else $1 = 0$), so necessarily $1 + a = b$. Similarly $1 + b = a$, and then $a + b = 1$.

Multiplication table. The nonzero elements form a group of order 3, so it is cyclic. In particular, for any $x \in \{a, b\}$, we have $x^3 = 1$ and $x \neq 1$. Thus if we pick $a \neq 1$, then $a^2 \neq 1$ and must equal the remaining nonzero element b. Then $a^3 = a \cdot a^2 = a \cdot b = 1$, so $ab = 1$. Commutativity gives $ba = 1$. Also $b = a^2$ implies $b^2 = a^4 = a$ (since $a^3 = 1$), and the row/column "Sudoku" constraint for multiplication by a nonzero element gives that each nonzero row/column is a permutation of $\{1, a, b\}$. Finally, 0 times anything is 0.

Proof.

We determine the operation tables using only field axioms and the hint.

Step 1: the addition table. Since F is a field with $|F| = 4$, its characteristic is a prime p and $|F| = p^n$ for some $n \in \mathbb{N}$. Hence $4 = p^n$ forces $p = 2$. Therefore

$$1 + 1 = 0.$$

More generally, for any $x \in F$ we have $x + x = (1 + 1)x = 0 \cdot x = 0$, so

$$1 + 1 = a + a = b + b = 0.$$

Now fix any $x \in F$. By Problem 5(a) (applied in the additive group), the map $T_x : F \to F$ given by $T_x(y) = x+y$ is bijective. Hence each row and column of the addition table contains every element of F exactly once.

We already know the 0-row and 0-column are identities: $0 + y = y$ and $x + 0 = x$. In the 1-row we have $1+0 = 1$ and $1+1 = 0$, so $1+a$ and $1+b$ must be the remaining two elements a and b. But $1 + a$ cannot equal $0, 1$, or a (each would force a contradiction), so $1 + a = b$. Then the remaining entry is $1 + b = a$. Finally, using commutativity, $a + b = b + a$, and using $1 + a = b$ we get

$a+b = 1$ (since adding a to both sides of $1 + a = b$ gives $1 + (a + a) = b + a$, i.e. $1 = b + a$).

Thus the addition table is

$+$	0	1	a	b
0	0	1	a	b
1	1	0	b	a
a	a	b	0	1
b	b	a	1	0

Step 2: the multiplication table. The set $F \setminus \{0\} = \{1, a, b\}$ is an abelian group under $\cdot$ of order 3, hence cyclic. So the two elements a and b are the two non-identity elements of this cyclic group. In particular, $a \neq 1$ and $a^3 = 1$. Since $a^2 \neq 1$ (otherwise a would have order 2, impossible in a group of order 3), we must have

$$a^2 = b.$$

Then

$$a \cdot b = a \cdot a^2 = a^3 = 1,$$

so $ab = 1$. By commutativity, $ba = 1$. Also $b = a^2$ implies

$$b^2 = a^4 = a \cdot a^3 = a.$$

As usual in a field, $0 \cdot x = 0$ for all $x \in F$. Therefore the multiplication table is

$\cdot$	0	1	a	b
0	0	0	0	0
1	0	1	a	b
a	0	a	b	1
b	0	b	1	a

These tables are uniquely determined by the field axioms and the "Sudoku" constraints in the hint. $\square$

Appendix: Toolbox

Week 1 Recap: Induction and Functions

Natural numbers and induction. We use

$$\mathbb{N} = \{1, 2, 3, \ldots\}.$$

Principle of Mathematical Induction. Let $P(n)$ be a statement defined for all $n \in \mathbb{N}$. If

(a) $P(1)$ is true, and

(b) for all $n \in \mathbb{N}$, $P(n) \Rightarrow P(n+1)$,

then $P(n)$ is true for all $n \in \mathbb{N}$.

Functions. Let A, B be nonempty sets. A *function* $f : A \to B$ assigns to each $a \in A$ exactly one element $f(a) \in B$. The set A is the *domain* and B is the *codomain*.

Composition. If $f : A \to B$ and $g : B \to C$ are functions, their *composition* is the function $g \circ f : A \to C$ given by

$$(g \circ f)(a) = g(f(a)) \qquad (a \in A).$$

Image and preimage. Let $f : A \to B$.

- For $X \subseteq A$, the *image* of X under f is
$$f(X) = \{f(x) : x \in X\}.$$
- For $Y \subseteq B$, the *preimage* of Y under f is
$$f^{-1}(Y) = \{a \in A : f(a) \in Y\}.$$

Remark. For a general function $f : A \to B$, the notation $f^{-1}(Y)$ denotes a preimage and does *not* require f to be invertible.

Injective / Surjective / Bijective. Let $f : A \to B$.

- f is *injective* if for all $a_1, a_2 \in A$,
$$f(a_1) = f(a_2) \Rightarrow a_1 = a_2.$$
- f is *surjective* if $f(A) = B$, i.e. for every $b \in B$ there exists $a \in A$ such that $f(a) = b$.
- f is *bijective* if it is both injective and surjective.

Remark. These properties depend on both the rule defining f and the choice of domain A and codomain B.

Countability

Finite and countable. A set S is *finite* if $S = \varnothing$ or there exists $n \in \mathbb{N}$ and a bijection $\{1, 2, \ldots, n\} \to S$. A set S is *countably infinite* if there exists a bijection $\mathbb{N} \to S$. A set S is *countable* if it is finite or countably infinite.

Equivalent characterizations. The following are standard and often useful:

- S is countable $\iff$ there exists an injection $S \hookrightarrow \mathbb{N}$.
- S is countable $\iff$ there exists a surjection $\mathbb{N} \twoheadrightarrow S$.

Countability of $\mathbb{N} \times \mathbb{N}$. There are many explicit bijections $\mathbb{N} \times \mathbb{N} \to \mathbb{N}$ (e.g. "diagonal" enumeration / pairing functions). In proofs, we often use the fact that

$$\mathbb{N} \times \mathbb{N} \text{ is countable.}$$

Disjoint unions. If A and B are countable and $A \cap B = \varnothing$, then $A \cup B$ is countable (one can interleave enumerations, or inject $A \cup B$ into $\mathbb{N} \times \{1, 2\}$). More generally, if $(A_n)_{n \in \mathbb{N}}$ are countable and pairwise disjoint, then

$$\bigcup_{n \in \mathbb{N}} A_n \text{ is countable}$$

by coding an element $x \in A_n$ as the pair (n, k) where k is its index in an enumeration of A_n.

Relations and Equivalence Relations

Binary relations. A (binary) relation $\mathcal{R}$ on a set X is a subset $\mathcal{R} \subseteq X \times X$. We write $x \,\mathcal{R}\, y$ instead of $(x, y) \in \mathcal{R}$.

Equivalence relation. A relation $\sim$ on X is an *equivalence relation* if it is:

- *Reflexive:* $x \sim x$ for all $x \in X$.
- *Symmetric:* $x \sim y \Rightarrow y \sim x$.
- *Transitive:* $x \sim y$ and $y \sim z \Rightarrow x \sim z$.

Equivalence classes. If $\sim$ is an equivalence relation on X and $x \in X$, the *equivalence class* of x is

$$[x] := \{y \in X : y \sim x\}.$$

The set of all equivalence classes is denoted $X/{\sim}$.

Partition property. Equivalence relations and partitions are two sides of the same coin:

- The equivalence classes $\{[x] : x \in X\}$ form a partition of X (classes are disjoint and their union is X).
- Conversely, any partition of X defines an equivalence relation (declare two elements equivalent iff they lie in the same block).

Example: congruence mod 2 on $\mathbb{Z}$. For $a, b \in \mathbb{Z}$, define $a \sim b$ iff $b - a$ is divisible by 2. Then $\sim$ is an equivalence relation and the equivalence classes are the *even* and *odd* integers:

$$[0] = 2\mathbb{Z}, \qquad [1] = 2\mathbb{Z} + 1.$$

Fields

Definition (Field). A triple $(F, +, \cdot)$ is a *field* if F is a set with at least two elements and $+$ and $\cdot$ are operations on F satisfying:

Addition axioms.

(A1) (Closure) If $a, b \in F$, then $a + b \in F$.

(A2) (Commutativity) If $a, b \in F$, then $a + b = b + a$.

(A3) (Associativity) If $a, b, c \in F$, then $(a + b) + c = a + (b + c)$.

(A4) (Identity) There exists $0 \in F$ such that $a + 0 = 0 + a = a$ for all $a \in F$.

(A5) (Inverses) For every $a \in F$ there exists $-a \in F$ such that $a + (-a) = (-a) + a = 0$.

Multiplication axioms.

(M1) (Closure) If $a, b \in F$, then $a \cdot b \in F$.

(M2) (Commutativity) If $a, b \in F$, then $a \cdot b = b \cdot a$.

(M3) (Associativity) If $a, b, c \in F$, then $(a \cdot b) \cdot c = a \cdot (b \cdot c)$.

(M4) (Identity) There exists $1 \in F$ such that $a \cdot 1 = 1 \cdot a = a$ for all $a \in F$.

(M5) (Inverses) For every $a \in F \setminus \{0\}$ there exists $a^{-1} \in F$ such that $a \cdot a^{-1} = a^{-1} \cdot a = 1$.

Distributivity.

(D) For all $a, b, c \in F$, $(a + b) \cdot c = a \cdot c + b \cdot c$.

Basic consequences (often used without comment). Let $(F, +, \cdot)$ be a field.

- The additive and multiplicative identities 0 and 1 are unique.
- Additive inverses and multiplicative inverses (for $a \neq 0$) are unique.
- *Additive cancellation:* if $a + b = a + c$, then $b = c$.
- *Multiplicative cancellation:* if $a \neq 0$ and $a \cdot b = a \cdot c$, then $b = c$.
- $a \cdot 0 = 0$ for all $a \in F$.
- $(-a)(-b) = ab$ and $a(-b) = -(ab)$ for all $a, b \in F$.

Translation and dilation maps Fix $x \in F$.

- The map $T_x : F \to F$ defined by $T_x(y) = x + y$ has inverse T_{-x}.
- If $x \neq 0$, the map $M_x : F \to F$ defined by $M_x(y) = x \cdot y$ has inverse $M_{x^{-1}}$.

A Note on Finite Field Tables

In a field, $(F, +)$ is an abelian group and $(F \setminus \{0\}, \cdot)$ is an abelian group. Consequently:

- In the addition table, each row/column is a permutation of F.
- In the multiplication table, each *nonzero* row/column is a permutation of $F \setminus \{0\}$.

This is the "Sudoku" constraint your hint refers to.

Math 521, Section 2
Homework 3

Problem 1. Let $(F, +, \cdot)$ be a field.

(a) Prove that the multiplicative identity is unique: If $b \in F$ satisfies $b \cdot a = a \cdot b = a$ for all $a \in F$, then $b = 1$.
(b) Prove that multiplicative inverses are unique: If $a \in F \setminus \{0\}$ and $b \in F$ satisfy $b \cdot a = a \cdot b = 1$, then $b = a^{-1}$.
(c) Prove that if $a \cdot b = a \cdot c$ and $a \neq 0$, then $b = c$.
(d) Prove that if $a, b \in F$ then $-a + (-b) = -(a + b)$.
(e) Prove that if $a, b \in F$ and $a, b \neq 0$, then $a^{-1} \cdot b^{-1} = (ab)^{-1}$.

Problem 2. Given two points (x_1, y_1) and (x_2, y_2) in $\mathbb{R}^2 = \{(x, y) : x, y \in \mathbb{R}\}$, we write

$$(x_1, y_1) < (x_2, y_2) \quad \Longleftrightarrow \quad x_1 < x_2 \quad \text{or} \quad (x_1 = x_2 \text{ and } y_1 < y_2).$$

(a) Prove that $<$ is an order relation on $\mathbb{R}^2$.
(b) Prove that $(\mathbb{R}^2, <)$ does not satisfy the least upper bound property. (Hint: Consider the set of all points on the y-axis.)

Problem 3. Recall that the set of complex numbers $\mathbb{C}$ consists of all numbers of the form $a + ib$ for $a, b \in \mathbb{R}$. We can add and multiply complex numbers as follows:

$$(a + ib) + (c + id) = (a + c) + i(b + d) \quad \text{and} \quad (a + ib) \cdot (c + id) = (ac - bd) + i(ad + bc).$$

With these operations, it turns out $(\mathbb{C}, +, \cdot)$ is a field (but you do not need to prove this).

(a) The definition of a field says that for any $a + ib \neq 0 + i0$, there exists $c + id \in \mathbb{C}$ so that $(a + ib) \cdot (c + id) = 1 + i0$. What are $c, d \in \mathbb{R}$ in terms of $a, b \in \mathbb{R}$? Prove your answer.
(b) Prove that there is no order relation $<$ on $\mathbb{C}$ that makes $(\mathbb{C}, +, \cdot, <)$ an ordered field. (Hint: Can $i = 0 + i1$ be positive? Negative? Zero?)

Problem 4. Let $A \subseteq \mathbb{R}$ be a nonempty and bounded set, let $c \in \mathbb{R}$ be a number, and define

$$B = \{c \cdot a : a \in A\}.$$

(a) Prove that if $c > 0$, then $\sup B = c \cdot \sup A$.
(b) Prove that if $c < 0$, then $\sup B = c \cdot \inf A$.

Problem 5. Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded sets, and define

$$S = \{a + b : a \in A \text{ and } b \in B\}.$$

Prove that $\sup S = \sup A + \sup B$.

Problem 6. Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded sets. Prove or disprove the following statements:

(a) If $a \leq b$ for all $a \in A$ and for all $b \in B$, then $\sup A \leq \inf B$.
(b) If $a < b$ for all $a \in A$ and for all $b \in B$, then $\sup A < \inf B$.

Homework 3

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Contents

Problem 1

Let $(F, +, \cdot)$ be a field.

(a) Prove that the multiplicative identity is unique: If $b \in F$ satisfies $b \cdot a = a \cdot b = a$ for all $a \in F$, then $b = 1$.

(b) Prove that multiplicative inverses are unique: If $a \in F \setminus \{0\}$ and $b \in F$ satisfy $b \cdot a = a \cdot b = 1$, then $b = a^{-1}$.

(c) Prove that if $a \cdot b = a \cdot c$ and $a \neq 0$, then $b = c$.

(d) Prove that if $a, b \in F$ then $-a + (-b) = -(a + b)$.

(e) Prove that if $a, b \in F$ and $a, b \neq 0$, then $a^{-1} \cdot b^{-1} = (ab)^{-1}$.

Discussion & Scratch Work

Main ideas (and why we cannot "just divide"). In a general field, "division" is not a primitive operation. Instead, we justify any division-like step by introducing multiplicative inverses via **(M5)** and then using associativity **(M3)** and the identity **(M4)**.

- **(a)** Compare $b \cdot 1$ computed from the hypothesis and from **(M4)**.
- **(b)** Insert $1 = a \cdot a^{-1}$ (from **(M5)**) and reassociate (by **(M3)**) to force $b = a^{-1}$.
- **(c)** Prove cancellation by multiplying $a \cdot b = a \cdot c$ on the left by a^{-1} (existence from **(M5)**), then use **(M3)** and **(M4)**.
- **(d)** Show $(-a) + (-b)$ is an additive inverse of $(a + b)$ by checking $(a + b) + ((-a) + (-b)) = 0$ using **(A2)**, **(A3)**, **(A4)**, **(A5)**. Conclude it equals $-(a + b)$ by uniqueness of additive inverses (proved from the axioms, as in lecture).
- **(e)** Show $a^{-1} \cdot b^{-1}$ is an inverse of ab using commutativity **(M2)**, associativity **(M3)**, and **(M5)**. Then apply (b) to conclude $a^{-1}b^{-1} = (ab)^{-1}$.

Proof.
Let $(F, +, \cdot)$ be a field.

(a) Suppose $b \in F$ satisfies

$$b \cdot a = a \cdot b = a \quad \text{for all } a \in F.$$

Let $1 \in F$ be the multiplicative identity whose existence is guaranteed by **(M4)**. Setting $a = 1$ in the hypothesis gives

$$b \cdot 1 = 1.$$

But by **(M4)** (applied with $a = b$), we also have $b \cdot 1 = b$. Therefore $b = 1$, so the multiplicative identity is unique.

(b) Suppose $a \in F \setminus \{0\}$ and $b \in F$ satisfy

$$b \cdot a = a \cdot b = 1.$$

By the multiplicative inverse axiom **(M5)**, there exists $a^{-1} \in F$ such that

$$a \cdot a^{-1} = a^{-1} \cdot a = 1.$$

We prove that $b = a^{-1}$. Using the multiplicative identity axiom **(M4)**, we have

$$b = b \cdot 1.$$

Substituting $1 = a \cdot a^{-1}$ from **(M5)** gives

$$b = b \cdot (a \cdot a^{-1}).$$

By associativity of multiplication **(M3)**,

$$b \cdot (a \cdot a^{-1}) = (b \cdot a) \cdot a^{-1}.$$

Using the hypothesis $b \cdot a = 1$, this becomes

$$(b \cdot a) \cdot a^{-1} = 1 \cdot a^{-1}.$$

Finally, by **(M4)** we have $1 \cdot a^{-1} = a^{-1}$. Therefore $b = a^{-1}$, so the multiplicative inverse of a is unique.

(c) Suppose $a, b, c \in F$ satisfy $a \cdot b = a \cdot c$ and $a \neq 0$. By **(M5)**, there exists $a^{-1} \in F$ such that $a^{-1} \cdot a = 1$. Multiply the equality $a \cdot b = a \cdot c$ on the left by a^{-1}:

$$a^{-1} \cdot (a \cdot b) = a^{-1} \cdot (a \cdot c).$$

By associativity **(M3)**,

$$(a^{-1} \cdot a) \cdot b = (a^{-1} \cdot a) \cdot c.$$

Using $a^{-1} \cdot a = 1$ (from **(M5)**) and the identity axiom **(M4)**, we get

$$1 \cdot b = 1 \cdot c \quad \Rightarrow \quad b = c.$$

(d) We claim that $-a + (-b)$ is the additive inverse of $a + b$, hence equals $-(a + b)$. First compute, using associativity **(A3)** and commutativity **(A2)** of $+$,

$$\begin{aligned}
(a+b) + \big((-a) + (-b)\big) &= a + \Big(b + \big((-a) + (-b)\big)\Big) \quad \text{by } \textbf{(A3)} \\
&= a + \Big(\big(b + (-b)\big) + (-a)\Big) \quad \text{by } \textbf{(A3)} \text{ and } \textbf{(A2)} \\
&= a + \big(0 + (-a)\big) \quad \text{by } \textbf{(A5)} \\
&= a + (-a) \quad \text{by } \textbf{(A4)} \\
&= 0 \quad \text{by } \textbf{(A5)}.
\end{aligned}$$

Thus $(-a)+(-b)$ is an additive inverse of $a+b$.

Uniqueness of additive inverses (from axioms). If $x, y \in F$ both satisfy $(a+b)+x=0$ and $(a+b)+y=0$, then adding $-(a+b)$ to both sides and using **(A3)**, **(A4)**, **(A5)** yields $x=y$ (as in lecture). Therefore the additive inverse of $a+b$ is unique, and since $-(a+b)$ denotes that unique inverse, we conclude

$$-a+(-b)=(-a)+(-b)=-(a+b).$$

(e) Assume $a, b \in F$ with $a \neq 0$ and $b \neq 0$. By **(M5)**, there exist $a^{-1}, b^{-1} \in F$ such that $a \cdot a^{-1}=1$ and $b \cdot b^{-1}=1$. Consider

$$(ab)\cdot(a^{-1}\cdot b^{-1}).$$

By associativity **(M3)** and commutativity **(M2)**,

$$(ab)\cdot(a^{-1}b^{-1})=a\cdot b\cdot a^{-1}\cdot b^{-1}=(a\cdot a^{-1})\cdot(b\cdot b^{-1}).$$

Using **(M5)** and then **(M4)**, this equals $1\cdot 1=1$. Similarly,

$$(a^{-1}b^{-1})\cdot(ab)=1.$$

Hence $a^{-1}b^{-1}$ is a multiplicative inverse of ab. By part **(b)** (uniqueness of inverses),

$$a^{-1}\cdot b^{-1}=(ab)^{-1}.$$

$\square$

Problem 2

Given two points (x_1, y_1) and (x_2, y_2) in $\mathbb{R}^2 = \{(x, y) : x, y \in \mathbb{R}\}$, we write

$$(x_1, y_1) < (x_2, y_2) \Leftrightarrow x_1 < x_2 \text{ or } (x_1 = x_2 \text{ and } y_1 < y_2).$$

(a) Prove that $<$ is an order relation on $\mathbb{R}^2$.

(b) Prove that $(\mathbb{R}^2, <)$ does not satisfy the least upper bound property. (Hint: Consider the set of all points on the y-axis.)

Discussion & Scratch Work

Main ideas. This is the *lexicographic order* on $\mathbb{R}^2$.

- **(a)** To prove $<$ is an order relation on $\mathbb{R}^2$, verify the two defining properties: **trichotomy** and **transitivity**. The proofs reduce to the corresponding properties of the usual order on $\mathbb{R}$ by splitting into cases on the first coordinates.
- **(b)** To show the least upper bound property fails, use the set $A = \{(0, y) : y \in \mathbb{R}\}$ (the y-axis). It is bounded above (e.g. by $(1, 0)$), but any upper bound must have first coordinate > 0, and then you can always find a strictly smaller upper bound by decreasing the first coordinate (e.g. replace (x, y) by $(x/2, 0)$). Hence no least upper bound exists.

Proof.
Let $p = (x_1, y_1)$, $q = (x_2, y_2)$, and $r = (x_3, y_3)$ be points in $\mathbb{R}^2$.

(a) We prove that $<$ is an order relation on $\mathbb{R}^2$ in the sense of the lecture definition.

Trichotomy. By trichotomy of the usual order on $\mathbb{R}$, exactly one of $x_1 < x_2$, $x_1 = x_2$, or $x_2 < x_1$ holds.

- If $x_1 < x_2$, then $p < q$ by definition.
- If $x_2 < x_1$, then $q < p$ by definition.
- If $x_1 = x_2$, then by trichotomy on $\mathbb{R}$ applied to y_1, y_2, exactly one of $y_1 < y_2$, $y_1 = y_2$, or $y_2 < y_1$ holds. In the first case, $p < q$; in the second, $p = q$; in the third, $q < p$.

Therefore exactly one of $p < q$, $p = q$, or $q < p$ holds.

Transitivity. Assume $p < q$ and $q < r$. We show $p < r$.

If $x_1 < x_2$, then from $q < r$ we have either $x_2 < x_3$ or $x_2 = x_3$. In both cases $x_2 \le x_3$, hence $x_1 < x_3$, so $p < r$ by definition.

If instead $x_1 = x_2$, then $p < q$ forces $y_1 < y_2$. Now consider $q < r$:

- If $x_2 < x_3$, then $x_1 = x_2 < x_3$, so $p < r$.

- If $x_2 = x_3$, then $q < r$ forces $y_2 < y_3$, and by transitivity in $\mathbb{R}$ we get $y_1 < y_3$, so (since $x_1 = x_3$) we conclude $p < r$.

Thus $<$ is transitive.

Since $<$ satisfies trichotomy and transitivity, it is an order relation on $\mathbb{R}^2$.

(b) Let

$$A := \{(0, y) : y \in \mathbb{R}\} \subseteq \mathbb{R}^2.$$

Then A is nonempty (for example $(0, 0) \in A$). We claim A is bounded above in $(\mathbb{R}^2, <)$. Indeed, $(1, 0)$ is an upper bound, because for any $(0, y) \in A$ we have $0 < 1$, hence $(0, y) < (1, 0)$ by the definition of $<$.

We show A has *no* least upper bound. First note that if (u, v) is any upper bound of A, then necessarily $u > 0$. If $u < 0$, then $(0, 0) \in A$ would satisfy $(u, v) < (0, 0)$, so (u, v) could not be an upper bound. If $u = 0$, then being an upper bound would require $y \leq v$ for all $y \in \mathbb{R}$, which is impossible since A contains all points $(0, y)$ with $y \in \mathbb{R}$ and $\mathbb{R}$ is unbounded above.

Now take any upper bound (u, v) of A; then $u > 0$. Define $w := (u/2, 0)$. For any $(0, y) \in A$ we have $0 < u/2$, hence $(0, y) < (u/2, 0) = w$, so w is also an upper bound of A. But $w < (u, v)$ because $u/2 < u$.

Therefore every upper bound of A is strictly larger than some other upper bound of A, so no least upper bound exists. Hence $(\mathbb{R}^2, <)$ does not satisfy the least upper bound property. $\square$

Problem 3

Recall that the set of complex numbers $\mathbb{C}$ consists of all numbers of the form $a + ib$ for $a, b \in \mathbb{R}$. We can add and multiply complex numbers as follows:

$$(a + ib) + (c + id) = (a + c) + i(b + d) \quad \text{and} \quad (a + ib) \cdot (c + id) = (ac - bd) + i(ad + bc).$$

With these operations, it turns out $(\mathbb{C}, +, \cdot)$ is a field (but you do not need to prove this).

(a) The definition of a field says that for any $a + ib \neq 0 + i0$, there exists $c + id \in \mathbb{C}$ so that $(a + ib) \cdot (c + id) = 1 + i0$. What are $c, d \in \mathbb{R}$ in terms of $a, b \in \mathbb{R}$? Prove your answer.

(b) Prove that there is no order relation $<$ on $\mathbb{C}$ that makes $(\mathbb{C}, +, \cdot, <)$ an ordered field. (Hint: Can $i = 0 + i1$ be positive? Negative? Zero?)

Discussion & Scratch Work

Main ideas.

- **(a)** To find $(a+ib)^{-1}$, write an unknown inverse as $c+id$ and expand $(a+ib)(c+id) = (ac - bd) + i(ad + bc)$. Matching real and imaginary parts with $1 + i0$ gives a 2×2 linear system. Solve it to get

$$c = \frac{a}{a^2 + b^2}, \qquad d = -\frac{b}{a^2 + b^2}.$$

  The key point is that $a^2 + b^2 \neq 0$ whenever $a + ib \neq 0$.

- **(b)** In an ordered field, we always have $1 > 0$ and $x \neq 0 \Rightarrow x^2 > 0$ (Appendix, **(P3)**). But in $\mathbb{C}$ we have $i^2 = -1$. Trichotomy forces $i > 0$, $i = 0$, or $i < 0$; each case contradicts $x^2 > 0$ together with $1 > 0$. So no order can make $\mathbb{C}$ into an ordered field.

Proof.

(a) Let $a + ib \in \mathbb{C}$ with $a + ib \neq 0 + i0$. We seek $c + id \in \mathbb{C}$ such that

$$(a + ib)(c + id) = 1 + i0.$$

Using the given multiplication rule,

$$(a + ib)(c + id) = (ac - bd) + i(ad + bc).$$

Thus we require

$$ac - bd = 1, \qquad ad + bc = 0. \tag{1}$$

Multiply the first equation in (1) by a and the second by b, then add:

$$a(ac - bd) + b(ad + bc) = a.$$

The left-hand side simplifies to $(a^2 + b^2)c$, so

$$(a^2 + b^2)c = a.$$

Similarly, multiply the second equation by a and the first by b, then subtract:

$$a(ad + bc) - b(ac - bd) = 0 - b,$$

whose left-hand side simplifies to $(a^2 + b^2)d$, so

$$(a^2 + b^2)d = -b.$$

Because $a + ib \neq 0$, we have $(a, b) \neq (0, 0)$, hence $a^2 + b^2 > 0$ (since $a^2 \geq 0$ and $b^2 \geq 0$, and not both are zero, we have $a^2 + b^2 > 0$) and in particular $a^2 + b^2 \neq 0$. Therefore we may divide by $a^2 + b^2$ to obtain

$$c = \frac{a}{a^2 + b^2}, \qquad d = -\frac{b}{a^2 + b^2}.$$

A direct substitution into (1) (or into $(ac - bd) + i(ad + bc)$) shows that these values indeed yield $(a+ib)(c+id) = 1+i0$. This proves the claimed formula for the multiplicative inverse.

(b) Suppose, for contradiction, that there exists an order relation $<$ on $\mathbb{C}$ such that $(\mathbb{C}, +, \cdot, <)$ is an ordered field. By Appendix **(P3)**, we have $1 > 0$ and, for any $x \in \mathbb{C}$, $x \neq 0 \Rightarrow x^2 > 0$. By trichotomy, exactly one of $i > 0$, $i = 0$, or $i < 0$ holds. We treat each case:

- If $i = 0$, then $i^2 = 0$, contradicting $i^2 = -1$.
- If $i > 0$, then $i \neq 0$, so Appendix **(P3)** implies $i^2 > 0$. But $i^2 = -1$, so $-1 > 0$, hence $1 < 0$, contradicting $1 > 0$.
- If $i < 0$, then $-i > 0$ (using Appendix **(P1)** with $a = i$). Since $-i \neq 0$, Appendix **(P3)** gives $(-i)^2 > 0$. But $(-i)^2 = i^2 = -1$, leading again to $-1 > 0$ and hence $1 < 0$, a contradiction.

All cases contradict the ordered-field consequences, so no such order relation exists. $\square$

Problem 4

Let $A \subseteq \mathbb{R}$ be a nonempty and bounded set, let $c \in \mathbb{R}$ be a number, and define

$$B = \{c \cdot a : a \in A\}.$$

(a) Prove that if $c > 0$, then $\sup B = c \cdot \sup A$.

(b) Prove that if $c < 0$, then $\sup B = c \cdot \inf A$.

Discussion & Scratch Work

> **Main idea.** We prove an equality of suprema by showing the candidate value is (i) an upper bound for B and (ii) the *least* such upper bound. The only subtlety is that we may multiply inequalities by a positive number without changing the direction (ordered-field axiom **(O2)**), while multiplying by a negative number reverses the inequality (Appendix **(P2)**).
>
> - If $c > 0$, then $a \leq \sup A$ implies $ca \leq c \sup A$, so $c \sup A$ is an upper bound for B. For leastness, any upper bound M of B yields $ca \leq M$ for all $a \in A$, hence $a \leq M/c$ after multiplying by $c^{-1} > 0$, so M/c is an upper bound for A and $\sup A \leq M/c$.
> - If $c < 0$, then $\inf A \leq a$ implies $c \inf A \geq ca$, so $c \inf A$ is an upper bound for B. For leastness, if $ca \leq M$ for all $a \in A$, multiplying by $c^{-1} < 0$ reverses the inequality and gives $a \geq M/c$, so M/c is a lower bound for A and hence $M/c \leq \inf A$.

Proof.
Let $A \subseteq \mathbb{R}$ be nonempty and bounded, let $c \in \mathbb{R}$, and set $B = \{ca : a \in A\}$. We use the definitions of sup and inf from the Appendix.

(a) Assume $c > 0$. We prove $\sup B = c \sup A$.

Step 1: $c \sup A$ is an upper bound for B. Let $b \in B$. Then $b = ca$ for some $a \in A$. Because $\sup A$ is an upper bound for A, we have $a \leq \sup A$. Since $c > 0$, multiplying by c preserves the inequality (ordered-field axiom **(O2)**), so

$$ca \leq c \sup A.$$

Thus $b \leq c \sup A$, and $c \sup A$ is an upper bound for B.

Step 2: $c \sup A$ is the least upper bound for B. Let $M \in \mathbb{R}$ be any upper bound for B. Then for every $a \in A$, we have $ca \in B$, hence

$$ca \leq M.$$

Because $c > 0$, we have $c \neq 0$ and $c^{-1} > 0$. Multiplying by c^{-1} preserves the inequality (**(O2)**), giving

$$a \leq c^{-1}M = \frac{M}{c} \quad \text{for all } a \in A.$$

So M/c is an upper bound for A. By definition of $\sup A$, this implies $\sup A \leq M/c$. Multiplying by $c > 0$ again preserves the inequality (**(O2)**), so

$$c \sup A \leq M.$$

Therefore $c \sup A$ is the least upper bound of B, i.e. $\sup B = c \sup A$.

(b) Assume $c < 0$. We prove $\sup B = c \inf A$.

Step 1: $c \inf A$ is an upper bound for B. Let $b \in B$, so $b = ca$ for some $a \in A$. Because $\inf A$ is a lower bound for A, we have $\inf A \leq a$. Since $c < 0$, multiplying by c reverses the inequality (Appendix **(P2)**), hence

$$c \inf A \geq ca.$$

Equivalently, $ca \leq c \inf A$, so $b \leq c \inf A$. Thus $c \inf A$ is an upper bound for B.

Step 2: $c \inf A$ is the least upper bound for B. Let $M \in \mathbb{R}$ be any upper bound for B. Then for every $a \in A$,

$$ca \leq M.$$

Because $c < 0$, we have $c \neq 0$ and $c^{-1} < 0$. Multiplying by c^{-1} reverses the inequality (Appendix **(P2)**), so

$$a \geq c^{-1}M = \frac{M}{c} \quad \text{for all } a \in A.$$

Thus M/c is a lower bound for A. By definition of $\inf A$ (greatest lower bound), we get $M/c \leq \inf A$. Multiplying by $c < 0$ reverses the inequality again (Appendix **(P2)**), yielding

$$M \geq c \inf A.$$

So every upper bound M of B satisfies $c \inf A \leq M$, hence $c \inf A$ is the least upper bound of B. Therefore $\sup B = c \inf A$. $\square$

Problem 5

Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded sets, and define

$$S = \{a + b : a \in A \text{ and } b \in B\}.$$

Prove that $\sup S = \sup A + \sup B$.

Discussion & Scratch Work

Main idea. Let $\alpha := \sup A$ and $\beta := \sup B$. We show

$$\sup S = \alpha + \beta$$

by the standard two-step pattern for suprema:

- **Upper bound:** For any $s \in S$, write $s = a + b$ with $a \in A$, $b \in B$. Since $a \le \alpha$ and $b \le \beta$, we get $s \le \alpha + \beta$. Hence $\alpha + \beta$ is an upper bound for S.
- **Leastness:** If M is any upper bound for S, then for each fixed $b \in B$ we have $a + b \le M$ for all $a \in A$, hence $a \le M - b$. Thus $M - b$ is an upper bound for A, so $\alpha \le M - b$, i.e. $\alpha + b \le M$ for all $b \in B$. Therefore $M - \alpha$ is an upper bound for B, so $\beta \le M - \alpha$, which gives $\alpha + \beta \le M$.

This proves $\alpha + \beta$ is the least upper bound of S.

Proof.
Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded, and set

$$S := \{a + b : a \in A \text{ and } b \in B\}.$$

Let $\alpha := \sup A$ and $\beta := \sup B$.

Let $s \in S$. Then there exist $a \in A$ and $b \in B$ with $s = a + b$. Since α is an upper bound for A and β is an upper bound for B, we have $a \le \alpha$ and $b \le \beta$. Adding these inequalities gives

$$a + b \le \alpha + \beta,$$

so $s \le \alpha + \beta$. Hence $\alpha + \beta$ is an upper bound for S.

Let $M \in \mathbb{R}$ be any upper bound for S. Fix $b \in B$. For every $a \in A$, we have $a + b \in S$, hence $a + b \le M$. Adding $-b$ to both sides (by **(O1)**) yields $a \le M - b$ for all $a \in A$, so $M - b$ is an upper bound for A. By definition of $\alpha = \sup A$, we get

$$\alpha \le M - b,$$

which implies $\alpha + b \le M$. Because this holds for every $b \in B$, we have $b \le M - \alpha$ for all $b \in B$, so $M - \alpha$ is an upper bound for B. By definition of $\beta = \sup B$, we obtain

$$\beta \le M - \alpha.$$

Adding α to both sides gives $\alpha + \beta \le M$.

Since $\alpha + \beta$ is an upper bound for S and is $\le$ every upper bound M of S, it is the least upper bound. Therefore

$$\sup S = \alpha + \beta = \sup A + \sup B.$$

$\square$

Problem 6

Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded sets. Prove or disprove the following statements:

(a) If $a \leq b$ for all $a \in A$ and for all $b \in B$, then $\sup A \leq \inf B$.

(b) If $a < b$ for all $a \in A$ and for all $b \in B$, then $\sup A < \inf B$.

Discussion & Scratch Work

Main ideas. Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded.

- **(a) True.** The hypothesis

$$a \leq b \quad \text{for all } a \in A \text{ and all } b \in B$$

says: *every element of B is an upper bound for A.* By the definition of supremum, $\sup A$ is *below every* upper bound of A, hence $\sup A \leq b$ for all $b \in B$. That means $\sup A$ is a *lower bound* for B, so by the definition of infimum (greatest lower bound), $\sup A \leq \inf B$.

- **(b) False in general.** Even if $a < b$ for all $a \in A$ and $b \in B$, it can still happen that $\sup A = \inf B$ because $\sup A$ need not belong to A. A clean counterexample is

$$A = (0, 1), \qquad B = \{1\}.$$

Then $a < 1$ for all $a \in (0, 1)$, but $\sup A = 1$ and $\inf B = 1$.

Proof.
Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded.

(a) Assume that $a \leq b$ for all $a \in A$ and all $b \in B$. We prove that $\sup A \leq \inf B$.

Fix any $b \in B$. By the hypothesis, $a \leq b$ for every $a \in A$, so b is an upper bound for A (Appendix, Definition **(B1)**). Let $\alpha := \sup A$. By the defining property of the supremum (Appendix, Definition **(S1)**), we have

$$\alpha \leq b.$$

Because $b \in B$ was arbitrary, we conclude that $\alpha \leq b$ for all $b \in B$, i.e. α is a lower bound for B (Appendix, Definition **(B2)**).

Let $\beta := \inf B$. By the defining property of the infimum (Appendix, Definition **(S2)**), β is the *greatest* lower bound of B, so every lower bound of B is $\leq \beta$. In particular, since α is a lower bound of B, we obtain

$$\sup A = \alpha \leq \beta = \inf B.$$

(b) The statement is *false*. Consider

$$A := (0, 1), \qquad B := \{1\}.$$

Then A and B are nonempty and bounded subsets of $\mathbb{R}$. Moreover, for every $a \in A$ and every $b \in B$ we have $b = 1$ and $0 < a < 1$, hence $a < b$.

However, $\sup A = 1$ (because 1 is an upper bound for $(0, 1)$ and any smaller number fails to be an upper bound), and clearly $\inf B = 1$ because B consists of the single element 1. Therefore

$$\sup A = 1 = \inf B,$$

so it is *not* true that $\sup A < \inf B$ in general. $\square$

Appendix: Toolbox

Basically a summary of lectures 6—9; something like a little cheatsheet review. Anyways, below is the good stuff.

Field axioms

(A1) **(Closure for $+$)** If $a, b \in F$, then $a + b \in F$.

(A2) **(Commutativity of $+$)** If $a, b \in F$, then $a + b = b + a$.

(A3) **(Associativity of $+$)** If $a, b, c \in F$, then $(a + b) + c = a + (b + c)$.

(A4) **(Additive identity)** There exists $0 \in F$ such that $a + 0 = 0 + a = a$ for all $a \in F$.

(A5) **(Additive inverses)** For each $a \in F$ there exists $-a \in F$ such that $a + (-a) = (-a) + a = 0$.

(M1) **(Closure for $\cdot$)** If $a, b \in F$, then $a \cdot b \in F$.

(M2) **(Commutativity of $\cdot$)** If $a, b \in F$, then $a \cdot b = b \cdot a$.

(M3) **(Associativity of $\cdot$)** If $a, b, c \in F$, then $(a \cdot b) \cdot c = a \cdot (b \cdot c)$.

(M4) **(Multiplicative identity)** There exists $1 \in F$ such that $a \cdot 1 = 1 \cdot a = a$ for all $a \in F$.

(M5) **(Multiplicative inverses)** For each $a \in F \setminus \{0\}$ there exists $a^{-1} \in F$ such that $a \cdot a^{-1} = a^{-1} \cdot a = 1$.

(D) **(Distributivity)** If $a, b, c \in F$, then $(a + b) \cdot c = a \cdot c + b \cdot c$.

Derived properties of fields (proved from the axioms in Lecture 7)

(1) The additive and multiplicative identities are unique.

(2) Additive and multiplicative inverses are unique.

(3) If $a, b, c \in F$ satisfy $a + b = a + c$, then $b = c$. If $a, b, c \in F$ satisfy $a \cdot b = a \cdot c$ and $a \neq 0$, then $b = c$.

(4) For all $a \in F$, we have $a \cdot 0 = 0 \cdot a = 0$.

(5) For all $a, b \in F$, we have $(-a) \cdot b = a \cdot (-b) = -(a \cdot b)$.

(6) For all $a, b \in F$, we have $(-a) \cdot (-b) = a \cdot b$.

(7) If $a \cdot b = 0$, then $a = 0$ or $b = 0$.

Order relations and ordered fields (Lectures 7–8)

Order relations

Definition (Order relation). Let A be a nonempty set. An *order relation* (or *inequality*) on A is a relation $<$ on A such that:

(OR1) **(Trichotomy)** If $a, b \in A$, then exactly one of the following holds: $a < b$, or $a = b$, or $b < a$.

(OR2) **(Transitivity)** If $a, b, c \in A$ satisfy $a < b$ and $b < c$, then $a < c$.

Notation.

- $a > b$ means $b < a$.
- $a \leq b$ means $(a < b)$ or $(a = b)$.
- $a \geq b$ means $(b < a)$ or $(a = b)$.

Ordered fields

Definition (Ordered field). Let $(F, +, \cdot)$ be a field. We say that $(F, +, \cdot, <)$ is an *ordered field* if it is equipped with an order relation $<$ on F such that:

(O1) If $a, b, c \in F$ satisfy $a < b$, then $a + c < b + c$.

(O2) If $a, b, c \in F$ satisfy $a < b$ and $0 < c$, then $a \cdot c < b \cdot c$.

Useful consequences (proved in Lecture 8). Let $(F, +, \cdot, <)$ be an ordered field.

(P1) $a > 0 \iff -a < 0$.

(P2) If $a, b, c \in F$ satisfy $a < b$ and $c < 0$, then $a \cdot c > b \cdot c$.

(P3) If $a \in F$ satisfies $a \neq 0$, then $a^2 = a \cdot a > 0$. In particular, $1 > 0$.

(P4) If $a, b \in F$ satisfy $0 < a < b$, then $0 < b^{-1} < a^{-1}$.

Bounds, maxima/minima, and suprema/infima (Lecture 8)

Bounds and extrema

Definition (Bounds; max/min). Let $(F, +, \cdot, <)$ be an ordered field, and let $A \subseteq F$ be nonempty.

(B1) A is *bounded above* if there exists $M \in F$ such that $a \leq M$ for all $a \in A$. Any such M is an *upper bound* for A. If, in addition, $M \in A$, then M is the *maximum* of A.

(B2) A is *bounded below* if there exists $m \in F$ such that $m \leq a$ for all $a \in A$. Any such m is a *lower bound* for A. If, in addition, $m \in A$, then m is the *minimum* of A.

(B3) A is *bounded* if it is both bounded above and bounded below.

Remark (Max/min need not exist; uniqueness). Some sets do not have a maximum (or a minimum). If a maximum exists, then it is unique: if $M, M' \in A$ both satisfy $a \leq M$ for all $a \in A$ and $a \leq M'$ for all $a \in A$, then $M \leq M'$ and $M' \leq M$, hence $M = M'$. (The same statement holds for minima.)

Remark. Even if a set does not have a maximum, it may still have (at least one) upper bound.

Least upper bounds and greatest lower bounds

Definition (Supremum/infimum). Let $(F, +, \cdot, <)$ be an ordered field, and let $A \subseteq F$ be nonempty.

(S1) A number $L \in F$ is the *least upper bound* (or *supremum*) of A if:

- L is an upper bound for A, and
- if M is an upper bound for A, then $L \leq M$.

When this is true, we write $L = \sup A$.

(S2) A number $\ell \in F$ is the *greatest lower bound* (or *infimum*) of A if:

- ℓ is a lower bound for A, and
- if m is a lower bound for A, then $m \leq \ell$.

When this is true, we write $\ell = \inf A$.

Lemma (Uniqueness of the supremum). If $A \subseteq F$ has a least upper bound, then it is unique.

Proof. Suppose $L, M \in F$ are both least upper bounds for A. Then L is an upper bound and, since M is a least upper bound, we have $M \leq L$. Similarly, M is an upper bound and, since L is a least upper bound, we have $L \leq M$. Therefore $L = M$. □

Completeness and the real numbers (Lectures 8–9)

Proposition 1 (Irrationality of $\sqrt{2}$)**.** *We have* $\sqrt{2} \notin \mathbb{Q}$.

Proof. Suppose, for contradiction, that $\sqrt{2} \in \mathbb{Q}$. Then we may write

$$\sqrt{2} = \frac{a}{b}$$

for some integers $a \in \mathbb{Z}$ and $b \in \mathbb{N}$ with no common factors (i.e. the fraction is in lowest terms). Squaring gives

$$2 = \frac{a^2}{b^2} \qquad \Rightarrow \qquad 2b^2 = a^2.$$

Thus a^2 is even, hence a is even. Write $a = 2k$ for some $k \in \mathbb{Z}$. Substituting yields

$$2b^2 = (2k)^2 = 4k^2 \qquad \Rightarrow \qquad b^2 = 2k^2.$$

So b^2 is even, hence b is even. Therefore both a and b are even, contradicting that a/b was in lowest terms. Hence $\sqrt{2} \notin \mathbb{Q}$. ☐

Definition 1 (Least upper bound property)**.** *Let $<$ be an order relation on a set F. We say that F has the* least upper bound property *if for every nonempty subset $A \subseteq F$ that is bounded above, there exists a least upper bound of A that belongs to F (i.e.* $\sup A \in F$*).*

Remark. *The ordered field $(\mathbb{Q}, +, \cdot, <)$ does* not *have the least upper bound property. For example,*

$$A := \{x \in \mathbb{Q} : x^2 < 2\}$$

is nonempty and bounded above in $\mathbb{Q}$, but A has no supremum in $\mathbb{Q}$ (its "expected" supremum would be $\sqrt{2}$, which is not rational).

Theorem 1 (Existence and uniqueness of $\mathbb{R}$ as a complete ordered field)**.** *There exists an ordered field with the least upper bound property. We denote it by*

$$(\mathbb{R}, +, \cdot, <).$$

This ordered field is unique up to order-preserving field isomorphism, and moreover $\mathbb{R}$ contains $\mathbb{Q}$ as a subfield.

Proposition 2 (Existence of infima in $\mathbb{R}$)**.** *If $A \subseteq \mathbb{R}$ is nonempty and bounded below, then* $\inf A$ *exists in $\mathbb{R}$.*

Proof. Assume $A \subseteq \mathbb{R}$ is nonempty and bounded below. Define

$$-A := \{-a : a \in A\}.$$

Then $-A$ is nonempty, and $-A$ is bounded above (if m is a lower bound for A, then $-m$ is an upper bound for $-A$). By the least upper bound property of $\mathbb{R}$, $\sup(-A)$ exists in $\mathbb{R}$. We claim that

$$\inf A = -\sup(-A).$$

Let $L := \sup(-A)$. For any $a \in A$ we have $-a \in -A$, hence $-a \leq L$, i.e. $-L \leq a$. So $-L$ is a lower bound for A. If m is any lower bound for A, then $m \leq a$ for all $a \in A$, so $-a \leq -m$ for all $a \in A$, meaning $-m$ is an upper bound for $-A$. By definition of $L = \sup(-A)$, we have $L \leq -m$, hence $m \leq -L$. Thus $-L$ is the greatest lower bound of A, so $\inf A = -L = -\sup(-A)$. ☐

Remark (A useful identity)**.** *For any nonempty $A \subseteq \mathbb{R}$ that is bounded below,*

$$\inf A = -\sup(-A), \qquad \textit{where } -A := \{-a : a \in A\}.$$

Conceptually: turning a lower-bound problem for A into an upper-bound problem for $-A$ lets us apply completeness (the LUB property).

Math 521, Section 2
Homework 4

Problem 1. Let $\{a_n\}_{n\geq 1}$ be a sequence.

(a) Prove that if $\{a_n\}_{n\geq 1}$ converges to a, then the sequence $\{|a_n|\}_{n\geq 1}$ converges to $|a|$. (Hint: $\big||x|-|y|\big| \leq |x-y|$ for any $x, y \in \mathbb{R}$.)
(b) Provide an example where $\{|a_n|\}_{n\geq 1}$ converges, but $\{a_n\}_{n\geq 1}$ diverges.

Problem 2 (Squeeze theorem)**.** Prove that if $\{a_n\}_{n\geq 1}$, $\{b_n\}_{n\geq 1}$, and $\{c_n\}_{n\geq 1}$ are sequences such that

$$a_n \leq b_n \leq c_n \text{ for all } n \geq 1 \quad \text{and} \quad \lim_{n\to\infty} a_n = L = \lim_{n\to\infty} c_n,$$

then $\{b_n\}_{n\geq 1}$ converges and $\lim_{n\to\infty} b_n = L$.

Problem 3. Suppose that the sequence $\{a_n\}_{n\geq 1}$ converges.

(a) Prove that if $a_n \geq m$ for all but finitely many n, then $\lim_{n\to\infty} a_n \geq m$.
(b) Prove that if $a_n \leq M$ for all but finitely many n, then $\lim_{n\to\infty} a_n \leq M$.

Problem 4. Suppose that the sequence $\{a_n\}_{n\geq 1}$ converges. Prove or disprove the following statements:

(a) If $\lim_{n\to\infty} a_n \geq m$, then there exists $N \in \mathbb{N}$ so that $a_n \geq m$ for all $n \geq N$.
(b) If $\lim_{n\to\infty} a_n > m$, then there exists $N \in \mathbb{N}$ so that $a_n > m$ for all $n \geq N$.

Problem 5. Prove that if $\{a_n\}_{n\geq 1}$ is a sequence of positive numbers that converges to $a > 0$, then the sequence $\{\sqrt{a_n}\}_{n\geq 1}$ converges to $\sqrt{a}$.

Problem 6. Consider the following sequence:

$$a_1 = \sqrt{2} \quad \text{and} \quad a_{n+1} = \sqrt{2 + a_n} \quad \text{for all } n \geq 1.$$

(a) Prove that the sequence $\{a_n\}_{n\geq 1}$ is bounded.
(b) Prove that the sequence is increasing.
(c) Conclude that the sequence converges, and find its limit.

Homework 4

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Problem 1

Let $\{a_n\}_{n\geq 1}$ be a sequence.

(a) Prove that if $\{a_n\}_{n\geq 1}$ converges to a, then the sequence $\{|a_n|\}_{n\geq 1}$ converges to $|a|$. (Hint: $||x| - |y|| \leq |x - y|$ for any $x, y \in \mathbb{R}$.)

(b) Provide an example where $\{|a_n|\}_{n\geq 1}$ converges, but $\{a_n\}_{n\geq 1}$ diverges.

Discussion & Scratch Work

- **(a)** The idea is: if $a_n \to a$, then eventually a_n is very close to a. The absolute value function is "distance from 0," so if two numbers are close, their distances from 0 are also close. The key inequality is

$$\big||x| - |y|\big| \leq |x - y|,$$

  which lets us bound $\big||a_n| - |a|\big|$ by $|a_n - a|$ and then use the definition of convergence.
- **(b)** To make $|a_n|$ converge while a_n diverges, we can alternate signs: take $a_n = (-1)^n$. Then $|a_n| = 1$ for all n (so it converges), but a_n oscillates between 1 and -1 (so it cannot converge).

Proof.
(a) Assume $a_n \to a$. Let $\varepsilon > 0$ be given. By convergence, there exists $N \in \mathbb{N}$ such that for all $n \geq N$ we have $|a_n - a| < \varepsilon$. Using the inequality $\big||x| - |y|\big| \leq |x - y|$ (valid for all $x, y \in \mathbb{R}$), with $x = a_n$ and $y = a$, we obtain for all $n \geq N$:

$$\big||a_n| - |a|\big| \leq |a_n - a| < \varepsilon.$$

Hence, by the definition of convergence, $|a_n|$ converges to $|a|$.

(b) Define $a_n := (-1)^n$. Then $|a_n| = 1$ for all n, so $\{|a_n|\}$ converges to 1. However, $\{a_n\}$ does not converge: if it converged to some $L \in \mathbb{R}$, then the subsequences $a_{2k} = 1$ and $a_{2k+1} = -1$ would both have to converge to L, forcing $1 = L = -1$, a contradiction. Therefore $\{|a_n|\}$ converges while $\{a_n\}$ diverges. $\square$

Problem 2

(Squeeze theorem). Prove that if $\{a_n\}_{n\geq 1}$, $\{b_n\}_{n\geq 1}$, and $\{c_n\}_{n\geq 1}$ are sequences such that $a_n \leq b_n \leq c_n$ for all $n \geq 1$ and $\lim_{n\to\infty} a_n = L = \lim_{n\to\infty} c_n$, then $\{b_n\}_{n\geq 1}$ converges and $\lim_{n\to\infty} b_n = L$.

Discussion & Scratch Work

- Intuition: b_n is "trapped" between a_n and c_n. If both outer sequences converge to the same limit L, then for any small tolerance $\varepsilon > 0$, eventually both a_n and c_n lie inside the interval $(L-\varepsilon,\, L+\varepsilon)$. Since $a_n \leq b_n \leq c_n$, the middle term must also lie in that same interval, forcing $b_n \to L$.
- We will use the definition of convergence:

$$a_n \to L \iff \forall \varepsilon > 0\ \exists N \in \mathbb{N} \text{ s.t. } \forall n \geq N,\ |a_n - L| < \varepsilon,$$

  (and similarly for c_n). The only "construction" step is choosing a single index after which *both* conditions hold; we do this by taking a maximum.

Proof.
Assume that $a_n \leq b_n \leq c_n$ for all $n \geq 1$, and that $\lim_{n\to\infty} a_n = L = \lim_{n\to\infty} c_n$. We prove that $b_n \to L$.

Let $\varepsilon > 0$ be given. Since $a_n \to L$, there exists $N_1 \in \mathbb{N}$ such that for all $n \geq N_1$,

$$|a_n - L| < \varepsilon \quad \Longrightarrow \quad L - \varepsilon < a_n < L + \varepsilon.$$

Similarly, since $c_n \to L$, there exists $N_2 \in \mathbb{N}$ such that for all $n \geq N_2$,

$$|c_n - L| < \varepsilon \quad \Longrightarrow \quad L - \varepsilon < c_n < L + \varepsilon.$$

Set $N := \max\{N_1, N_2\}$. Then for all $n \geq N$, both inequalities hold, and we have

$$L - \varepsilon < a_n \leq b_n \leq c_n < L + \varepsilon.$$

In particular, for all $n \geq N$,

$$L - \varepsilon < b_n < L + \varepsilon,$$

which is equivalent to $|b_n - L| < \varepsilon$. Since $\varepsilon > 0$ was arbitrary, this proves that $b_n \to L$. $\square$

Problem 3

Suppose that the sequence $\{a_n\}_{n\geq 1}$ converges.

(a) Prove that if $a_n \geq m$ for all but finitely many n, then $\lim_{n\to\infty} a_n \geq m$.

(b) Prove that if $a_n \leq M$ for all but finitely many n, then $\lim_{n\to\infty} a_n \leq M$.

Discussion & Scratch Work

- The key idea is: convergence prevents the sequence from staying *strictly below* a number that it is eventually always above. If $a_n \to \ell$ and (eventually) $a_n \geq m$, then the limit *cannot* satisfy $\ell < m$, because choosing $\varepsilon = \frac{m-\ell}{2}$ would force $a_n < \ell + \varepsilon < m$ for all large n, contradicting $a_n \geq m$ for all large n.
- Part (b) is the symmetric argument: if (eventually) $a_n \leq M$, then the limit cannot be $> M$.

Proof.
Let $\ell := \lim_{n\to\infty} a_n$.
(a) Assume that $a_n \geq m$ for all but finitely many n. Then there exists $N_0 \in \mathbb{N}$ such that

$$a_n \geq m \quad \text{for all } n \geq N_0.$$

We claim that $\ell \geq m$. Suppose for contradiction that $\ell < m$. Set

$$\varepsilon := \frac{m-\ell}{2} > 0.$$

Since $a_n \to \ell$, there exists $N_1 \in \mathbb{N}$ such that for all $n \geq N_1$,

$$|a_n - \ell| < \varepsilon \quad \Longrightarrow \quad a_n < \ell + \varepsilon = \frac{\ell + m}{2} < m.$$

Let $N := \max\{N_0, N_1\}$. Then for all $n \geq N$ we have simultaneously $a_n \geq m$ and $a_n < m$, a contradiction. Hence $\ell \geq m$.

(b) Assume that $a_n \leq M$ for all but finitely many n. Then there exists $N_0 \in \mathbb{N}$ such that $a_n \leq M$ for all $n \geq N_0$. We claim that $\ell \leq M$. Suppose for contradiction that $\ell > M$. Set $\varepsilon := \frac{\ell - M}{2} > 0$. Since $a_n \to \ell$, there exists $N_1 \in \mathbb{N}$ such that for all $n \geq N_1$,

$$|a_n - \ell| < \varepsilon \quad \Longrightarrow \quad a_n > \ell - \varepsilon = \frac{\ell + M}{2} > M.$$

Let $N := \max\{N_0, N_1\}$. Then for all $n \geq N$ we have simultaneously $a_n \leq M$ and $a_n > M$, a contradiction. Hence $\ell \leq M$. $\square$

Problem 4

Suppose that the sequence $\{a_n\}_{n\geq 1}$ converges. Prove or disprove the following statements:

(a) If $\lim_{n\to\infty} a_n \geq m$, then there exists $N \in \mathbb{N}$ so that $a_n \geq m$ for all $n \geq N$.

(b) If $\lim_{n\to\infty} a_n > m$, then there exists $N \in \mathbb{N}$ so that $a_n > m$ for all $n \geq N$.

Discussion & Scratch Work

- **Key idea.** Part (a) smells suspicious because the hypothesis allows equality: $\lim a_n \geq m$ includes the borderline case $\lim a_n = m$. Convergence to m does *not* force the tail of the sequence to lie on one side of m; it only forces the tail to lie *near* m. So we should try to build a convergent sequence with limit m that stays strictly below m forever.
- Part (b) replaces $\geq$ by $>$, which creates a positive gap $L - m$. If $a_n \to L$ and $L > m$, then choosing ε smaller than this gap forces a_n to eventually lie in $(L - \varepsilon, L + \varepsilon)$, which sits entirely above m. So (b) should be true, and the proof will be a direct ε–N argument.

Proof.
Let $\ell := \lim_{n\to\infty} a_n$.

(a) Disprove. The statement is false. Fix $m \in \mathbb{R}$ and define

$$a_n := m - \frac{1}{n} \qquad (n \geq 1).$$

Then $a_n \to m$ as $n \to \infty$, so $\lim_{n\to\infty} a_n = m \geq m$ holds. However, for every $n \geq 1$ we have $a_n = m - \frac{1}{n} < m$, so there is *no* $N \in \mathbb{N}$ such that $a_n \geq m$ for all $n \geq N$. Hence (a) is false.

(b) Prove. Assume $\ell > m$. Set

$$\varepsilon := \frac{\ell - m}{2} > 0.$$

Since $a_n \to \ell$, there exists $N \in \mathbb{N}$ such that for all $n \geq N$,

$$|a_n - \ell| < \varepsilon \quad \Longrightarrow \quad a_n > \ell - \varepsilon = \ell - \frac{\ell - m}{2} = \frac{\ell + m}{2} > m.$$

Therefore $a_n > m$ for all $n \geq N$, as desired. □

Problem 5

Prove that if $\{a_n\}_{n\geq 1}$ is a sequence of positive numbers that converges to $a > 0$, then the sequence $\{\sqrt{a_n}\}_{n\geq 1}$ converges to $\sqrt{a}$.

Discussion & Scratch Work

- The main trick is to rewrite the difference of square-roots by "rationalizing":

$$|\sqrt{x} - \sqrt{y}| = \frac{|x-y|}{\sqrt{x}+\sqrt{y}} \qquad (x, y \geq 0).$$

  Here we will take $x = a_n$ and $y = a$.
- Because $a > 0$, the denominator $\sqrt{a_n} + \sqrt{a}$ is always at least $\sqrt{a}$ (since $a_n > 0 \Rightarrow \sqrt{a_n} \geq 0$). So we can control $|\sqrt{a_n} - \sqrt{a}|$ directly by controlling $|a_n - a|$.
- Concretely, given $\varepsilon > 0$, it suffices to make $|a_n - a| < \varepsilon\sqrt{a}$.

Proof.
Let $\varepsilon > 0$ be given. Since $a_n \to a$, there exists $N \in \mathbb{N}$ such that for all $n \geq N$,

$$|a_n - a| < \varepsilon\sqrt{a}.$$

For $n \geq N$, we have $a_n > 0$ and $a > 0$, so $\sqrt{a_n}$ and $\sqrt{a}$ are defined and nonnegative. Using

$$(\sqrt{a_n} - \sqrt{a})(\sqrt{a_n} + \sqrt{a}) = a_n - a,$$

we obtain

$$|\sqrt{a_n} - \sqrt{a}| = \frac{|a_n - a|}{\sqrt{a_n} + \sqrt{a}} \leq \frac{|a_n - a|}{\sqrt{a}}.$$

Therefore, for all $n \geq N$,

$$|\sqrt{a_n} - \sqrt{a}| \leq \frac{|a_n - a|}{\sqrt{a}} < \frac{\varepsilon\sqrt{a}}{\sqrt{a}} = \varepsilon.$$

Since $\varepsilon > 0$ was arbitrary, it follows that $\sqrt{a_n} \to \sqrt{a}$. □

Problem 6

Consider the following sequence: $a_1 = \sqrt{2}$ and $a_{n+1} = \sqrt{2 + a_n}$ for all $n \geq 1$.

(a) Prove that the sequence $\{a_n\}_{n\geq 1}$ is bounded.

(b) Prove that the sequence is increasing.

(c) Conclude that the sequence converges, and find its limit.

Discussion & Scratch Work

- **What we need.** To show the sequence is *bounded*, we need to find real numbers m, M such that $m \leq a_n \leq M$ for all $n \geq 1$ (equivalently: bounded below and bounded above). To show it is *increasing*, we need $a_n \leq a_{n+1}$ for all $n \geq 1$.
- **Boundedness idea.** Because $a_{n+1} = \sqrt{2 + a_n}$, a natural guess is that the sequence stays below 2: if $a_n \leq 2$, then $a_{n+1} = \sqrt{2 + a_n} \leq \sqrt{4} = 2$. Also $a_n \geq 0$ implies $a_{n+1} \geq 0$. Since $a_1 = \sqrt{2} \in (0, 2)$, induction should give $0 < a_n \leq 2$ for all n.
- **Monotonicity idea.** Since all terms are positive, we can compare a_{n+1} and a_n by squaring:
$$a_{n+1} \geq a_n \iff \sqrt{2 + a_n} \geq a_n \iff 2 + a_n \geq a_n^2.$$
The last inequality is $(a_n - 2)(a_n + 1) \leq 0$, which is true whenever $-1 \leq a_n \leq 2$. Our boundedness work gives $0 < a_n \leq 2$, so this will force $a_{n+1} \geq a_n$.
- **Convergence + limit.** Once we know $\{a_n\}$ is increasing and bounded, we may invoke the Monotone Convergence Theorem (proved in lecture): the sequence converges. If $a_n \to L$, then passing to the limit in the recursion gives $L = \sqrt{2 + L}$, so $L^2 - L - 2 = 0$ and $L \in \{2, -1\}$. Since $L \geq 0$, we conclude $L = 2$.

Proof.
(a) Boundedness. We first show that $0 < a_n \leq 2$ for all $n \geq 1$.

Base case. We have $a_1 = \sqrt{2}$, hence $0 < a_1 \leq 2$.

Inductive step. Assume $0 < a_n \leq 2$ for some $n \geq 1$. Then $2 + a_n \leq 4$, so

$$a_{n+1} = \sqrt{2 + a_n} \leq \sqrt{4} = 2.$$

Also $2 + a_n > 0$, so $a_{n+1} = \sqrt{2 + a_n} > 0$. Thus $0 < a_{n+1} \leq 2$.

By induction, $0 < a_n \leq 2$ for all $n \geq 1$. In particular, the sequence is bounded (below by 0 and above by 2).

(b) Increasing. Fix $n \geq 1$. From part (a), $a_n > 0$, so we may square inequalities without changing direction. We compute:

$$a_{n+1} \geq a_n \iff \sqrt{2 + a_n} \geq a_n \iff 2 + a_n \geq a_n^2 \iff a_n^2 - a_n - 2 \leq 0 \iff (a_n - 2)(a_n + 1) \leq 0.$$

But part (a) gives $0 < a_n \leq 2$, hence $a_n - 2 \leq 0$ and $a_n + 1 > 0$, so indeed $(a_n - 2)(a_n + 1) \leq 0$. Therefore $a_{n+1} \geq a_n$ for all $n \geq 1$, and the sequence is increasing.

(c) Convergence and limit. By parts (a) and (b), $\{a_n\}$ is bounded and increasing, so by the Monotone Convergence Theorem it converges. Let

$$L := \lim_{n\to\infty} a_n.$$

Because $a_{n+1} = \sqrt{2+a_n}$ and the square-root function is continuous on $(0,\infty)$, we may take limits on both sides to obtain

$$L = \lim_{n\to\infty} a_{n+1} = \lim_{n\to\infty} \sqrt{2+a_n} = \sqrt{2+\lim_{n\to\infty} a_n} = \sqrt{2+L}.$$

Squaring gives $L^2 = 2+L$, i.e.

$$L^2 - L - 2 = 0 \quad \Longrightarrow \quad (L-2)(L+1) = 0.$$

Thus $L \in \{2,-1\}$. Since each $a_n > 0$, we have $L \geq 0$, so $L \neq -1$ and therefore $L = 2$. $\square$

Appendix: Toolbox

Completeness / Real numbers

Proposition 1 (Archimedean property)**.** *For every $x \in \mathbb{R}$, there exists $n \in \mathbb{N}$ such that $n > x$.*

Corollary 1. *For every $\varepsilon > 0$, there exists $n \in \mathbb{N}$ such that $\frac{1}{n} < \varepsilon$.*

Lemma (Integer "sandwich")**.** *For every $a \in \mathbb{R}$, there exists $N \in \mathbb{Z}$ such that*

$$N \leq a < N + 1.$$

Proposition 2 ($\mathbb{Q}$ is dense in $\mathbb{R}$)**.** *If $x, y \in \mathbb{R}$ satisfy $x < y$, then there exists $q \in \mathbb{Q}$ such that*

$$x < q < y.$$

Corollary 2 ($\mathbb{R} \setminus \mathbb{Q}$ is dense in $\mathbb{R}$)**.** *If $x, y \in \mathbb{R}$ satisfy $x < y$, then there exists $r \in \mathbb{R} \setminus \mathbb{Q}$ such that*

$$x < r < y.$$

Countability

Proposition 3 (Uncountability via Cantor diagonal)**.** *Let*

$$A := \{f : \mathbb{N} \to \{0, 1\}\}.$$

Then A is uncountable.

Remark (Cantor's diagonal argument (what is happening conceptually))**.** *If we (incorrectly) assume A is countable, we can list its elements as*

$$A = \{f_1, f_2, f_3, \dots\}, \qquad f_k : \mathbb{N} \to \{0, 1\}.$$

Think of this as an infinite table whose (k, n)-entry is $f_k(n) \in \{0, 1\}$. We then define a new function

$$g : \mathbb{N} \to \{0, 1\}, \qquad g(n) := 1 - f_n(n).$$

This flips the diagonal entries. For each $n \in \mathbb{N}$ we have

$$g(n) \neq f_n(n),$$

so $g \neq f_n$ for every n (they differ at input n). Thus $g \in A$ but g is not in our purported enumeration, contradicting that the list contained all of A.

Corollary 3. *The interval $(0, 1) = \{x \in \mathbb{R} : 0 < x < 1\}$ is uncountable.*

Sequences and limits

Definition 1 (Sequence). *A* sequence of real numbers *is a function* $f : \mathbb{N} \to \mathbb{R}$. *We write* $a_n := f(n)$ *and denote the sequence by*

$$(a_n)_{n\geq 1} \quad \textit{or} \quad \{a_n\}_{n\geq 1}.$$

Definition 2 (Absolute value). *For* $x \in \mathbb{R}$,

$$|x| = \begin{cases} x & \text{if } x \geq 0, \\ -x & \text{if } x < 0. \end{cases}$$

Exercise 1 (Basic properties of $|\cdot|$). *For all* $x, y \in \mathbb{R}$*:*

(a) $|x| \geq 0$.

(b) $|x| = 0 \iff x = 0$.

(c) $|x + y| \leq |x| + |y|$ *(triangle inequality).*

(d) $\big||x| - |y|\big| \leq |x - y|$.

(e) $|xy| = |x||y|$.

Definition 3 (Limit / convergence). *We say that the sequence* $\{a_n\}_{n\geq 1}$ converges *to* $a \in \mathbb{R}$ *if*

$$\forall \varepsilon > 0 \ \exists N \in \mathbb{N} \text{ s.t. } \forall n \geq N, \ |a_n - a| < \varepsilon.$$

In this case we write $\lim_{n\to\infty} a_n = a$ *or* $a_n \to a$ *as* $n \to \infty$.

Definition 4 (Divergence). *A sequence* diverges *if it does not converge.*

Lemma (Uniqueness of limits). *If* $\{a_n\}_{n\geq 1}$ *converges, then its limit is unique.*

Lemma (Convergent $\Rightarrow$ bounded). *If* $\{a_n\}_{n\geq 1}$ *converges, then it is bounded.*

Definition 5 (Bounded sequence). *We say* $\{a_n\}_{n\geq 1}$ *is bounded above / bounded below / bounded if the set* $\{a_n : n \geq 1\}$ *is bounded above / bounded below / bounded in* $\mathbb{R}$.

Theorem 1 (Scalar multiple). *If* $a_n \to a$ *in* $\mathbb{R}$ *and* $k \in \mathbb{R}$, *then* $(ka_n) \to ka$.

Theorem 2 (Limit laws). *If* $a_n \to a$ *and* $b_n \to b$, *then:*

(a) $a_n + b_n \to a + b$.

(b) $a_n b_n \to ab$.

(c) *If* $b \neq 0$ *and* $b_n \neq 0$ *for all* n, *then* $\dfrac{a_n}{b_n} \to \dfrac{a}{b}$.

Definition 6 (Monotone sequences). *A sequence* $\{a_n\}_{n\geq 1}$ *is*

(a) increasing *if* $a_n \leq a_{n+1}$ *for all* $n \geq 1$,

(b) strictly increasing *if* $a_n < a_{n+1}$ *for all* $n \geq 1$,

(c) decreasing *if* $a_n \geq a_{n+1}$ *for all* $n \geq 1$,

(d) strictly decreasing *if* $a_n > a_{n+1}$ *for all* $n \geq 1$,

(e) monotone *if it is increasing or decreasing.*

Theorem 3 (Monotone convergence theorem). *If* $\{a_n\}_{n\geq 1}$ *is monotone and bounded, then it converges.*

Cauchy sequences and subsequences

Definition 7 (Cauchy sequence). *A sequence* $\{a_n\}_{n\geq 1}$ *is* Cauchy *if*

$$\forall \varepsilon > 0\ \exists N \in \mathbb{N}\ s.t.\ \forall n, m \geq N,\ |a_n - a_m| < \varepsilon.$$

Theorem 4 (Cauchy criterion). *A sequence of real numbers is Cauchy if and only if it converges.*

Lemma (Cauchy $\Rightarrow$ bounded). *If* $\{a_n\}_{n\geq 1}$ *is Cauchy, then it is bounded.*

Definition 8 (Subsequence). *Let* $\{a_n\}_{n\geq 1}$ *be a sequence. We say* $\{a_{h_n}\}_{n\geq 1}$ *is a* subsequence *of* $\{a_n\}_{n\geq 1}$ *if* $\{h_n\}_{n\geq 1}$ *is a strictly increasing sequence of natural numbers (i.e.* $h_n \in \mathbb{N}$ *and* $h_n < h_{n+1}$ *for all* n*).*

Lemma. *If* $\{h_n\}_{n\geq 1}$ *is strictly increasing in* $\mathbb{N}$*, then* $h_n \geq n$ *for all* $n \geq 1$.

Example 1 (Subsequences). ***(a)*** *If* $h_n = n$*, then* $\{a_{h_n}\}$ *is the original sequence.*

(b) *If* $h_n = 2n$*, then* $\{a_{2n}\}$ *is the even-index subsequence.*

(c) *If* $a_n = (-1)^n$*, then the subsequence* $a_{2n} = 1$ *converges (even though* a_n *diverges).*

Theorem 5 (Subsequence of a convergent sequence). *If* $a_n \to a$ *and* $\{a_{h_n}\}$ *is any subsequence, then* $a_{h_n} \to a$.

Theorem 6 (Convergence detected by subsequences). *A sequence* $\{a_n\}$ *converges to* a *if and only if every subsequence of* $\{a_n\}$ *converges to* a.

Math 521, Section 2
Homework 5

Problem 1. Let $\{a_n\}_{n\geq 1}$ be a sequence of positive real numbers. Prove that $\lim\limits_{n\to\infty} a_n = +\infty$ if and only if $\lim\limits_{n\to\infty} \frac{1}{a_n} = 0$.

Problem 2. Let $\{a_n\}_{n\geq 1}$ be a sequence. Prove that if $\{a_{k_n}\}_{n\geq 1}$ is a subsequence of $\{a_n\}_{n\geq 1}$ that converges to $L \in \mathbb{R}$, then

$$\liminf_{n\to\infty} a_n \leq L \leq \limsup_{n\to\infty} a_n.$$

(Hint: As $\{a_n\}_{n\geq 1}$ is not necessarily bounded, we have to consider the cases $\liminf\limits_{n\to\infty} a_n = \pm\infty$ and $\limsup\limits_{n\to\infty} a_n = \pm\infty$. However, the proof is much easier in these cases.)

Problem 3. Let $\{a_n\}_{n\geq 1}$ and $\{b_n\}_{n\geq 1}$ be two bounded sequences. Prove that

$$\limsup_{n\to\infty}(a_n + b_n) \leq \limsup_{n\to\infty} a_n + \limsup_{n\to\infty} b_n.$$

Problem 4. Let $\{a_n\}_{n\geq 1}$ be a sequence. Prove that $\limsup\limits_{n\to\infty} |a_n| = 0$ if and only if $\lim\limits_{n\to\infty} a_n = 0$.

Homework 5

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February 27, 2026 @1:20 PM CDT

Contents

Problem 1

Let $\{a_n\}_{n\geq 1}$ be a sequence of positive real numbers. Prove that $\lim_{n\to\infty} a_n = +\infty$ if and only if $\lim_{n\to\infty} \frac{1}{a_n} = 0$.

Discussion & Scratch Work

This is an "if and only if" statement, so we prove both directions using the definitions.

Because $a_n > 0$ for all n, the reciprocal $\frac{1}{a_n}$ is well-defined and positive. The key observation is that for positive numbers, taking reciprocals reverses inequalities: for $x > 0$ and $M > 0$,

$$x > M \quad \Longleftrightarrow \quad \frac{1}{x} < \frac{1}{M}.$$

So "a_n eventually exceeds every M" is the same as "$\frac{1}{a_n}$ eventually falls below every ε" (after choosing $\varepsilon = 1/M$ or $M = 1/\varepsilon$).

Proof.
($\Rightarrow$) Assume $\lim_{n\to\infty} a_n = +\infty$. By definition, for every $M > 0$ there exists $N \in \mathbb{N}$ such that $a_n > M$ for all $n \geq N$. Let $\varepsilon > 0$ be given and choose $M = 1/\varepsilon$ (note $M > 0$). Then there exists N such that for all $n \geq N$,

$$a_n > \frac{1}{\varepsilon}.$$

Since $a_n > 0$, taking reciprocals yields

$$\frac{1}{a_n} < \varepsilon \qquad \text{for all } n \geq N.$$

Thus for all $n \geq N$ we have $\left|\frac{1}{a_n} - 0\right| = \frac{1}{a_n} < \varepsilon$, which shows $\lim_{n\to\infty} \frac{1}{a_n} = 0$.

($\Leftarrow$) Assume $\lim_{n\to\infty} \frac{1}{a_n} = 0$. By definition, for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that for all $n \geq N$,

$$\left|\frac{1}{a_n} - 0\right| < \varepsilon.$$

Because $\frac{1}{a_n} > 0$, this is the same as $\frac{1}{a_n} < \varepsilon$. Now let $M > 0$ be given and choose $\varepsilon = 1/M$ (note $\varepsilon > 0$). Then there exists N such that for all $n \geq N$,

$$\frac{1}{a_n} < \frac{1}{M}.$$

Since $a_n > 0$, taking reciprocals reverses the inequality and gives

$$a_n > M \qquad \text{for all } n \geq N.$$

This is exactly the definition of $\lim_{n\to\infty} a_n = +\infty$. $\square$

Problem 2

Let $\{a_n\}_{n\geq 1}$ be a sequence. Prove that if $\{a_{k_n}\}_{n\geq 1}$ is a subsequence of $\{a_n\}_{n\geq 1}$ that converges to $L \in \mathbb{R}$, then

$$\liminf_{n\to\infty} a_n \leq L \leq \limsup_{n\to\infty} a_n.$$

(Hint: As $\{a_n\}_{n\geq 1}$ is not necessarily bounded, we have to consider the cases $\liminf_{n\to\infty} a_n = -\infty$ and $\limsup_{n\to\infty} a_n = +\infty$. However, the proof is much easier in these cases.)

Discussion & Scratch Work

The core idea is: compare each subsequence term a_{k_n} to the *tail* bounds of the original sequence. Fix a tail starting at N. Then

$$u_N := \inf\{a_m : m \geq N\} \quad \text{and} \quad v_N := \sup\{a_m : m \geq N\}$$

satisfy $u_N \leq a_m \leq v_N$ for every $m \geq N$.

Now use the fact that $k_n \to \infty$ (because $\{k_n\}$ is strictly increasing), so for any fixed N, eventually $k_n \geq N$. Hence for all large n,

$$u_N \leq a_{k_n} \leq v_N.$$

Taking $n \to \infty$ gives $u_N \leq L \leq v_N$ for each N. Finally, let $N \to \infty$: by definition,

$$\liminf_{n\to\infty} a_n = \lim_{N\to\infty} u_N, \qquad \limsup_{n\to\infty} a_n = \lim_{N\to\infty} v_N,$$

and passing to the limit preserves the inequalities. (If $\liminf = -\infty$ or $\limsup = +\infty$, the corresponding inequality is automatic.)

Proof.
Let $\{a_{k_n}\}_{n\geq 1}$ be a subsequence of $\{a_n\}_{n\geq 1}$ and suppose

$$\lim_{n\to\infty} a_{k_n} = L \in \mathbb{R}.$$

Case 1: $\liminf_{n\to\infty} a_n = -\infty$. Then $\liminf_{n\to\infty} a_n \leq L$ holds automatically, since $-\infty \leq L$.

Case 2: $\limsup_{n\to\infty} a_n = +\infty$. Then $L \leq \limsup_{n\to\infty} a_n$ holds automatically, since $L \leq +\infty$.

So assume from now on that $\{a_n\}$ is bounded (equivalently, both $\liminf a_n$ and $\limsup a_n$ are finite real numbers). For each $N \in \mathbb{N}$ define the tail infimum and tail supremum

$$u_N = \inf\{a_m : m \geq N\}, \qquad v_N = \sup\{a_m : m \geq N\}.$$

By definition of infimum and supremum, for every $m \geq N$ we have

$$u_N \leq a_m \leq v_N.$$

Fix $N \in \mathbb{N}$. Because $\{k_n\}$ is strictly increasing, we have $k_n \geq n$ for all n, and in particular there exists n_0 such that $k_n \geq N$ for all $n \geq n_0$. Therefore, for all $n \geq n_0$,

$$u_N \leq a_{k_n} \leq v_N.$$

Now take the limit as $n \to \infty$. Since u_N and v_N are constants (in n) and $a_{k_n} \to L$, we obtain

$$u_N \le L \le v_N \qquad \text{for every } N \in \mathbb{N}.$$

Finally, let $N \to \infty$. By definition,

$$\liminf_{n\to\infty} a_n = \lim_{N\to\infty} u_N, \qquad \limsup_{n\to\infty} a_n = \lim_{N\to\infty} v_N.$$

Taking limits in the inequality $u_N \le L \le v_N$ gives

$$\liminf_{n\to\infty} a_n \le L \le \limsup_{n\to\infty} a_n,$$

as desired. $\square$

Problem 3

Let $\{a_n\}_{n\geq 1}$ and $\{b_n\}_{n\geq 1}$ be two bounded sequences. Prove that

$$\limsup_{n\to\infty}(a_n + b_n) \leq \limsup_{n\to\infty} a_n + \limsup_{n\to\infty} b_n.$$

Discussion & Scratch Work

We work directly from the definition of lim sup via tail suprema. We know that they exist for a_n and b_n because the sequences are bounded. We also know they exist for $a_n + b_n$ because the sum of two bounded sequences is bounded (which I think we prove by considering the definition of bounded; if a_n is bounded, then there exists $M_1 > 0$ such that $|a_n| \leq M$ for all n, and similarly for b_n which is bounded by M_2, so $|a_n + b_n| \leq |a_n| + |b_n| \leq$ by the triangle inequality so so $a_n + b_n$ is bounded by $M_1 + M_2$ which we could call $K \in \mathbb{R}$ s.t. $\forall n \in \mathbb{N}, |a_n + b_n| \leq K$ and thus $a_n + b_n$ is bounded). Anyway, check out these definitions:

Since $\{a_n\}$ and $\{b_n\}$ are bounded, for each $N \in \mathbb{N}$ we define

$$v_N^{(a)} = \sup\{a_n : n \geq N\}, \qquad v_N^{(b)} = \sup\{b_n : n \geq N\},$$

and

$$v_N^{(a+b)} = \sup\{a_n + b_n : n \geq N\}.$$

For every $n \geq N$, we have

$$a_n \leq v_N^{(a)} \quad \text{and} \quad b_n \leq v_N^{(b)}.$$

Adding these inequalities gives

$$a_n + b_n \leq v_N^{(a)} + v_N^{(b)}.$$

Since this holds for all $n \geq N$, the supremum satisfies

$$v_N^{(a+b)} \leq v_N^{(a)} + v_N^{(b)}.$$

Now take limits as $N \to \infty$ and use the definition of lim sup.

(Please don't make us prove this again this was so saucey)

Proof.

Since $\{a_n\}$ and $\{b_n\}$ are bounded sequences, for each $N \in \mathbb{N}$ the sets $\{a_n : n \geq N\}$ and $\{b_n : n \geq N\}$ are nonempty and bounded. Define

$$v_N^{(a)} = \sup\{a_n : n \geq N\}, \qquad v_N^{(b)} = \sup\{b_n : n \geq N\},$$

and

$$v_N^{(a+b)} = \sup\{a_n + b_n : n \geq N\}.$$

For every $n \geq N$, we have

$$a_n \leq v_N^{(a)} \quad \text{and} \quad b_n \leq v_N^{(b)}.$$

Therefore

$$a_n + b_n \leq v_N^{(a)} + v_N^{(b)}.$$

Since this inequality holds for all $n \geq N$, taking the supremum over $n \geq N$ gives

$$v_N^{(a+b)} \leq v_N^{(a)} + v_N^{(b)}.$$

Now take limits as $N \to \infty$. By definition,

$$\limsup_{n\to\infty}(a_n + b_n) = \lim_{N\to\infty} v_N^{(a+b)},$$

$$\limsup_{n\to\infty} a_n = \lim_{N\to\infty} v_N^{(a)}, \qquad \limsup_{n\to\infty} b_n = \lim_{N\to\infty} v_N^{(b)}.$$

Passing to the limit in the inequality $v_N^{(a+b)} \leq v_N^{(a)} + v_N^{(b)}$ yields

$$\limsup_{n\to\infty}(a_n + b_n) \leq \limsup_{n\to\infty} a_n + \limsup_{n\to\infty} b_n.$$

□

Problem 4

Let $\{a_n\}_{n\geq 1}$ be a sequence. Prove that $\limsup_{n\to\infty} |a_n| = 0$ if and only if $\lim_{n\to\infty} a_n = 0$.

Discussion & Scratch Work

This is an "if and only if" statement, so I will prove both directions.

A helpful way for me to think about lim sup is via tail suprema:

$$\limsup_{n\to\infty} x_n = \lim_{N\to\infty} \Big(\sup\{x_n : n \geq N\}\Big)$$

whenever the sequence $\{x_n\}$ is bounded above (and if it is not bounded above, then $\limsup x_n = +\infty$).

Here I take $x_n = |a_n| \geq 0$. Intuitively:

- If $a_n \to 0$, then eventually $|a_n|$ is small; therefore every tail supremum $\sup\{|a_n| : n \geq N\}$ must also be small, forcing $\limsup |a_n| = 0$.
- Conversely, if the tail suprema of $|a_n|$ tend to 0, then (in particular) every term in a sufficiently far tail must be smaller than ε, which is exactly the definition of $a_n \to 0$.

Proof.
($\Rightarrow$) Assume $\lim_{n\to\infty} a_n = 0$. Let $\varepsilon > 0$ be given. By definition of convergence, there exists $N_0 \in \mathbb{N}$ such that

$$|a_n - 0| = |a_n| < \varepsilon \qquad \text{for all } n \geq N_0.$$

Now fix any $N \geq N_0$. Then for every $n \geq N$ I have $|a_n| < \varepsilon$, and hence

$$\sup\{|a_n| : n \geq N\} \leq \varepsilon.$$

Define $v_N := \sup\{|a_n| : n \geq N\}$. The inequality above shows that for all $N \geq N_0$, $v_N \leq \varepsilon$. Taking $N \to \infty$ yields

$$\limsup_{n\to\infty} |a_n| = \lim_{N\to\infty} v_N \leq \varepsilon.$$

Because $\varepsilon > 0$ was arbitrary and $\limsup_{n\to\infty} |a_n| \geq 0$ (since $|a_n| \geq 0$), I conclude $\limsup_{n\to\infty} |a_n| = 0$.

($\Leftarrow$) Assume $\limsup_{n\to\infty} |a_n| = 0$. Set again

$$v_N := \sup\{|a_n| : n \geq N\}.$$

By definition of lim sup (and since $|a_n| \geq 0$), I have $\lim_{N\to\infty} v_N = 0$. Let $\varepsilon > 0$ be given. Then there exists $N_0 \in \mathbb{N}$ such that

$$v_{N_0} < \varepsilon.$$

But for every $n \geq N_0$ I have $|a_n| \leq v_{N_0}$ (because v_{N_0} is the supremum of the tail $\{|a_n| : n \geq N_0\}$). Therefore, for all $n \geq N_0$,

$$|a_n| \leq v_{N_0} < \varepsilon.$$

This is exactly the definition of $a_n \to 0$. Hence $\lim_{n\to\infty} a_n = 0$. $\square$

Appendix: Toolbox

Sequences and Cauchy

Definition 1. *We say that $\{a_n\}_{n\geq 1}$ is a Cauchy sequence if: for all $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that $|a_n - a_m| < \varepsilon$ for all $m, n \geq N$.*

Theorem 1 (Cauchy criterion)**.** *Let $\{a_n\}_{n\geq 1}$ be a sequence. The sequence $\{a_n\}_{n\geq 1}$ is Cauchy if and only if it converges.*

Proposition 1. *If $\{a_n\}_{n\geq 1}$ converges, then it is Cauchy.*

Lemma. *If $\{a_n\}_{n\geq 1}$ is a Cauchy sequence, then it is bounded.*

Proposition 2. *If $\{a_n\}_{n\geq 1}$ is a Cauchy sequence, then it converges.*

Subsequences

Definition 2. *Let $\{a_n\}_{n\geq 1}$ be a sequence. We say $\{a_{k_n}\}_{n\geq 1}$ is a subsequence of $\{a_n\}_{n\geq 1}$ if $\{k_n\}_{n\geq 1}$ is a strictly increasing sequence of natural numbers: $k_n \in \mathbb{N}$ and $k_n < k_{n+1}$ for all $n \in \mathbb{N}$.*

Remark. *Any such $\{k_n\}_{n\geq 1}$ must satisfy $k_n \geq n$ for all $n \geq 1$. This is easy to prove by induction. (Exercise.)*

Theorem 2. *Let $\{a_n\}_{n\geq 1}$ be a sequence and $a \in \mathbb{R}$. The sequence $\{a_n\}_{n\geq 1}$ converges to a if and only if every subsequence $\{a_{k_n}\}_{n\geq 1}$ converges to a.*

Monotone Subsequences

Proposition 3. *For any sequence, there is a subsequence that is monotone.*

Theorem 3 (Bolzano–Weierstrass)**.** *For any bounded sequence, there exists a subsequence that converges.*

Divergence to $\pm\infty$

Definition 3. *Let $\{a_n\}_{n\geq 1}$ be a sequence. We say:*

(1) *$\{a_n\}_{n\geq 1}$ diverges to $+\infty$ if for all $M > 0$ there exists $N \in \mathbb{N}$ such that $a_n > M$ for all $n \geq N$. When this is true, we write $\lim_{n\to\infty} a_n = +\infty$.*

(2) *$\{a_n\}_{n\geq 1}$ diverges to $-\infty$ if for all $m < 0$ there exists $N \in \mathbb{N}$ such that $a_n < m$ for all $n \geq N$. When this is true, we write $\lim_{n\to\infty} a_n = -\infty$.*

Proposition 4. *If $\{a_n\}_{n\geq 1}$ is unbounded and increasing (decreasing), then $\{a_n\}_{n\geq 1}$ diverges to $+\infty$ ($-\infty$, respectively).*

Remark. *If a sequence is increasing, there are only two possibilities:*

(1) *bounded $\Rightarrow$ converges;*

(2) *unbounded $\Rightarrow$ diverges to $+\infty$.*

Remark. *"$\lim a_n$" only sometimes makes sense:*

(1) *converges $\Rightarrow \lim_{n\to\infty} a_n = a \in \mathbb{R}$;*

(2) *diverges: diverges to $\pm\infty \Rightarrow \lim_{n\to\infty} a_n = \pm\infty$, else $\lim_{n\to\infty} a_n$ does not exist.*

Limit Superior and Limit Inferior

Definition 4. *Let $\{a_n\}_{n\geq 1}$ be a sequence (convergent or not). Case 1: $\{a_n\}_{n\geq 1}$ is bounded. Fix any $N \in \mathbb{N}$. Then the set $\{a_n : n \geq N\}$ is nonempty and bounded, so*

$$u_N = \inf\{a_n : n \geq N\}, \qquad v_N = \sup\{a_n : n \geq N\}$$

both exist. We automatically have $u_N \leq v_N$ for all N.

Proposition 5. *With u_N and v_N as above, $\{u_N\}_{N\geq 1}$ is increasing and $\{v_N\}_{N\geq 1}$ is decreasing. In particular,*

$$u_1 \leq u_2 \leq \cdots \leq u_N \leq v_N \leq \cdots \leq v_2 \leq v_1 \quad \text{for all } N \geq 1.$$

Proposition 6. *The sequences $\{u_N\}_{N\geq 1}$ and $\{v_N\}_{N\geq 1}$ are both monotone and bounded, and therefore converge. Set*

$$u = \lim_{N\to\infty} u_N \qquad \text{and} \qquad v = \lim_{N\to\infty} v_N.$$

Then $u \leq v$. Moreover, if $\lim_{n\to\infty} a_n = a$ exists, then $u \leq a \leq v$.

Definition 5. *Let $\{a_n\}_{n\geq 1}$ be a bounded sequence. We define*

$$\limsup_{n\to\infty} a_n = \lim_{N\to\infty} v_N = \lim_{N\to\infty} \Big(\sup\{a_n : n \geq N\} \Big)$$

and

$$\liminf_{n\to\infty} a_n = \lim_{N\to\infty} u_N = \lim_{N\to\infty} \Big(\inf\{a_n : n \geq N\} \Big).$$

Corollary 1. *If $\{a_n\}_{n\geq 1}$ is bounded, then*

$$\liminf_{n\to\infty} a_n \leq \limsup_{n\to\infty} a_n.$$

Definition 6. *If $\{a_n\}_{n\geq 1}$ is not bounded above, then we define*

$$\limsup_{n\to\infty} a_n = +\infty.$$

If $\{a_n\}_{n\geq 1}$ is not bounded below, then we define

$$\liminf_{n\to\infty} a_n = -\infty.$$

Remark. *So* $\limsup$ *and* $\liminf$ *make sense for any sequence, but their values are in* $\mathbb{R} \cup \{\pm\infty\}$.

Corollary 2. *For any sequence* $\{a_n\}_{n\geq 1}$*, we have*

$$\liminf_{n\to\infty} a_n \leq \limsup_{n\to\infty} a_n.$$

Theorem 4. *Let* $\{a_n\}_{n\geq 1}$ *be a sequence.*

(1) *If* $\lim_{n\to\infty} a_n$ *exists in* $\mathbb{R} \cup \{\pm\infty\}$*, then*

$$\liminf_{n\to\infty} a_n = \lim_{n\to\infty} a_n = \limsup_{n\to\infty} a_n.$$

(2) *If* $\liminf_{n\to\infty} a_n = \limsup_{n\to\infty} a_n$ *in* $\mathbb{R} \cup \{\pm\infty\}$*, then* $\lim_{n\to\infty} a_n$ *exists and equals this value.*

Remark. *Here, "*$\lim_{n\to\infty} a_n$ *exists in* $\mathbb{R}\cup\{\pm\infty\}$*" just means "*$\{a_n\}_{n\geq 1}$ *converges, or diverges to* $+\infty$*, or diverges to* $-\infty$*."*

Math 521, Section 2
Homework 6

Problem 1. Prove that the series

$$\sum_{n=1}^{\infty} \frac{1}{(3n-2)(3n+1)}$$

converges, and find its value. (Hint: You computed the partial sums on Homework 1.)

Problem 2. Let $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ be convergent series.

(a) Prove that for any $k \in \mathbb{R}$, the series $\sum_{n=1}^{\infty} ka_n$ converges and

$$\sum_{n=1}^{\infty} ka_n = k \sum_{n=1}^{\infty} a_n.$$

(b) Prove that the series $\sum_{n=1}^{\infty}(a_n + b_n)$ converges and

$$\sum_{n=1}^{\infty}(a_n + b_n) = \sum_{n=1}^{\infty} a_n + \sum_{n=1}^{\infty} b_n.$$

Problem 3. (a) Prove that

$$\sum_{n=1}^{\infty} \frac{n-1}{2^{n+1}} = \frac{1}{2}.$$

(Hint: $\frac{k-1}{2^{k+1}} = \frac{k}{2^k} - \frac{k+1}{2^{k+1}}$.)

(b) Use part (a) to compute

$$\sum_{n=1}^{\infty} \frac{n}{2^n}.$$

Problem 4. Prove that if $\sum_{n=1}^{\infty} a_n$ converges absolutely and $\{b_n\}_{n\geq 1}$ is a bounded sequence, then the series $\sum_{n=1}^{\infty} a_n b_n$ also converges. (Hint: Use the Cauchy criterion.)

Problem 5. (a) Prove that if $\sum_{n=1}^{\infty} a_n$ converges absolutely, then the series $\sum_{n=1}^{\infty} a_n^2$ also converges.

(b) Provide an example where $\sum_{n=1}^{\infty} a_n$ diverges and $\sum_{n=1}^{\infty} a_n^2$ converges.

Problem 6. For each of the following series, determine if it converges and prove your answer.

(a) $\displaystyle\sum_{n=2}^{\infty} \frac{1}{[n+(-1)^n]^2}$ (b) $\displaystyle\sum_{n=1}^{\infty} \left(\sqrt{n+1} - \sqrt{n}\right)$ (c) $\displaystyle\sum_{n=1}^{\infty} (-1)^n$

Homework 6

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Contents

Problem 1

Prove that the series
$$\sum_{n=1}^{\infty} \frac{1}{(3n-2)(3n+1)}$$
converges, and find its value. (Hint: You computed the partial sums on Homework 1.)

Discussion & Scratch Work

> We want to show the series converges and compute its sum. The hint suggests computing the partial sums and looking for a telescoping form. A standard move is to use partial fractions:
> $$\frac{1}{(3n-2)(3n+1)} = \frac{1}{3}\left(\frac{1}{3n-2} - \frac{1}{3n+1}\right),$$
> which should make the partial sums collapse.

Proof.
Let
$$a_n := \frac{1}{(3n-2)(3n+1)} \qquad (n \in \mathbb{N}).$$
For $N \in \mathbb{N}$ define the Nth partial sum
$$S_N := \sum_{n=1}^{N} a_n.$$
Using partial fractions,
$$\frac{1}{(3n-2)(3n+1)} = \frac{1}{3}\left(\frac{1}{3n-2} - \frac{1}{3n+1}\right).$$
Hence
$$S_N = \frac{1}{3}\sum_{n=1}^{N}\left(\frac{1}{3n-2} - \frac{1}{3n+1}\right) = \frac{1}{3}\left[\left(1-\frac{1}{4}\right) + \left(\frac{1}{4}-\frac{1}{7}\right) + \cdots + \left(\frac{1}{3N-2} - \frac{1}{3N+1}\right)\right].$$
All intermediate terms cancel, so
$$S_N = \frac{1}{3}\left(1 - \frac{1}{3N+1}\right) = \frac{N}{3N+1}.$$
Therefore
$$\lim_{N\to\infty} S_N = \lim_{N\to\infty} \frac{N}{3N+1} = \frac{1}{3}.$$
Since the sequence of partial sums (S_N) converges, the series $\sum_{n=1}^{\infty} \frac{1}{(3n-2)(3n+1)}$ converges, and its value is $\frac{1}{3}$. □

Problem 2

Let $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ be convergent series.

(a) Prove that for any $k \in \mathbb{R}$, the series $\sum_{n=1}^{\infty} ka_n$ converges and

$$\sum_{n=1}^{\infty} ka_n = k \sum_{n=1}^{\infty} a_n.$$

(b) Prove that the series $\sum_{n=1}^{\infty}(a_n + b_n)$ converges and

$$\sum_{n=1}^{\infty}(a_n + b_n) = \sum_{n=1}^{\infty} a_n + \sum_{n=1}^{\infty} b_n.$$

Discussion & Scratch Work

We will work entirely with partial sums.

Let

$$s_n := \sum_{k=1}^{n} a_k, \qquad t_n := \sum_{k=1}^{n} b_k \qquad (n \in \mathbb{N}).$$

Because $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ converge, there exist $s, t \in \mathbb{R}$ such that

$$s_n \to s \quad \text{and} \quad t_n \to t.$$

(a) For fixed $k \in \mathbb{R}$, the partial sums of $\sum_{n=1}^{\infty} ka_n$ are

$$p_n := \sum_{j=1}^{n} ka_j = k \sum_{j=1}^{n} a_j = ks_n.$$

Hence $p_n \to ks$ by the limit laws, so $\sum ka_n$ converges and its sum is ks.

(b) The partial sums of $\sum_{n=1}^{\infty}(a_n + b_n)$ are

$$u_n := \sum_{j=1}^{n}(a_j + b_j) = \sum_{j=1}^{n} a_j + \sum_{j=1}^{n} b_j = s_n + t_n.$$

Hence $u_n \to s + t$, so the series converges and its sum is $s + t$.

Proof.
Let $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ be convergent series, and define their partial sums

$$s_n := \sum_{k=1}^{n} a_k, \qquad t_n := \sum_{k=1}^{n} b_k \qquad (n \in \mathbb{N}).$$

By convergence, there exist $s, t \in \mathbb{R}$ such that $s_n \to s$ and $t_n \to t$.

(a) Fix $k \in \mathbb{R}$ and define

$$p_n := \sum_{j=1}^{n} k a_j \qquad (n \in \mathbb{N}).$$

Then for every n,

$$p_n = \sum_{j=1}^{n} k a_j = k \sum_{j=1}^{n} a_j = k s_n.$$

Since $s_n \to s$, the limit laws imply $p_n = k s_n \to ks$. Hence the sequence of partial sums (p_n) converges, so the series $\sum_{n=1}^{\infty} k a_n$ converges. Moreover, by definition of the sum of a series,

$$\sum_{n=1}^{\infty} k a_n := \lim_{n\to\infty} p_n = \lim_{n\to\infty} (k s_n) = k \lim_{n\to\infty} s_n = ks = k \sum_{n=1}^{\infty} a_n.$$

(b) Define

$$u_n := \sum_{j=1}^{n} (a_j + b_j) \qquad (n \in \mathbb{N}).$$

Then for every n,

$$u_n = \sum_{j=1}^{n} (a_j + b_j) = \sum_{j=1}^{n} a_j + \sum_{j=1}^{n} b_j = s_n + t_n.$$

Since $s_n \to s$ and $t_n \to t$, again by the limit laws we have $u_n = s_n + t_n \to s + t$. Therefore $\sum_{n=1}^{\infty} (a_n + b_n)$ converges, and

$$\sum_{n=1}^{\infty} (a_n + b_n) := \lim_{n\to\infty} u_n = \lim_{n\to\infty} (s_n + t_n) = \lim_{n\to\infty} s_n + \lim_{n\to\infty} t_n = s + t = \sum_{n=1}^{\infty} a_n + \sum_{n=1}^{\infty} b_n.$$

$\square$

Problem 3

(a) Prove that

$$\sum_{n=1}^{\infty} \frac{n-1}{2^{n+1}} = \frac{1}{2}.$$

(Hint: $\frac{k-1}{2^{k+1}} = \frac{k}{2^k} - \frac{k+1}{2^{k+1}}$.)

(b) Use part (a) to compute

$$\sum_{n=1}^{\infty} \frac{n}{2^n}.$$

Discussion & Scratch Work

For part (a), I use the given identity to rewrite the summand in a telescoping form. After writing out the first several terms of the corresponding partial sums, everything cancels except the first positive term and the last negative term. This gives an explicit formula for the partial sums, and then I take the limit.

For part (b), I rewrite

$$\frac{n}{2^n} = 2 \cdot \frac{n-1}{2^{n+1}} + \frac{1}{2^n}.$$

Then I use part (a) together with the geometric series

$$\sum_{n=1}^{\infty} \frac{1}{2^n} = 1$$

to compute the sum.

Proof.
(a) Consider the series

$$\sum_{n=1}^{\infty} a_n := \sum_{n=1}^{\infty} \frac{n-1}{2^{n+1}}, \qquad \text{where } a_n = \frac{n-1}{2^{n+1}}.$$

Let the sequence of partial sums be

$$S_N := \sum_{n=1}^{N} \frac{n-1}{2^{n+1}}.$$

Using the given identity with $k = n$, we may rewrite the summand as

$$\frac{n-1}{2^{n+1}} = \frac{n}{2^n} - \frac{n+1}{2^{n+1}}.$$

Hence

$$S_N = \sum_{n=1}^{N} \left(\frac{n}{2^n} - \frac{n+1}{2^{n+1}} \right).$$

Writing out the first several terms, we obtain

$$S_N = \left(\frac{1}{2} - \frac{2}{2^2}\right) + \left(\frac{2}{2^2} - \frac{3}{2^3}\right) + \left(\frac{3}{2^3} - \frac{4}{2^4}\right) + \cdots + \left(\frac{N}{2^N} - \frac{N+1}{2^{N+1}}\right).$$

This is a telescoping sum, so all intermediate terms cancel and we are left with

$$S_N = \frac{1}{2} - \frac{N+1}{2^{N+1}}.$$

Now take the limit as $N \to \infty$:

$$\lim_{N\to\infty} S_N = \lim_{N\to\infty}\left(\frac{1}{2} - \frac{N+1}{2^{N+1}}\right) = \frac{1}{2} - \lim_{N\to\infty}\frac{N+1}{2^{N+1}} = \frac{1}{2}.$$

Therefore the sequence of partial sums $\{S_N\}_{N\geq 1}$ converges, so by definition the series converges. Since

$$\sum_{n=1}^{\infty} \frac{n-1}{2^{n+1}} := \lim_{N\to\infty} S_N,$$

we conclude that

$$\sum_{n=1}^{\infty} \frac{n-1}{2^{n+1}} = \frac{1}{2}.$$

(b) For each $n \geq 1$,

$$\frac{n}{2^n} = \frac{(n-1)+1}{2^n} = \frac{n-1}{2^n} + \frac{1}{2^n} = 2\cdot\frac{n-1}{2^{n+1}} + \frac{1}{2^n}.$$

Therefore,

$$\sum_{n=1}^{\infty} \frac{n}{2^n} = 2\sum_{n=1}^{\infty} \frac{n-1}{2^{n+1}} + \sum_{n=1}^{\infty} \frac{1}{2^n}.$$

By part (a),

$$\sum_{n=1}^{\infty} \frac{n-1}{2^{n+1}} = \frac{1}{2},$$

and since $\sum_{n=1}^{\infty} \frac{1}{2^n}$ is a geometric series with ratio $\frac{1}{2}$,

$$\sum_{n=1}^{\infty} \frac{1}{2^n} = 1.$$

Hence

$$\sum_{n=1}^{\infty} \frac{n}{2^n} = 2\left(\frac{1}{2}\right) + 1 = 2.$$

$\square$

Problem 4

Prove that if $\sum_{n=1}^{\infty} a_n$ converges absolutely and $\{b_n\}_{n\geq 1}$ is a bounded sequence, then the series $\sum_{n=1}^{\infty} a_n b_n$ also converges. (Hint: Use the Cauchy criterion.)

Discussion & Scratch Work

The goal is to use the Cauchy criterion for series.

Since $\sum_{n=1}^{\infty} a_n$ converges absolutely, the series

$$\sum_{n=1}^{\infty} |a_n|$$

converges. Therefore, by the Cauchy criterion, for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that

$$\sum_{k=m}^{n} |a_k| < \varepsilon \qquad \text{for all } n \geq m \geq N.$$

Also, since $\{b_n\}_{n\geq 1}$ is bounded, there exists $M \geq 0$ such that

$$|b_n| \leq M \qquad \text{for all } n \in \mathbb{N}.$$

To prove that $\sum_{n=1}^{\infty} a_n b_n$ converges, I apply the Cauchy criterion again. So I must estimate

$$\left| \sum_{k=m}^{n} a_k b_k \right|$$

in terms of the convergent tail sums of $\sum |a_n|$. The boundedness of b_n gives

$$|a_k b_k| = |a_k|\,|b_k| \leq M|a_k|,$$

so the tail of $\sum a_n b_n$ is controlled by the tail of $\sum |a_n|$.

Proof.

Since $\sum_{n=1}^{\infty} a_n$ converges absolutely, the series

$$\sum_{n=1}^{\infty} |a_n|$$

converges. By the Cauchy criterion for series, given $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$\sum_{k=m}^{n} |a_k| < \frac{\varepsilon}{M+1} \qquad \text{for all } n \geq m \geq N,$$

where $M \geq 0$ is chosen so that

$$|b_n| \leq M \qquad \text{for all } n \in \mathbb{N}.$$

(Such an M exists because $\{b_n\}$ is bounded.)

Now let $n \geq m \geq N$. Then

$$\left|\sum_{k=m}^{n} a_k b_k\right| \leq \sum_{k=m}^{n} |a_k b_k| = \sum_{k=m}^{n} |a_k|\,|b_k| \leq \sum_{k=m}^{n} |a_k|\,M = M\sum_{k=m}^{n} |a_k|.$$

Therefore,

$$\left|\sum_{k=m}^{n} a_k b_k\right| \leq M\sum_{k=m}^{n} |a_k| < M \cdot \frac{\varepsilon}{M+1} < \varepsilon.$$

Thus, for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$\left|\sum_{k=m}^{n} a_k b_k\right| < \varepsilon \qquad \text{for all } n \geq m \geq N.$$

By the Cauchy criterion for series, it follows that

$$\sum_{n=1}^{\infty} a_n b_n$$

converges. $\square$

Problem 5

(a) Prove that if $\sum_{n=1}^{\infty} a_n$ converges absolutely, then the series $\sum_{n=1}^{\infty} a_n^2$ also converges.

(b) Provide an example where $\sum_{n=1}^{\infty} a_n$ diverges and $\sum_{n=1}^{\infty} a_n^2$ converges.

Discussion & Scratch Work

For part (a), I want to compare $|a_n|^2$ to $|a_n|$. Since $\sum |a_n|$ converges, I know that $a_n \to 0$, so eventually $|a_n| < 1$. For all sufficiently large n, this gives

$$|a_n|^2 \leq |a_n|.$$

Then the comparison test shows that

$$\sum |a_n|^2$$

converges, which means exactly that

$$\sum a_n^2$$

converges absolutely, hence converges.

For part (b), I need an example where $\sum a_n$ diverges but $\sum a_n^2$ converges. A standard choice is

$$a_n = \frac{1}{n}.$$

Then

$$\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} \frac{1}{n}$$

diverges (harmonic series), while

$$\sum_{n=1}^{\infty} a_n^2 = \sum_{n=1}^{\infty} \frac{1}{n^2}$$

converges, since this is a p-series with $p = 2 > 1$.

Proof.
(a) Assume that $\sum_{n=1}^{\infty} a_n$ converges absolutely. Then by definition,

$$\sum_{n=1}^{\infty} |a_n|$$

converges. Since every absolutely convergent series converges, it follows that

$$\sum_{n=1}^{\infty} a_n$$

converges, and therefore

$$a_n \to 0.$$

Hence there exists $N \in \mathbb{N}$ such that

$$|a_n| < 1 \qquad \text{for all } n \geq N.$$

For such n, we have

$$|a_n|^2 \leq |a_n|.$$

Now split the series

$$\sum_{n=1}^{\infty} |a_n|^2 = \sum_{n=1}^{N-1} |a_n|^2 + \sum_{n=N}^{\infty} |a_n|^2.$$

The first sum is finite, and for the tail we have

$$0 \leq |a_n|^2 \leq |a_n| \qquad \text{for all } n \geq N.$$

Since $\sum_{n=N}^{\infty} |a_n|$ converges, the comparison test implies that

$$\sum_{n=N}^{\infty} |a_n|^2$$

converges. Therefore

$$\sum_{n=1}^{\infty} |a_n|^2$$

converges, so the series

$$\sum_{n=1}^{\infty} a_n^2$$

converges absolutely. In particular, it converges.

(b) Take

$$a_n := \frac{1}{n} \qquad (n \in \mathbb{N}).$$

Then

$$\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} \frac{1}{n}$$

diverges, since the harmonic series diverges. However,

$$\sum_{n=1}^{\infty} a_n^2 = \sum_{n=1}^{\infty} \frac{1}{n^2}$$

converges, since by the p-series criterion the series

$$\sum_{n=1}^{\infty} \frac{1}{n^\alpha}$$

converges whenever $\alpha > 1$, and here $\alpha = 2$. □

Problem 6

For each of the following series, determine if it converges and prove your answer.

(a)

$$\sum_{n=2}^{\infty} \frac{1}{[n+(-1)^n]^2}$$

(b)

$$\sum_{n=1}^{\infty} \left(\sqrt{n+1}-\sqrt{n}\right)$$

(c)

$$\sum_{n=1}^{\infty} (-1)^n$$

Discussion & Scratch Work

For part (a), the denominator looks messy because of the $(-1)^n$, but for $n \geq 2$ I still have a simple lower bound:

$$n+(-1)^n \geq n-1.$$

Since $n \geq 2$, I also have $n-1 \geq \frac{n}{2}$, so

$$\frac{1}{[n+(-1)^n]^2} \leq \frac{1}{(n/2)^2} = \frac{4}{n^2}.$$

This suggests a direct comparison with the convergent p-series $\sum 1/n^2$.

For part (b), the terms do not need a comparison test at all. The expression

$$\sqrt{n+1}-\sqrt{n}$$

is already in telescoping form, since consecutive terms cancel in the partial sums.

For part (c), I should first check the necessary condition for convergence of a series: if $\sum a_n$ converges, then $a_n \to 0$. Here the terms are $a_n = (-1)^n$, which do not tend to 0, so the series must diverge.

Proof.
(a) Consider the series

$$\sum_{n=2}^{\infty} \frac{1}{[n+(-1)^n]^2}.$$

For every $n \geq 2$, we have

$$n+(-1)^n \geq n-1.$$

Also, since $n \geq 2$, we have

$$n-1 \geq \frac{n}{2}.$$

Therefore, for all $n \geq 2$,

$$n + (-1)^n \geq \frac{n}{2},$$

and hence

$$0 < \frac{1}{[n + (-1)^n]^2} \leq \frac{1}{(n/2)^2} = \frac{4}{n^2}.$$

Now the series

$$\sum_{n=2}^{\infty} \frac{4}{n^2} = 4 \sum_{n=2}^{\infty} \frac{1}{n^2}$$

converges, since $\sum 1/n^2$ is a p-series with $p = 2 > 1$. By the comparison test, it follows that

$$\sum_{n=2}^{\infty} \frac{1}{[n + (-1)^n]^2}$$

converges.

(b) Consider the series

$$\sum_{n=1}^{\infty} \left(\sqrt{n+1} - \sqrt{n} \right).$$

Let

$$S_N := \sum_{n=1}^{N} \left(\sqrt{n+1} - \sqrt{n} \right)$$

be the Nth partial sum. Then

$$S_N = (\sqrt{2} - 1) + (\sqrt{3} - \sqrt{2}) + (\sqrt{4} - \sqrt{3}) + \cdots + (\sqrt{N+1} - \sqrt{N}).$$

This is a telescoping sum, so all intermediate terms cancel and we obtain

$$S_N = \sqrt{N+1} - 1.$$

Now,

$$\lim_{N\to\infty} S_N = \lim_{N\to\infty} (\sqrt{N+1} - 1) = +\infty.$$

Therefore the sequence of partial sums does not converge to a real number, so the series

$$\sum_{n=1}^{\infty} \left(\sqrt{n+1} - \sqrt{n} \right)$$

diverges.

(c) Consider the series

$$\sum_{n=1}^{\infty} (-1)^n.$$

Let

$$a_n := (-1)^n.$$

If the series $\sum_{n=1}^{\infty}(-1)^n$ converged, then by the basic necessary condition for convergence of a series, we would have

$$a_n \to 0.$$

But the sequence $\{(-1)^n\}$ oscillates between -1 and 1, so it does not converge to 0. Therefore the series

$$\sum_{n=1}^{\infty}(-1)^n$$

diverges. $\square$

Appendix: Toolbox

Cesàro–Stolz

Theorem (Cesàro–Stolz). Let $\{a_n\}_{n\geq 1}$ be a sequence of nonzero real numbers. Then

$$\liminf_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| \leq \liminf_{n\to\infty}|a_n|^{1/n} \leq \limsup_{n\to\infty}|a_n|^{1/n} \leq \limsup_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right|.$$

Lemma. If $r > 0$, then $\lim_{n\to\infty} r^{1/n} = 1$.

Series: Definitions and Basic Facts

Definition (Series). Let $\{a_n\}_{n\geq 1}$ be a sequence of real numbers. The series $\sum_{n=1}^{\infty} a_n$ converges/diverges if the sequence of partial sums $s_n = a_1 + \cdots + a_n$ converges/diverges.

Definition (Absolute Convergence). The series $\sum_{n=1}^{\infty} a_n$ converges absolutely if the series $\sum_{n=1}^{\infty} |a_n|$ converges.

Remark. As $|a_n| \geq 0$, the sequence $s_n = |a_1| + \cdots + |a_n|$ is increasing, so either $\sum_{n=1}^{\infty} |a_n|$ converges or diverges to $+\infty$.

Corollary. If $\sum_{n=1}^{\infty} a_n$ converges, then $\lim_{n\to\infty} a_n = 0$.

Remark (Geometric Series). The series $\sum_{n=0}^{\infty} r^n$ converges if $|r| < 1$ and

$$\sum_{n=0}^{\infty} r^n = \frac{1}{1-r}.$$

If $|r| \geq 1$, then $\sum_{n=0}^{\infty} r^n$ diverges.

Cauchy Criterion for Series

Theorem (Cauchy criterion). The series $\sum_{n=1}^{\infty} a_n$ converges if and only if for all $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that

$$\left|\sum_{k=m}^{n} a_k\right| < \varepsilon \quad \text{for all } n \geq m \geq N.$$

Absolute Convergence

Corollary. If $\sum_{n=1}^{\infty} a_n$ converges absolutely, then it converges.

Comparison Test

Theorem (Comparison test). Let $\sum_{n=1}^{\infty} a_n$ be a series with $a_n \geq 0$ for all $n \geq 1$.

(1) If $\sum_{n=1}^{\infty} a_n$ converges and $|b_n| \leq a_n$ for all $n \geq 1$, then $\sum_{n=1}^{\infty} b_n$ converges.

(2) If $\sum_{n=1}^{\infty} a_n$ diverges and $b_n \geq a_n$ for all $n \geq 1$, then $\sum_{n=1}^{\infty} b_n$ diverges.

Dyadic Criterion

Theorem (Dyadic criterion). Let $\{a_n\}_{n\geq 1}$ be a decreasing sequence with $a_n \geq 0$ for all $n \geq 1$. The series $\sum_{n=1}^{\infty} a_n$ converges if and only if the series $\sum_{n=0}^{\infty} 2^n a_{2^n}$ converges.

Corollary. The series $\sum_{n=1}^{\infty} \frac{1}{n^{\alpha}}$ converges if and only if $\alpha > 1$.

Math 521, Section 2
Homework 7

Problem 1. For each of the following series, determine if it converges and prove your answer.

$$\text{(a)} \quad \sum_{n=2}^{\infty} \frac{1}{\log n} \qquad\qquad \text{(b)} \quad \sum_{n=1}^{\infty} \frac{n-1}{n^2} \qquad\qquad \text{(c)} \quad \sum_{n=1}^{\infty} \frac{n^2}{n!}$$

Problem 2. For each of the following series, determine if it converges and prove your answer.

$$\text{(a)} \quad \sum_{n=1}^{\infty} \frac{n^4}{4^n} \qquad\qquad \text{(b)} \quad \sum_{n=1}^{\infty} \frac{n!}{n^4+3} \qquad\qquad \text{(c)} \quad \sum_{n=1}^{\infty} \frac{2^n}{n!}$$

Problem 3. For each of the following series, determine if it converges and prove your answer.

$$\text{(a)} \quad \sum_{n=1}^{\infty} \frac{1}{n^n} \qquad\qquad \text{(b)} \quad \sum_{n=1}^{\infty} \frac{n!}{n^n} \qquad\qquad \text{(c)} \quad \sum_{n=1}^{\infty} \frac{(-1)^n n!}{2^n}$$

Problem 4. Consider the series

$$\sum_{n=1}^{\infty} 2^{(-1)^n - n} = \frac{1}{4} + \frac{1}{2} + \frac{1}{16} + \frac{1}{8} + \frac{1}{64} + \frac{1}{32} + \dots.$$

(a) Show that the ratio test is inconclusive.
(b) Prove that the series converges using the root test.

Problem 5. Below are two series $\sum_{n=1}^{\infty} a_n$ and the numbers $s \in \mathbb{R}$ to which they converge. For each of these series, what is the set of all numbers $\tilde{s} \in \mathbb{R}$ such that there exists a rearrangement that satisfies $\sum_{n=1}^{\infty} \tilde{a}_n = \tilde{s}$? Prove your answer.

$$\text{(a)} \quad \sum_{n=1}^{\infty} \frac{(-1)^n}{n} = -\log 2 \qquad\qquad \text{(b)} \quad \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} = -\frac{\pi^2}{12}$$

(Hint: Your proof should be quite short, because we already did most of the work in lecture.)

Problem 6. Prove or disprove the following statements:

(a) If $\sum_{n=1}^{\infty} a_n$ converges, then $\sum_{n=1}^{\infty} a_{2n}$ converges.
(b) If $\sum_{n=1}^{\infty} a_n$ converges absolutely, then $\sum_{n=1}^{\infty} a_{2n}$ converges absolutely.

Homework 7

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Contents

Problem 1

For each of the following series, determine if it converges and prove your answer.

(a)

$$\sum_{n=2}^{\infty} \frac{1}{\log n}$$

(b)

$$\sum_{n=1}^{\infty} \frac{n-1}{n^2}$$

(c)

$$\sum_{n=1}^{\infty} \frac{n^2}{n!}$$

Discussion & Scratch Work

For each part, I first identify the general type of positive-term series and then choose the test that best matches its dominant behavior.

(a) The terms

$$\frac{1}{\log n}$$

go to zero extremely slowly, and they are larger than the harmonic terms $1/n$ for $n \geq 2$. This strongly suggests divergence. A direct comparison with the harmonic series is the cleanest method here, because it proves divergence immediately. The dyadic criterion could also be used, but comparison is shorter and more transparent.

(b) The terms

$$\frac{n-1}{n^2}$$

behave like $1/n$ for large n, so I again expect divergence. I could rewrite the term as

$$\frac{1}{n} - \frac{1}{n^2},$$

which suggests that the harmonic part should dominate, but a direct lower comparison with $1/(2n)$ is cleaner. This makes the comparison test the most straightforward choice.

(c) The factorial in the denominator grows much faster than the polynomial n^2 in the numerator, so I expect the terms to shrink very rapidly and the series to converge. Because the expression involves a factorial, the ratio test is the natural tool: successive terms simplify nicely, whereas a direct comparison would be less immediate. The ratio tends to 0, which confirms convergence.

Proof.
(a) Consider the series
$$\sum_{n=2}^{\infty} \frac{1}{\log n}.$$
For every $n \geq 2$, we have
$$\log n \leq n.$$
Since both quantities are positive, taking reciprocals yields
$$\frac{1}{\log n} \geq \frac{1}{n}.$$
Now the harmonic series
$$\sum_{n=2}^{\infty} \frac{1}{n}$$
diverges. Therefore, by the comparison test, the series
$$\sum_{n=2}^{\infty} \frac{1}{\log n}$$
also diverges.
(b) Consider the series
$$\sum_{n=1}^{\infty} \frac{n-1}{n^2}.$$
For every $n \geq 2$, we have
$$\frac{n-1}{n^2} \geq \frac{1}{2n}.$$
Indeed,
$$\frac{n-1}{n^2} \geq \frac{1}{2n} \quad \Longleftrightarrow \quad 2(n-1) \geq n \quad \Longleftrightarrow \quad n \geq 2.$$
Now the series
$$\sum_{n=2}^{\infty} \frac{1}{2n} = \frac{1}{2} \sum_{n=2}^{\infty} \frac{1}{n}$$
diverges, since the harmonic series diverges. Therefore, by the comparison test,
$$\sum_{n=2}^{\infty} \frac{n-1}{n^2}$$
diverges. Adding the first term $\frac{1-1}{1^2} = 0$ does not affect convergence, so
$$\sum_{n=1}^{\infty} \frac{n-1}{n^2}$$
diverges.

(c) Consider the series

$$\sum_{n=1}^{\infty} \frac{n^2}{n!}.$$

Set

$$a_n := \frac{n^2}{n!} \qquad (n \geq 1).$$

Then

$$\left| \frac{a_{n+1}}{a_n} \right| = \frac{(n+1)^2}{(n+1)!} \cdot \frac{n!}{n^2} = \frac{(n+1)^2}{(n+1)n^2} = \frac{n+1}{n^2}.$$

Therefore,

$$\lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n\to\infty} \frac{n+1}{n^2} = 0 < 1.$$

By the ratio test, the series

$$\sum_{n=1}^{\infty} \frac{n^2}{n!}$$

converges absolutely. In particular, it converges. $\square$

Problem 2

For each of the following series, determine if it converges and prove your answer.

(a)

$$\sum_{n=1}^{\infty} \frac{n^4}{4^n}$$

(b)

$$\sum_{n=1}^{\infty} \frac{n!}{n^4+3}$$

(c)

$$\sum_{n=1}^{\infty} \frac{2^n}{n!}$$

Discussion & Scratch Work

For each part, I first look at the dominant growth in the numerator and denominator, and then choose the test that makes that growth comparison easiest to formalize.

(a) The terms

$$\frac{n^4}{4^n}$$

have polynomial growth in the numerator and exponential growth in the denominator. Exponential growth should dominate, so I expect convergence. Since the expression is of the form "polynomial over exponential," the ratio test is natural: passing from a_n to a_{n+1} simplifies cleanly and the ratio should tend to $\frac{1}{4} < 1$. A comparison test could also work, but the ratio test is more direct.

(b) The terms

$$\frac{n!}{n^4+3}$$

have factorial growth in the numerator and only polynomial growth in the denominator. Since factorial growth is much faster, I expect the terms not even to go to 0, which would force divergence immediately. The ratio test is the cleanest way to confirm this expectation, because it captures the factorial growth very efficiently. A direct term test would also be possible after enough asymptotic work, but the ratio test is more systematic.

(c) The terms

$$\frac{2^n}{n!}$$

have exponential growth in the numerator but factorial growth in the denominator. Since factorial growth dominates exponential growth, I expect convergence. Again, the ratio test is the most natural choice because the factorial causes strong cancellation between successive terms, and the ratio should go to $0 < 1$.

Proof.

(a) Consider the series
$$\sum_{n=1}^{\infty} \frac{n^4}{4^n}.$$
Set
$$a_n := \frac{n^4}{4^n} \qquad (n \geq 1).$$
Then
$$\left|\frac{a_{n+1}}{a_n}\right| = \frac{(n+1)^4}{4^{n+1}} \cdot \frac{4^n}{n^4} = \frac{(n+1)^4}{4n^4} = \frac{1}{4}\left(\frac{n+1}{n}\right)^4.$$
Therefore,
$$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \frac{1}{4}\lim_{n\to\infty}\left(\frac{n+1}{n}\right)^4 = \frac{1}{4} < 1.$$
By the ratio test, the series
$$\sum_{n=1}^{\infty} \frac{n^4}{4^n}$$
converges absolutely. In particular, it converges.

(b) Consider the series
$$\sum_{n=1}^{\infty} \frac{n!}{n^4+3}.$$
Set
$$b_n := \frac{n!}{n^4+3} \qquad (n \geq 1).$$
Then
$$\left|\frac{b_{n+1}}{b_n}\right| = \frac{(n+1)!}{(n+1)^4+3} \cdot \frac{n^4+3}{n!} = (n+1)\frac{n^4+3}{(n+1)^4+3}.$$
Now,
$$\frac{n^4+3}{(n+1)^4+3} \to 1 \qquad \text{as } n \to \infty,$$
so
$$\left|\frac{b_{n+1}}{b_n}\right| \sim n+1,$$
and in particular,
$$\lim_{n\to\infty}\left|\frac{b_{n+1}}{b_n}\right| = +\infty > 1.$$
By the ratio test, the series
$$\sum_{n=1}^{\infty} \frac{n!}{n^4+3}$$
diverges.

(c) Consider the series
$$\sum_{n=1}^{\infty} \frac{2^n}{n!}.$$
Set
$$c_n := \frac{2^n}{n!} \qquad (n \geq 1).$$
Then
$$\left|\frac{c_{n+1}}{c_n}\right| = \frac{2^{n+1}}{(n+1)!} \cdot \frac{n!}{2^n} = \frac{2}{n+1}.$$
Therefore,
$$\lim_{n\to\infty} \left|\frac{c_{n+1}}{c_n}\right| = \lim_{n\to\infty} \frac{2}{n+1} = 0 < 1.$$
By the ratio test, the series
$$\sum_{n=1}^{\infty} \frac{2^n}{n!}$$
converges absolutely. In particular, it converges. □

Problem 3

For each of the following series, determine if it converges and prove your answer.

(a)

$$\sum_{n=1}^{\infty} \frac{1}{n^n}$$

(b)

$$\sum_{n=1}^{\infty} \frac{n!}{n^n}$$

(c)

$$\sum_{n=1}^{\infty} \frac{(-1)^n n!}{2^n}$$

Discussion & Scratch Work

For each part, I first compare the general size of the terms to the type of test that is best adapted to them.

(a) The terms

$$\frac{1}{n^n}$$

decay extremely fast, much faster than an ordinary geometric or p-series. Since the exponent itself depends on n, the root test is the most natural choice: taking the nth root removes the varying exponent immediately. A comparison test could be made to work, but the root test is much cleaner.

(b) The terms

$$\frac{n!}{n^n}$$

also involve competing growth rates, now factorial versus power. Because factorial expressions simplify very well under the ratio test, that is the most efficient method here. The ratio should approach e^{-1}-type behavior intuitively, and in fact it will tend to a number strictly less than 1, which proves convergence. The root test is possible too, but the ratio test is more direct.

(c) The terms

$$\frac{(-1)^n n!}{2^n}$$

have alternating sign, but the factorial in the numerator grows much faster than the exponential in the denominator. So before thinking about alternating-series ideas, I should check whether the terms even go to 0. The ratio test is the cleanest way to capture this growth. Since the absolute ratio will tend to something larger than 1, the terms cannot go to 0, so the series must diverge.

Proof.
(a) Consider the series

$$\sum_{n=1}^{\infty} \frac{1}{n^n}.$$

Set

$$a_n := \frac{1}{n^n} \qquad (n \geq 1).$$

Then

$$|a_n|^{1/n} = \left(\frac{1}{n^n}\right)^{1/n} = \frac{1}{n}.$$

Therefore,

$$\lim_{n\to\infty} |a_n|^{1/n} = \lim_{n\to\infty} \frac{1}{n} = 0 < 1.$$

By the root test, the series

$$\sum_{n=1}^{\infty} \frac{1}{n^n}$$

converges absolutely. In particular, it converges.
(b) Consider the series

$$\sum_{n=1}^{\infty} \frac{n!}{n^n}.$$

Set

$$b_n := \frac{n!}{n^n} \qquad (n \geq 1).$$

Then

$$\left|\frac{b_{n+1}}{b_n}\right| = \frac{(n+1)!}{(n+1)^{n+1}} \cdot \frac{n^n}{n!} = \frac{(n+1)n^n}{(n+1)^{n+1}} = \left(\frac{n}{n+1}\right)^n.$$

Now

$$\left(\frac{n}{n+1}\right)^n = \left(1 - \frac{1}{n+1}\right)^n \to e^{-1} < 1 \qquad \text{as } n \to \infty.$$

Hence

$$\lim_{n\to\infty} \left|\frac{b_{n+1}}{b_n}\right| = e^{-1} < 1.$$

By the ratio test, the series

$$\sum_{n=1}^{\infty} \frac{n!}{n^n}$$

converges absolutely. In particular, it converges.
(c) Consider the series

$$\sum_{n=1}^{\infty} \frac{(-1)^n n!}{2^n}.$$

Set
$$c_n := \frac{(-1)^n n!}{2^n} \qquad (n \geq 1).$$
Then
$$\left|\frac{c_{n+1}}{c_n}\right| = \left|\frac{(-1)^{n+1}(n+1)!}{2^{n+1}} \cdot \frac{2^n}{(-1)^n n!}\right| = \frac{n+1}{2}.$$
Therefore,
$$\lim_{n\to\infty}\left|\frac{c_{n+1}}{c_n}\right| = \lim_{n\to\infty}\frac{n+1}{2} = +\infty > 1.$$
By the ratio test, the series
$$\sum_{n=1}^{\infty} \frac{(-1)^n n!}{2^n}$$
diverges. □

Problem 4

Consider the series

$$\sum_{n=1}^{\infty} 2^{(-1)^n - n} = \frac{1}{4} + \frac{1}{2} + \frac{1}{16} + \frac{1}{8} + \frac{1}{64} + \frac{1}{32} + \cdots .$$

(a) Show that the ratio test is inconclusive.

(b) Prove that the series converges using the root test.

Discussion & Scratch Work

> This problem tells me which tests to use, so the main task is to apply them carefully and explicitly.
>
> **(a)** To show that the ratio test is inconclusive, I should not just say that the limit does not exist. I need to compute the ratio of consecutive terms and show that its tail behavior crosses both sides of 1. Because the exponent $(-1)^n$ depends on parity, the ratio will behave differently for even and odd n. So the cleanest way is to compute a_{n+1}/a_n and then split into the two parity cases.
>
> **(b)** Once the ratio test fails, the root test is more natural because the nth root interacts well with the factor 2^{-n}. After taking the nth root, the only remaining subtlety is the factor $2^{(-1)^n/n}$, which tends to 1 since $(-1)^n/n \to 0$. This should give a limit strictly less than 1, so the series converges absolutely.

Proof.
Let

$$a_n := 2^{(-1)^n - n} \qquad (n \geq 1).$$

Then the given series is

$$\sum_{n=1}^{\infty} a_n.$$

(a) We show that the ratio test is inconclusive. For each $n \geq 1$,

$$\left| \frac{a_{n+1}}{a_n} \right| = 2^{(-1)^{n+1} - (n+1) - \left((-1)^n - n \right)} = 2^{(-1)^{n+1} - (-1)^n - 1}.$$

Now split into cases.

If n is even, then $(-1)^n = 1$ and $(-1)^{n+1} = -1$, so

$$\left| \frac{a_{n+1}}{a_n} \right| = 2^{-1-1-1} = 2^{-3} = \frac{1}{8}.$$

If n is odd, then $(-1)^n = -1$ and $(-1)^{n+1} = 1$, so

$$\left| \frac{a_{n+1}}{a_n} \right| = 2^{1-(-1)-1} = 2^1 = 2.$$

Therefore the sequence

$$\left\{\left|\frac{a_{n+1}}{a_n}\right|\right\}_{n\geq 1}$$

alternates between $\frac{1}{8}$ and 2. In particular,

$$\liminf_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \frac{1}{8} \qquad \text{and} \qquad \limsup_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = 2.$$

Hence

$$\liminf_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| \leq 1 \leq \limsup_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right|,$$

so by the remark following the ratio test, the ratio test is inconclusive.

(b) We now apply the root test. For each $n \geq 1$,

$$|a_n|^{1/n} = \left(2^{(-1)^n-n}\right)^{1/n} = 2^{\frac{(-1)^n-n}{n}} = 2^{\frac{(-1)^n}{n}-1}.$$

Since

$$\frac{(-1)^n}{n} \to 0 \qquad \text{as } n\to\infty,$$

it follows that

$$\lim_{n\to\infty}|a_n|^{1/n} = 2^{-1} = \frac{1}{2} < 1.$$

Therefore,

$$\limsup_{n\to\infty}|a_n|^{1/n} = \frac{1}{2} < 1.$$

By the root test, the series

$$\sum_{n=1}^{\infty} 2^{(-1)^n-n}$$

converges absolutely. In particular, it converges. □

Problem 5

Below are two series $\sum_{n=1}^{\infty} a_n$ and the numbers $s \in \mathbb{R}$ to which they converge. For each of these series, what is the set of all numbers $\tilde{s} \in \mathbb{R}$ such that there exists a rearrangement that satisfies $\sum_{n=1}^{\infty} \tilde{a}_n = \tilde{s}$? Prove your answer.

(a)

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{n} = -\log 2$$

(b)

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} = -\frac{\pi^2}{12}$$

(Hint: Your proof should be quite short, because we already did most of the work in lecture.)

Discussion & Scratch Work

This problem is almost entirely a matter of identifying whether each series is conditionally convergent or absolutely convergent, because Lecture 20 proves that rearrangements behave very differently in those two cases.

(a) The series

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$$

is the alternating harmonic series up to an overall minus sign. It converges by the alternating series test, but it does not converge absolutely because

$$\sum_{n=1}^{\infty} \left| \frac{(-1)^n}{n} \right| = \sum_{n=1}^{\infty} \frac{1}{n}$$

diverges. So this is a conditionally convergent series. By Riemann's rearrangement result from Lecture 20, a conditionally convergent series can be rearranged so that it converges to any prescribed real number. Hence the set of all possible rearranged sums is $\mathbb{R}$.

(b) The series

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}$$

converges absolutely because

$$\sum_{n=1}^{\infty} \left| \frac{(-1)^n}{n^2} \right| = \sum_{n=1}^{\infty} \frac{1}{n^2}$$

converges. Lecture 20 also proves that every rearrangement of an absolutely convergent series converges to the same sum as the original series. Therefore the only possible rearranged sum is the original value.

Proof.
(a) Let
$$a_n := \frac{(-1)^n}{n} \qquad (n \geq 1).$$
Then
$$\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} \frac{(-1)^n}{n}$$
converges, since it is the alternating harmonic series up to an overall minus sign. On the other hand,
$$\sum_{n=1}^{\infty} |a_n| = \sum_{n=1}^{\infty} \frac{1}{n}$$
diverges, because the harmonic series diverges. Hence
$$\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$$
converges, but does not converge absolutely. Thus it is conditionally convergent.

By Riemann's rearrangement result, if a series converges but does not converge absolutely, then for any real number $r \in \mathbb{R}$ there exists a rearrangement whose sum is r. Therefore the set of all numbers $\tilde{s} \in \mathbb{R}$ such that there exists a rearrangement with
$$\sum_{n=1}^{\infty} \tilde{a}_n = \tilde{s}$$
is exactly
$$\mathbb{R}.$$

(b) Let
$$b_n := \frac{(-1)^n}{n^2} \qquad (n \geq 1).$$
Then
$$\sum_{n=1}^{\infty} |b_n| = \sum_{n=1}^{\infty} \frac{1}{n^2}$$
converges, since this is a p-series with $p = 2 > 1$. Hence
$$\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}$$
converges absolutely.

Now Lecture 20 proves that if a series converges absolutely, then every rearrangement converges to the same sum as the original series. Since the original sum is given to be
$$\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} = -\frac{\pi^2}{12},$$

it follows that the set of all possible rearranged sums is

$$\left\{-\frac{\pi^2}{12}\right\}.$$

□

Problem 6

Prove or disprove the following statements:

(a) If $\sum_{n=1}^{\infty} a_n$ converges, then $\sum_{n=1}^{\infty} a_{2n}$ converges.

(b) If $\sum_{n=1}^{\infty} a_n$ converges absolutely, then $\sum_{n=1}^{\infty} a_{2n}$ converges absolutely.

Discussion & Scratch Work

The key point in both parts is that the series

$$\sum_{n=1}^{\infty} a_{2n}$$

keeps only the even-indexed terms of the original series. So I should ask whether passing to a subsequence of terms preserves convergence.

(a) I do *not* expect ordinary convergence to be preserved, because a convergent series may rely on cancellation between its odd and even terms. If I keep only the even terms, I may destroy that cancellation completely. The alternating harmonic series is the standard example to test this suspicion: the full series converges, but the even terms alone look like a constant multiple of the harmonic series. So this statement should be false, and a counterexample is the right method.

(b) Absolute convergence is much stronger. If

$$\sum_{n=1}^{\infty} |a_n|$$

converges, then throwing away terms should not break convergence, because the remaining terms form a positive subseries of a convergent positive series. So I expect this statement to be true. The cleanest proof is a direct comparison:

$$0 \leq |a_{2n}| \leq (\text{the full positive series } \sum |a_n|),$$

which shows that the even-indexed absolute-value series also converges.

Proof.
(a) The statement is *false*. Consider the series

$$\sum_{n=1}^{\infty} a_n \qquad \text{with} \qquad a_n := \frac{(-1)^{n+1}}{n}.$$

Then

$$\sum_{n=1}^{\infty} a_n = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots$$

converges by the alternating series test. However, the even-indexed subseries is

$$\sum_{n=1}^{\infty} a_{2n} = \sum_{n=1}^{\infty} \frac{(-1)^{2n+1}}{2n} = -\sum_{n=1}^{\infty} \frac{1}{2n} = -\frac{1}{2}\sum_{n=1}^{\infty} \frac{1}{n}.$$

Since the harmonic series diverges, it follows that

$$\sum_{n=1}^{\infty} a_{2n}$$

diverges. Therefore the statement in part (a) is false.

(b) The statement is *true*. Assume that

$$\sum_{n=1}^{\infty} a_n$$

converges absolutely. By definition, this means that

$$\sum_{n=1}^{\infty} |a_n|$$

converges.

For each $N \in \mathbb{N}$, set

$$S_N := \sum_{n=1}^{N} |a_n|, \qquad T_N := \sum_{n=1}^{N} |a_{2n}|.$$

Since all terms are nonnegative, both $\{S_N\}$ and $\{T_N\}$ are increasing sequences. Moreover, for every N we have

$$T_N = |a_2| + |a_4| + \cdots + |a_{2N}| \le |a_1| + |a_2| + \cdots + |a_{2N}| = S_{2N}.$$

Because $\sum_{n=1}^{\infty} |a_n|$ converges, the sequence $\{S_N\}$ is bounded. Hence the subsequence $\{S_{2N}\}$ is also bounded, and therefore $\{T_N\}$ is bounded above as well. Since $\{T_N\}$ is increasing and bounded above, it converges.

Thus the series

$$\sum_{n=1}^{\infty} |a_{2n}|$$

converges, which means exactly that

$$\sum_{n=1}^{\infty} a_{2n}$$

converges absolutely. $\square$

Appendix: Toolbox

Series and Convergence

Definition 1 (Convergence of a series). *Let $\sum_{n=1}^{\infty} a_n$ be a series, and for $n \geq 1$ set*

$$s_n := \sum_{k=1}^{n} a_k = a_1 + \cdots + a_n.$$

We say that $\sum_{n=1}^{\infty} a_n$ converges to s if $s_n \to s$ as $n \to \infty$. Equivalently, $\sum_{n=1}^{\infty} a_n$ converges if and only if its sequence of partial sums converges.

Theorem 1 (Comparison test). *Let $\sum_{n=1}^{\infty} a_n$ be a series with $a_n \geq 0$ for all $n \geq 1$.*

(1) If $\sum_{n=1}^{\infty} a_n$ converges and $|b_n| \leq a_n$ for all $n \geq 1$, then $\sum_{n=1}^{\infty} b_n$ converges.

(2) If $\sum_{n=1}^{\infty} a_n$ diverges and $b_n \geq a_n$ for all $n \geq 1$, then $\sum_{n=1}^{\infty} b_n$ diverges.

Theorem 2 (Dyadic criterion). *Let $\{a_n\}_{n\geq 1}$ be a decreasing sequence with $a_n \geq 0$ for all $n \geq 1$. The series $\sum_{n=1}^{\infty} a_n$ converges if and only if the series*

$$\sum_{n=0}^{\infty} 2^n a_{2^n}$$

converges.

Root and Ratio Tests

Theorem 3 (Root test). *Let $\sum_{n=1}^{\infty} a_n$ be a series.*

(1) If $\limsup_{n\to\infty} |a_n|^{1/n} < 1$, then $\sum_{n=1}^{\infty} a_n$ converges absolutely.

(2) If $\liminf_{n\to\infty} |a_n|^{1/n} > 1$, then $\sum_{n=1}^{\infty} a_n$ diverges.

Remark. *If*

$$\liminf_{n\to\infty} |a_n|^{1/n} \leq 1 \leq \limsup_{n\to\infty} |a_n|^{1/n},$$

then the root test is inconclusive.

Theorem 4 (Ratio test). *Let $\sum_{n=1}^{\infty} a_n$ be a series of nonzero numbers.*

(1) If $\limsup_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| < 1$, then $\sum_{n=1}^{\infty} a_n$ converges absolutely.

(2) If $\liminf_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| > 1$, then $\sum_{n=1}^{\infty} a_n$ diverges.

Remark. *If*

$$\liminf_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| \leq 1 \leq \limsup_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right|,$$

then the ratio test is inconclusive.

Dirichlet and Alternating Series

Theorem 5 (Dirichlet's criterion). *Let $\{a_n\}_{n\geq 1}$ and $\{b_n\}_{n\geq 1}$ be sequences. If $\{a_n\}_{n\geq 1}$ is decreasing, $\lim_{n\to\infty} a_n = 0$, and the sequence*

$$\left\{\sum_{k=1}^{n} b_k\right\}_{n\geq 1}$$

is bounded, then

$$\sum_{n=1}^{\infty} a_n b_n$$

converges.

Remark (Summation by parts). *If $t_n := \sum_{k=1}^{n} b_k$ for $n \geq 1$ and $t_0 := 0$, then*

$$\sum_{k=m}^{n} a_k b_k = -\sum_{k=m}^{n-1} (a_{k+1} - a_k) t_k + a_n t_n - a_m t_{m-1}.$$

This identity is sometimes called "summation by parts."

Corollary 1 (Alternating Series Test). *If $\{a_n\}_{n\geq 1}$ is a decreasing sequence and $\lim_{n\to\infty} a_n = 0$, then*

$$\sum_{n=1}^{\infty} (-1)^n a_n$$

converges.

Rearrangements of Series

Definition 2 (Rearrangement of a series). *Let $\{a_n\}_{n\geq 1}$ be a sequence. We say*

$$\sum_{n=1}^{\infty} \tilde{a}_n$$

is a rearrangement of

$$\sum_{n=1}^{\infty} a_n$$

if there exists a bijection $k : \mathbb{N} \to \mathbb{N}$ such that

$$\tilde{a}_n = a_{k(n)} \qquad \text{for all } n \geq 1.$$

Theorem 6 (Riemann's rearrangement result). *Let $\sum_{n=1}^{\infty} a_n$ be a series that converges, but does not converge absolutely. For any α, β satisfying*

$$-\infty \leq \alpha \leq \beta \leq \infty,$$

there exists a rearrangement $\sum_{n=1}^{\infty} \tilde{a}_n$ *so that the partial sums*

$$\tilde{s}_n := \sum_{k=1}^{n} \tilde{a}_k$$

obey

$$\liminf_{n\to\infty} \tilde{s}_n = \alpha \qquad \text{and} \qquad \limsup_{n\to\infty} \tilde{s}_n = \beta.$$

Theorem 7. *If* $\sum_{n=1}^{\infty} a_n$ *converges absolutely, then any rearrangement* $\sum_{n=1}^{\infty} \tilde{a}_n$ *converges to* $\sum_{n=1}^{\infty} a_n$.

Metric Spaces

Definition 3 (Metric)**.** *Let* X *be a nonempty set. A metric on* X *is a function*

$$d : X \times X \to \mathbb{R}$$

such that:

(1) $d(x, y) \geq 0$ *for all* $x, y \in X$.

(2) $d(x, y) = 0$ *if and only if* $x = y$.

(3) $d(x, y) = d(y, x)$ *for all* $x, y \in X$.

(4) *(Triangle inequality)* $d(x, z) \leq d(x, y) + d(y, z)$ *for all* $x, y, z \in X$.

Definition 4 (Metric space)**.** *Let* X *be a nonempty set and let* d *be a metric on* X. *Together, we call* (X, d) *a metric space.*

Math 521, Section 2
Homework 8

Problem 1. Consider the metric space $(\mathbb{R}, d)$ with the usual metric $d(x, y) = |x - y|$. For each of the following sets $A \subseteq \mathbb{R}$, find its interior and prove your answer.

(a) $A = [0, 1)$
(b) $A = \mathbb{Z}$
(c) $A = \{x \in \mathbb{Q} : 0 < x < 1\}$

Problem 2. Recall that d_1 and d_∞ are both metrics on $\mathbb{R}^2$.

(a) Prove that $d_\infty(x, y) \leq d_1(x, y) \leq 2d_\infty(x, y)$ for all $x, y \in \mathbb{R}^2$.
(b) Prove that for any $x \in \mathbb{R}^2$ and $r > 0$ we have

$$\{y \in \mathbb{R}^2 : d_\infty(x, y) < \tfrac{r}{2}\} \subseteq \{y \in \mathbb{R}^2 : d_1(x, y) < r\} \subseteq \{y \in \mathbb{R}^2 : d_\infty(x, y) < r\}.$$

I recommend that you draw a picture of these three sets.
(c) Prove that a set $A \subseteq \mathbb{R}^2$ is open in the metric space $(\mathbb{R}^2, d_1)$ if and only if it is open in the metric space $(\mathbb{R}^2, d_\infty)$.

Problem 3. Let X be a nonempty set, and recall that the discrete metric d is a metric on X. Prove that every set $A \subseteq X$ is open in the metric space (X, d).

Problem 4. Let (X, d) be any metric space. Define the function $\tilde{d} : X \times X \to \mathbb{R}$ by

$$\tilde{d}(x, y) = \min\{d(x, y), 1\}.$$

(a) Prove that $\tilde{d}$ is also a metric on X.
(b) Prove that a set $A \subseteq X$ is open in (X, d) if and only if it is open in $(X, \tilde{d})$.

Problem 5. Let (X, d) be a metric space, $A \subseteq X$, and $x \in X$. Prove that x is in A° if and only if there exists an open set U so that $x \in U \subseteq A$.

Problem 6. Let (X, d) be a metric space and let A be a nonempty subset of X. Prove that A is open if and only if there exists a set I, positive numbers $r_i > 0$, and points $a_i \in X$ so that $A = \bigcup_{i \in I} B_{r_i}(a_i)$.

Homework 8

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Contents

Problem 1

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(a) $A = [0, 1)$

(b) $A = \mathbb{Z}$

(c) $A = \{x \in \mathbb{Q} : 0 < x < 1\}$

Discussion & Scratch Work

For part (a), I want to identify exactly which points of $A = [0, 1)$ admit a small open ball that stays entirely inside A.

Because the metric is the usual one on $\mathbb{R}$,

$$B_r(x) = \{y \in \mathbb{R} : |x - y| < r\} = (x - r, x + r).$$

So the problem becomes: for which points $x \in [0, 1)$ does there exist $r > 0$ such that

$$(x - r, x + r) \subseteq [0, 1)?$$

The natural guess is that

$$A^\circ = (0, 1).$$

Indeed, if $x \in (0, 1)$, then x has positive distance from both endpoints 0 and 1, so I should be able to choose a small radius that stays inside the interval. On the other hand, the endpoint 0 should fail to be interior, because every open ball around 0 extends to the left of 0, hence leaves A.

So the proof has two parts:

(1) show that every $x \in (0, 1)$ is an interior point of A by constructing an explicit radius $r > 0$;

(2) show that 0 is not an interior point of A.

Since $A = [0, 1)$ contains only the points in $(0, 1)$ together with 0, these two steps will identify the interior completely.

For part (b), I again translate the definition of interior point into balls for the usual metric. Since

$$B_r(n) = (n - r, n + r)$$

for any integer $n \in \mathbb{Z}$, the question is whether there can be some radius $r > 0$ for which this entire interval stays inside $\mathbb{Z}$. The natural guess is that this is impossible, because every open interval in $\mathbb{R}$ contains non-integer real numbers. So I expect that no integer is an interior point, and hence

$$\mathbb{Z}^\circ = \varnothing.$$

To prove this, I take an arbitrary integer n and an arbitrary radius $r > 0$, and then I exhibit a specific point in $B_r(n)$ that is not an integer. This shows that no open ball centered at an integer can stay inside $\mathbb{Z}$, so no point of $\mathbb{Z}$ is interior.

For part (c), I consider

$$A = \{x \in \mathbb{Q} : 0 < x < 1\}.$$

Although the rationals are dense in $\mathbb{R}$, that does not help with interior points. In fact, every open interval in $\mathbb{R}$ contains irrational numbers as well. This suggests that no open ball centered at a rational point can stay entirely inside A, since A contains only rationals.

So I suspect that

$$A^\circ = \varnothing.$$

To prove this, I take an arbitrary point $x \in A$ and an arbitrary radius $r > 0$, and show that $B_r(x)$ is not contained in A. Using the density of the irrationals, I can always find an irrational number inside $(x - r, x + r)$, which will not belong to A. This shows that no point of A is an interior point.

Proof.

(a) Let $A = [0, 1)$. I claim that

$$A^\circ = (0, 1).$$

First, let $x \in (0, 1)$. Define

$$r := \frac{1}{2}\min\{x, 1 - x\}.$$

Since $x > 0$ and $1 - x > 0$, we have $r > 0$. Also,

$$r \le \frac{x}{2} \qquad \text{and} \qquad r \le \frac{1 - x}{2},$$

so

$$x - r > 0 \qquad \text{and} \qquad x + r < 1.$$

Therefore,

$$B_r(x) = (x - r, x + r) \subseteq (0, 1) \subseteq [0, 1) = A.$$

Hence x is an interior point of A. Since $x \in (0, 1)$ was arbitrary, it follows that

$$(0, 1) \subseteq A^\circ.$$

Now we show that 0 is not an interior point of A. Let $r > 0$. Then

$$B_r(0) = (-r, r).$$

But

$$-\frac{r}{2} \in (-r, r) \qquad \text{and} \qquad -\frac{r}{2} \notin [0, 1).$$

So

$$B_r(0) \nsubseteq A.$$

Since this happens for every $r > 0$, the point 0 is not an interior point of A.

Because the only points of $A = [0, 1)$ are the points of $(0, 1)$ together with 0, we conclude that the interior of A is exactly

$$A^\circ = (0, 1).$$

(b) Let $A = \mathbb{Z}$. I claim that

$$A^\circ = \varnothing.$$

Let $n \in \mathbb{Z}$. We show that n is not an interior point of A. Let $r > 0$. Then

$$B_r(n) = (n - r, n + r).$$

Since $(n - r, n + r)$ is an open interval in $\mathbb{R}$, it contains points that are not integers. In particular, we can choose some $y \in (n - r, n + r)$ with $y \notin \mathbb{Z}$. Then

$$y \in B_r(n) \quad \text{but} \quad y \notin \mathbb{Z},$$

so

$$B_r(n) \not\subseteq \mathbb{Z}.$$

Since this happens for every $r > 0$, the point n is not an interior point of $\mathbb{Z}$.

Because $n \in \mathbb{Z}$ was arbitrary, no point of $\mathbb{Z}$ is an interior point. Therefore

$$\mathbb{Z}^\circ = \varnothing.$$

(c) Let

$$A = \{x \in \mathbb{Q} : 0 < x < 1\}.$$

I claim that

$$A^\circ = \varnothing.$$

Let $x \in A$. We show that x is not an interior point of A. Let $r > 0$. Then

$$B_r(x) = (x - r, x + r).$$

Since $\mathbb{R} \setminus \mathbb{Q}$ is dense in $\mathbb{R}$, there exists an irrational number y such that

$$x - r < y < x + r.$$

Hence

$$y \in B_r(x),$$

but $y \notin A$, since $A \subseteq \mathbb{Q}$. Therefore

$$B_r(x) \not\subseteq A.$$

Since this happens for every $r > 0$, the point x is not an interior point of A.

Because $x \in A$ was arbitrary, no point of A is an interior point. Thus

$$A^\circ = \varnothing.$$

□

Problem 2

Recall that d_1 and d_∞ are both metrics on $\mathbb{R}^2$.

(a) Prove that $d_\infty(x, y) \le d_1(x, y) \le 2d_\infty(x, y)$ for all $x, y \in \mathbb{R}^2$.

(b) Prove that for any $x \in \mathbb{R}^2$ and $r > 0$ we have

$$\{y \in \mathbb{R}^2 : d_\infty(x, y) < \tfrac{r}{2}\} \subseteq \{y \in \mathbb{R}^2 : d_1(x, y) < r\} \subseteq \{y \in \mathbb{R}^2 : d_\infty(x, y) < r\}.$$

I recommend that you draw a picture of these three sets.

(c) Prove that a set $A \subseteq \mathbb{R}^2$ is open in the metric space $(\mathbb{R}^2, d_1)$ if and only if it is open in the metric space $(\mathbb{R}^2, d_\infty)$.

Discussion & Scratch Work

(a) This part is asking me to compare the two metrics on $\mathbb{R}^2$ pointwise. So for arbitrary points $x = (x_1, x_2)$ and $y = (y_1, y_2)$, I should write out both definitions:

$$d_\infty(x, y) = \max\{|x_1 - y_1|, |x_2 - y_2|\}, \qquad d_1(x, y) = |x_1 - y_1| + |x_2 - y_2|.$$

The first inequality should follow because the maximum of two nonnegative numbers is at most their sum, and the second should follow because each term is at most the maximum. So this is mostly an algebraic comparison of two formulas.

(b) This part translates the inequalities from part (a) into containments of open balls. To prove one set is contained in another, I should take an arbitrary point in the first set and use the distance inequality from part (a) to show it belongs to the second set. So each inclusion should be a short direct argument using the definitions of the balls and the inequalities already proved. Check out the diagram below (it's pretty and awesome):

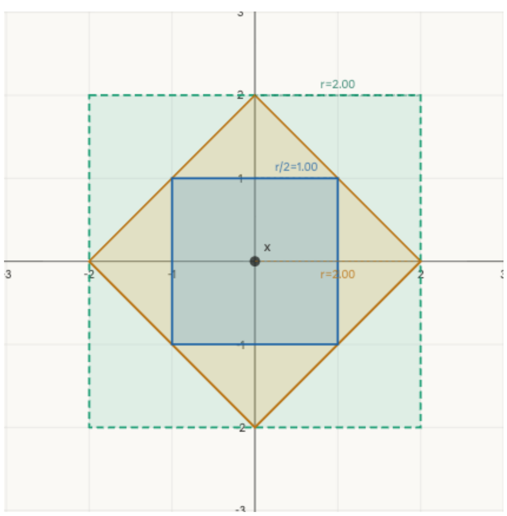

Figure 1: The three nested sets centered at x with parameter $r = 2$: the inner blue square is $\{d_\infty(x, y) < r/2\}$, the amber diamond is $\{d_1(x, y) < r\}$, and the outer teal square is $\{d_\infty(x, y) < r\}$.

(c) This part is asking me to show that the notions of open set coming from d_1 and d_∞ are the same. The key idea is that part (b) tells me that around any point, a sufficiently small d_∞-ball sits inside a given d_1-ball, and conversely a d_1-ball sits inside a corresponding d_∞-ball. So to prove openness in one metric implies openness in the other, I take a point of the set, start with the open ball guaranteed by one metric, and then use part (b) to find a smaller ball for the other metric that is still contained in the set.

Proof.
(a) Let $x = (x_1, x_2)$ and $y = (y_1, y_2)$ be points in $\mathbb{R}^2$. Then

$$d_\infty(x, y) = \max\{|x_1 - y_1|, |x_2 - y_2|\}$$

and

$$d_1(x, y) = |x_1 - y_1| + |x_2 - y_2|.$$

Because $|x_1 - y_1|$ and $|x_2 - y_2|$ are nonnegative, we have

$$\max\{|x_1 - y_1|, |x_2 - y_2|\} \leq |x_1 - y_1| + |x_2 - y_2|,$$

so

$$d_\infty(x, y) \leq d_1(x, y).$$

Also,

$$|x_1 - y_1| \leq \max\{|x_1 - y_1|, |x_2 - y_2|\} = d_\infty(x, y)$$

and similarly

$$|x_2 - y_2| \leq d_\infty(x, y).$$

Adding these inequalities gives

$$d_1(x, y) = |x_1 - y_1| + |x_2 - y_2| \leq 2d_\infty(x, y).$$

Therefore,

$$d_\infty(x, y) \leq d_1(x, y) \leq 2d_\infty(x, y)$$

for all $x, y \in \mathbb{R}^2$.
(b) Let $x \in \mathbb{R}^2$ and let $r > 0$.
We first prove that

$$\{y \in \mathbb{R}^2 : d_\infty(x, y) < r/2\} \subseteq \{y \in \mathbb{R}^2 : d_1(x, y) < r\}.$$

Let $y \in \mathbb{R}^2$ satisfy $d_\infty(x, y) < r/2$. By part (a),

$$d_1(x, y) \leq 2d_\infty(x, y) < 2 \cdot \frac{r}{2} = r.$$

Hence $y \in \{y \in \mathbb{R}^2 : d_1(x, y) < r\}$. This proves the first inclusion.
Now we prove that

$$\{y \in \mathbb{R}^2 : d_1(x, y) < r\} \subseteq \{y \in \mathbb{R}^2 : d_\infty(x, y) < r\}.$$

Let $y \in \mathbb{R}^2$ satisfy $d_1(x,y) < r$. By part (a),

$$d_\infty(x,y) \le d_1(x,y) < r.$$

Hence $y \in \{y \in \mathbb{R}^2 : d_\infty(x,y) < r\}$. This proves the second inclusion.
Therefore,

$$\{y \in \mathbb{R}^2 : d_\infty(x,y) < r/2\} \subseteq \{y \in \mathbb{R}^2 : d_1(x,y) < r\} \subseteq \{y \in \mathbb{R}^2 : d_\infty(x,y) < r\}.$$

(c) We prove that a set $A \subseteq \mathbb{R}^2$ is open in $(\mathbb{R}^2, d_1)$ if and only if it is open in $(\mathbb{R}^2, d_\infty)$.
Suppose first that A is open in $(\mathbb{R}^2, d_1)$. Let $x \in A$. Since A is open with respect to d_1, there exists $r > 0$ such that

$$\{y \in \mathbb{R}^2 : d_1(x,y) < r\} \subseteq A.$$

By part (b),

$$\{y \in \mathbb{R}^2 : d_\infty(x,y) < r/2\} \subseteq \{y \in \mathbb{R}^2 : d_1(x,y) < r\}.$$

Therefore,

$$\{y \in \mathbb{R}^2 : d_\infty(x,y) < r/2\} \subseteq A.$$

So x is an interior point of A with respect to d_∞. Since $x \in A$ was arbitrary, A is open in $(\mathbb{R}^2, d_\infty)$.
Conversely, suppose that A is open in $(\mathbb{R}^2, d_\infty)$. Let $x \in A$. Since A is open with respect to d_∞, there exists $r > 0$ such that

$$\{y \in \mathbb{R}^2 : d_\infty(x,y) < r\} \subseteq A.$$

By part (b),

$$\{y \in \mathbb{R}^2 : d_1(x,y) < r\} \subseteq \{y \in \mathbb{R}^2 : d_\infty(x,y) < r\}.$$

Therefore,

$$\{y \in \mathbb{R}^2 : d_1(x,y) < r\} \subseteq A.$$

So x is an interior point of A with respect to d_1. Since $x \in A$ was arbitrary, A is open in $(\mathbb{R}^2, d_1)$.
Hence a set $A \subseteq \mathbb{R}^2$ is open in $(\mathbb{R}^2, d_1)$ if and only if it is open in $(\mathbb{R}^2, d_\infty)$. $\square$

Problem 3

Let X be a nonempty set, and recall that the discrete metric d is a metric on X. Prove that every set $A \subseteq X$ is open in the metric space (X, d).

Discussion & Scratch Work

> This problem is asking me to use the definition of openness in a very specific metric: the discrete metric. In the discrete metric, distinct points are always distance 1 apart, so the key observation is that a sufficiently small open ball around any point should contain only that point.
>
> So to prove that an arbitrary set $A \subseteq X$ is open, I take an arbitrary point $a \in A$ and try to find a radius $r > 0$ such that
>
> $$B_r(a) \subseteq A.$$
>
> The natural choice is any radius with $0 < r \leq 1$, since then the ball around a should be just $\{a\}$. Once I show that, it follows immediately that every point of A is an interior point, and hence A is open.

Proof.
Let $A \subseteq X$. We will show that A is open in (X, d).

Let $a \in A$. Since d is the discrete metric, we have

$$d(a, x) = \begin{cases} 0 & \text{if } x = a, \\ 1 & \text{if } x \neq a. \end{cases}$$

Consider the open ball of radius $\frac{1}{2}$ centered at a:

$$B_{1/2}(a) = \{x \in X : d(a, x) < 1/2\}.$$

If $x = a$, then $d(a, x) = 0 < 1/2$, so $a \in B_{1/2}(a)$. On the other hand, if $x \neq a$, then $d(a, x) = 1 \not< 1/2$, so $x \notin B_{1/2}(a)$.
Therefore,

$$B_{1/2}(a) = \{a\}.$$

Since $a \in A$, we have $\{a\} \subseteq A$, and hence

$$B_{1/2}(a) \subseteq A.$$

So a is an interior point of A.

Because $a \in A$ was arbitrary, every point of A is an interior point. Therefore

$$A^\circ = A,$$

so A is open in (X, d). □

Problem 4

Let (X, d) be any metric space. Define the function $\tilde{d} : X \times X \to \mathbb{R}$ by

$$\tilde{d}(x, y) = \min\{d(x, y), 1\}.$$

(a) Prove that $\tilde{d}$ is also a metric on X.

(b) Prove that a set $A \subseteq X$ is open in (X, d) if and only if it is open in $(X, \tilde{d})$.

Discussion & Scratch Work

(a) To prove that

$$\tilde{d}(x, y) = \min\{d(x, y), 1\}$$

is a metric, I need to verify the four metric axioms from the definition in the appendix: nonnegativity, definiteness, symmetry, and the triangle inequality. The first three should follow immediately from the corresponding properties of d. The only nontrivial part is the triangle inequality, and there the key idea is that truncating distances at 1 cannot break the inequality. Concretely, I can start from

$$d(x, z) \leq d(x, y) + d(y, z)$$

and then compare both sides after applying the function $t \mapsto \min\{t, 1\}$.

(b) This part asks me to compare the open sets defined by d and by $\tilde{d}$. The key observation is that for distances strictly less than 1, the two metrics agree exactly. So small enough balls in one metric should be the same as balls in the other metric. To prove the two notions of openness coincide, I take a point of the set and use the open ball guaranteed by one metric, then shrink the radius if necessary to make it less than 1. Once the radius is below 1, the corresponding d-ball and $\tilde{d}$-ball are identical, so openness transfers in both directions.

Proof.

(a) We prove that $\tilde{d}$ is a metric on X.

First, since $d(x, y) \geq 0$ for all $x, y \in X$, we have

$$\tilde{d}(x, y) = \min\{d(x, y), 1\} \geq 0.$$

Thus $\tilde{d}$ is nonnegative.

Next, we show definiteness. If $x = y$, then $d(x, y) = 0$, so

$$\tilde{d}(x, y) = \min\{0, 1\} = 0.$$

Conversely, if $\tilde{d}(x, y) = 0$, then

$$\min\{d(x, y), 1\} = 0.$$

Since both $d(x, y)$ and 1 are nonnegative and $1 \neq 0$, this forces

$$d(x, y) = 0.$$

Because d is a metric, it follows that $x = y$.

Symmetry is immediate, since

$$\tilde{d}(x, y) = \min\{d(x, y), 1\} = \min\{d(y, x), 1\} = \tilde{d}(y, x).$$

It remains to prove the triangle inequality. Let $x, y, z \in X$. Since d is a metric,

$$d(x, z) \leq d(x, y) + d(y, z).$$

Therefore,

$$\tilde{d}(x, z) = \min\{d(x, z), 1\} \leq \min\{d(x, y) + d(y, z), 1\}.$$

So it is enough to show that

$$\min\{d(x, y) + d(y, z), 1\} \leq \min\{d(x, y), 1\} + \min\{d(y, z), 1\}.$$

Set

$$a := d(x, y), \qquad b := d(y, z),$$

so $a, b \geq 0$. We claim that

$$\min\{a + b, 1\} \leq \min\{a, 1\} + \min\{b, 1\}.$$

If $a \geq 1$ or $b \geq 1$, then the right-hand side is at least 1, while the left-hand side is at most 1, so the inequality holds. If instead $a < 1$ and $b < 1$, then

$$\min\{a, 1\} = a, \qquad \min\{b, 1\} = b,$$

and hence

$$\min\{a + b, 1\} \leq a + b = \min\{a, 1\} + \min\{b, 1\}.$$

Thus the claim holds in all cases, and therefore

$$\tilde{d}(x, z) \leq \tilde{d}(x, y) + \tilde{d}(y, z).$$

So $\tilde{d}$ satisfies the triangle inequality.

Hence $\tilde{d}$ is a metric on X.

(b) We prove that a set $A \subseteq X$ is open in (X, d) if and only if it is open in $(X, \tilde{d})$.

Suppose first that A is open in (X, d). Let $x \in A$. Since A is open with respect to d, there exists $r > 0$ such that

$$B_r^d(x) := \{y \in X : d(x, y) < r\} \subseteq A.$$

Set

$$\rho := \min\left\{r, \frac{1}{2}\right\} > 0.$$

We claim that

$$B_\rho^{\tilde{d}}(x) = B_\rho^d(x).$$

Indeed, if $y \in B_\rho^{\tilde{d}}(x)$, then

$$\tilde{d}(x,y) < \rho \leq \frac{1}{2} < 1.$$

Since $\tilde{d}(x,y) = \min\{d(x,y),1\}$ and the value is strictly less than 1, we must have

$$\tilde{d}(x,y) = d(x,y),$$

so $d(x,y) < \rho$ and hence $y \in B_\rho^d(x)$. Conversely, if $y \in B_\rho^d(x)$, then $d(x,y) < \rho \leq \frac{1}{2} < 1$, so

$$\tilde{d}(x,y) = d(x,y) < \rho,$$

and thus $y \in B_\rho^{\tilde{d}}(x)$. Therefore the two balls are equal.

Since $\rho \leq r$, we have

$$B_\rho^d(x) \subseteq B_r^d(x) \subseteq A.$$

Hence

$$B_\rho^{\tilde{d}}(x) = B_\rho^d(x) \subseteq A.$$

So x is an interior point of A with respect to $\tilde{d}$. Since $x \in A$ was arbitrary, A is open in $(X,\tilde{d})$.

Conversely, suppose that A is open in $(X,\tilde{d})$. Let $x \in A$. Then there exists $r > 0$ such that

$$B_r^{\tilde{d}}(x) := \{y \in X : \tilde{d}(x,y) < r\} \subseteq A.$$

Set

$$\rho := \min\left\{r, \frac{1}{2}\right\} > 0.$$

As above, because $\rho < 1$, we have

$$B_\rho^d(x) = B_\rho^{\tilde{d}}(x).$$

Since $\rho \leq r$,

$$B_\rho^{\tilde{d}}(x) \subseteq B_r^{\tilde{d}}(x) \subseteq A.$$

Therefore

$$B_\rho^d(x) = B_\rho^{\tilde{d}}(x) \subseteq A.$$

So x is an interior point of A with respect to d. Since $x \in A$ was arbitrary, A is open in (X,d).

Hence A is open in (X,d) if and only if it is open in $(X,\tilde{d})$. $\square$

Problem 5

Let (X, d) be a metric space, $A \subseteq X$, and $x \in X$. Prove that x is in A° if and only if there exists an open set U so that $x \in U \subseteq A$.

Discussion & Scratch Work

> This problem is asking me to connect two ways of saying that a point lies in the interior of a set. By definition, $x \in A^\circ$ means that there is an open ball around x contained in A. On the other hand, the statement involving an open set U says there is some open neighborhood of x contained in A.
>
> So the proof should go in both directions. If $x \in A^\circ$, then I can simply take the open ball from the definition and call it U. Conversely, if I already have an open set U with $x \in U \subseteq A$, then because U is open and contains x, there must be an open ball around x contained in U, hence contained in A. That is exactly the definition of $x \in A^\circ$.

Proof.
We prove that

$$x \in A^\circ \quad \Longleftrightarrow \quad \text{there exists an open set } U \text{ such that } x \in U \subseteq A.$$

Suppose first that $x \in A^\circ$. By definition of interior, there exists $r > 0$ such that

$$B_r(x) \subseteq A.$$

Since every open ball is an open set, we may take

$$U := B_r(x).$$

Then U is open, $x \in U$, and $U \subseteq A$. Hence there exists an open set U such that $x \in U \subseteq A$.
Conversely, suppose there exists an open set U such that

$$x \in U \subseteq A.$$

Because U is open and $x \in U$, by definition of openness there exists $r > 0$ such that

$$B_r(x) \subseteq U.$$

Since $U \subseteq A$, it follows that

$$B_r(x) \subseteq A.$$

Therefore, by definition of interior, we have

$$x \in A^\circ.$$

This proves the equivalence. $\square$

Problem 6

Let (X, d) be a metric space and let A be a nonempty subset of X. Prove that A is open if and only if there exists a set I, positive numbers $r_i > 0$, and points $a_i \in X$ so that

$$A = \bigcup_{i \in I} B_{r_i}(a_i).$$

Discussion & Scratch Work

This problem is really asking me to recognize the structure of open sets in a metric space. From the lecture notes, a set is open exactly when every point in it has some open ball around it still contained in the set. So the statement here says that open sets are precisely unions of open balls.

To prove this, I need to show both directions. If A is open, then for each point $a \in A$ I can choose a radius $r_a > 0$ with

$$B_{r_a}(a) \subseteq A.$$

Then taking the union of all those balls should recover A. Conversely, if A is already given as a union of open balls, then each ball is open by the lecture result, and an arbitrary union of open sets is open. So the proof is short: one direction uses the definition of openness pointwise, and the other uses the fact that unions of open sets are open.

Proof.
We prove that A is open if and only if there exist a set I, positive numbers $r_i > 0$, and points $a_i \in X$ such that

$$A = \bigcup_{i \in I} B_{r_i}(a_i).$$

Suppose first that A is open. Since A is nonempty, for each $a \in A$ there exists $r_a > 0$ such that

$$B_{r_a}(a) \subseteq A.$$

For each $a \in A$, choose $r_a > 0$ such that

$$B_{r_a}(a) \subseteq A.$$

Then

$$A = \bigcup_{a \in A} B_{r_a}(a).$$

Conversely, suppose there exist a set I, positive numbers $r_i > 0$, and points $a_i \in X$ such that

$$A = \bigcup_{i \in I} B_{r_i}(a_i).$$

Each ball $B_{r_i}(a_i)$ is open in (X, d) by the lecture result that every open ball is an open set. Since an arbitrary union of open sets is open, it follows that

$$A = \bigcup_{i \in I} B_{r_i}(a_i)$$

is open.

Therefore A is open if and only if it can be written as a union of open balls. $\square$

Appendix: Toolbox

Metrics and Metric Spaces

Definition 1. *Let X be a nonempty set. A metric on X is a function $d : X \times X \to \mathbb{R}$ such that*

(1) $d(x, y) \geq 0$ *for all* $x, y \in X$.

(2) $d(x, y) = 0$ *if and only if* $x = y$.

(3) $d(x, y) = d(y, x)$ *for all* $x, y \in X$.

(4) $d(x, z) \leq d(x, y) + d(y, z)$ *for all* $x, y, z \in X$.

Definition 2. *If d is a metric on X, then (X, d) is called a metric space.*

ℓ^p Metrics on $\mathbb{R}^n$

Proposition 1 (Hölder's inequality)**.** *Fix $1 \leq p \leq \infty$, and let $1 \leq q \leq \infty$ satisfy*

$$\frac{1}{p} + \frac{1}{q} = 1.$$

For any $(x_1, \dots, x_n), (y_1, \dots, y_n) \in \mathbb{R}^n$ we have

$$\sum_{k=1}^{n} |x_k y_k| \leq \left(\sum_{k=1}^{n} |x_k|^p \right)^{1/p} \left(\sum_{k=1}^{n} |y_k|^q \right)^{1/q}.$$

When $p = \infty$, replace $\left(\sum_{k=1}^{n} |x_k|^p\right)^{1/p}$ by $\max_{k=1,\dots,n} |x_k|$ above.

Remark. *When $p = 2$ (and so $q = 2$), we call this the Cauchy–Schwarz inequality.*

Corollary 1 (Minkowski's inequality)**.** *For any $1 \leq p \leq \infty$ and $(x_1, \dots, x_n), (y_1, \dots, y_n) \in \mathbb{R}^n$, we have*

$$\left(\sum_{k=1}^{n} |x_k + y_k|^p \right)^{1/p} \leq \left(\sum_{k=1}^{n} |x_k|^p \right)^{1/p} + \left(\sum_{k=1}^{n} |y_k|^p \right)^{1/p}.$$

Corollary 2. *For any $1 \leq p \leq \infty$, the function $d_p : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$,*

$$d_p(x, y) = \begin{cases} \left(\sum_{k=1}^{n} |x_k - y_k|^p\right)^{1/p} & \text{if } 1 \leq p < \infty, \\ \max_{k=1,\dots,n} |x_k - y_k| & \text{if } p = \infty, \end{cases}$$

is a metric on $\mathbb{R}^n$.

Open Balls and Open Sets

Definition 3. *Let (X, d) be a metric space. The open ball (or neighborhood) of radius $r > 0$ centered at $a \in X$ is*

$$B_r(a) = \{x \in X : d(a, x) < r\}.$$

Definition 4. *Let (X, d) be a metric space and $A \subseteq X$. We say:*

(1) *$a \in X$ is an interior point of A if there exists $r > 0$ such that $B_r(a) \subseteq A$.*

(2)

$$A^\circ = \{a \in X : a \text{ is an interior point of } A\}$$

is the interior of A.

(3) *A is open if $A^\circ = A$.*

Remark. *Note that $A^\circ \subseteq A$ for any set A.*

Basic Properties of Interior and Open Sets

Proposition 2. *Let (X, d) be a metric space and $A, B \subseteq X$.*

(1) *If $A \subseteq B$, then $A^\circ \subseteq B^\circ$.*

(2) *$A^\circ \cup B^\circ \subseteq (A \cup B)^\circ$.*

(3) *$A^\circ \cap B^\circ = (A \cap B)^\circ$.*

(4) *$(A^\circ)^\circ = A^\circ$. In particular, A° is an open set.*

(5) *A° is the largest open set contained in A.*

(6) *A finite intersection of open sets is open.*

(7) *An arbitrary union of open sets is open.*

Remark. *An arbitrary intersection of open sets need not be open.*

Math 521, Section 2
Homework 9

Problem 1. Consider the metric space $(\mathbb{R}, d)$ with the usual metric $d(x, y) = |x - y|$. For each of the following sets $A \subseteq \mathbb{R}$, find its closure, isolated points, and limit points. Prove your answer.

(a) $A = [0, 1)$
(b) $A = \mathbb{Z}$
(c) $A = \{x \in \mathbb{Q} : 0 < x < 1\}$

Problem 2. Let (X, d) be a metric space, let A be a nonempty subset of X, and let U be an open subset of X.

(a) Prove that $U \cap \overline{A} \subseteq \overline{U \cap A}$.
(b) Prove that $\overline{U \cap \overline{A}} = \overline{U \cap A}$. (Hint: Use part (a) and properties of closure from lecture.)
(c) Conclude that if $U \cap A = \emptyset$, then $U \cap \overline{A} = \emptyset$.

Problem 3. Recall that d_2 is a metric on $\mathbb{R}^2$. Consider the subspace

$$Y = \{(x, 0) : x \in \mathbb{R}\}$$

with the induced metric d_Y.

(a) Provide an example of a set $A \subseteq Y$ where A is open in (Y, d_Y) but not open in $(\mathbb{R}^2, d_2)$.
(b) Prove that for any $B \subseteq Y$, if B is closed in (Y, d_Y) then B is closed in $(\mathbb{R}^2, d_2)$.

Problem 4. Let (X, d) be a metric space, $\{x_n\}_{n\geq 1}$ a sequence in X, and $x \in X$. Prove that the sequence $\{x_n\}_{n\geq 1}$ converges to x if and only if for every open set U that contains x, we have $x_n \in U$ for all but finitely many $n \in \mathbb{N}$.

Problem 5. Recall that d_2 is a metric on $\mathbb{R}^2$. Suppose that $x^n = (x_1^n, x_2^n)$ is a sequence of points in $\mathbb{R}^2$.

(a) Prove that $|x_1^n - x_1| \leq d_2(x^n, x)$ for any $x = (x_1, x_2) \in \mathbb{R}^2$.
(b) Prove that the sequence $\{x^n\}_{n\geq 1}$ converges in $(\mathbb{R}^2, d_2)$ if and only if the sequences $\{x_1^n\}_{n\geq 1}$ and $\{x_2^n\}_{n\geq 1}$ of real numbers both converge.

Problem 6. Let (X, d) be a metric space and $\{x_n\}_{n\geq 1}$ a sequence in X. Prove that if $\{x_n\}_{n\geq 1}$ converges to $x \in X$, then for any $y \in X$, the sequence $\{d(x_n, y)\}_{n\geq 1}$ of real numbers converges to $d(x, y)$.

Homework 9

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Contents

Problem 1

Consider the metric space $(\mathbb{R}, d)$ with the usual metric $d(x, y) = |x - y|$. For each of the following sets $A \subseteq \mathbb{R}$, find its closure, isolated points, and limit points. Prove your answer.

(a) $A = [0, 1)$

(b) $A = \mathbb{Z}$

(c) $A = \{x \in \mathbb{Q} : 0 < x < 1\}$

Discussion & Scratch Work

For this problem, the main job is to identify the adherent points of each set, because once I know those, I know the closure. By definition,

$$x \in \overline{A} \quad \Longleftrightarrow \quad \forall r > 0,\ B_r(x) \cap A \neq \varnothing.$$

So for each set, I first ask: which points can I never separate from A by a small ball?

Then I classify the points of A themselves. A point of A is isolated if some small ball meets A only at that point, and it is a limit point if every ball contains another point of A. So the proof pattern is:

(1) identify all adherent points, hence determine $\overline{A}$;

(2) check whether points of A can be isolated;

(3) determine the limit points using the definition directly.

For $[0, 1)$, I expect the closure to be $[0, 1]$: the interval contributes all of its own points, and the endpoint 1 is still touched by points of A. I do not expect isolated points, since points of the interval are never alone in a small ball.

For $\mathbb{Z}$, I expect the closure to be exactly $\mathbb{Z}$, because non-integers can be separated from the integers by a small ball. I also expect every integer to be isolated, since a ball of radius less than $\frac{1}{2}$ contains no other integers.

For $\mathbb{Q} \cap (0, 1)$, I expect the closure to be $[0, 1]$ again, but for a different reason: density of $\mathbb{Q}$ in $\mathbb{R}$ means every ball around a point of $[0, 1]$ contains rationals from $(0, 1)$. I do not expect isolated points, because rational points in $(0, 1)$ always have other rational points arbitrarily close by.

Proof.
We work in $(\mathbb{R}, d)$ with $d(x, y) = |x - y|$.

(a) Let $A = [0, 1)$. I claim that

$$\overline{A} = [0, 1], \qquad A' = [0, 1], \qquad \text{and } A \text{ has no isolated points.}$$

First, let $x \in [0, 1]$. If $x < 1$, then $x \in A$, so for every $r > 0$ we have

$$x \in B_r(x) \cap A,$$

and hence x is adherent to A. If $x = 1$, then for every $r > 0$ the point $1 - \frac{r}{2}$ lies in $B_r(1) \cap A$, so 1 is also adherent to A. Thus

$$[0, 1] \subseteq \overline{A}.$$

Now let $x \notin [0, 1]$. If $x < 0$, set $r = \frac{|x|}{2} > 0$. Then

$$B_r(x) \subseteq (-\infty, 0),$$

so $B_r(x) \cap A = \varnothing$. If $x > 1$, set $r = \frac{x-1}{2} > 0$. Then

$$B_r(x) \subseteq (1, \infty),$$

so again $B_r(x) \cap A = \varnothing$. Hence no point outside $[0, 1]$ is adherent to A. Therefore

$$\overline{A} = [0, 1].$$

Next, let $x \in A = [0, 1)$. We show that x is not isolated. Fix $r > 0$. If $x = 0$, set $y = \frac{r}{2}$. Then $y \in A$, $y \neq x$, and $|y - x| < r$. If $0 < x < 1$, set

$$y = x + \min\left\{\frac{r}{2}, \frac{1-x}{2}\right\}.$$

Then $y \in [0, 1) = A$, $y \neq x$, and $|y - x| < r$. So every ball $B_r(x)$ contains a point of A different from x, and therefore no point of A is isolated.

Finally, we identify the limit points. Let $x \in [0, 1]$ and let $r > 0$. If $x < 1$, then the argument above gives a point $y \in A$ with $y \neq x$ and $y \in B_r(x)$. If $x = 1$, then $1 - \frac{r}{2} \in A \cap B_r(1)$. Thus every point of $[0, 1]$ is a limit point of A. Conversely, every limit point is adherent, so it must lie in $\overline{A} = [0, 1]$. Hence

$$A' = [0, 1].$$

(b) Let $A = \mathbb{Z}$. I claim that

$$\overline{A} = \mathbb{Z}, \qquad A' = \varnothing, \qquad \text{and every point of } A \text{ is isolated.}$$

If $n \in \mathbb{Z}$, then for every $r > 0$ we have

$$n \in B_r(n) \cap \mathbb{Z},$$

so n is adherent to A. Thus

$$\mathbb{Z} \subseteq \overline{A}.$$

Now let $x \notin \mathbb{Z}$. Write $n = \lfloor x \rfloor$. Then either $n < x < n + 1$ or $x = n + 1$ with $n + 1 \in \mathbb{Z}$; since $x \notin \mathbb{Z}$, we are in the first case. Set

$$\delta := \min\{x - n, (n + 1) - x\} > 0, \qquad r := \frac{\delta}{2}.$$

Then the ball $B_r(x)$ lies inside the interval $(n, n + 1)$, which contains no integers. Hence

$$B_r(x) \cap \mathbb{Z} = \varnothing,$$

so x is not adherent to A. Therefore

$$\overline{A} = \mathbb{Z}.$$

Next, let $n \in \mathbb{Z}$ and choose $r = \frac{1}{2}$. If $m \in \mathbb{Z}$ and $m \neq n$, then $|m - n| \geq 1$, so $m \notin B_{1/2}(n)$. Thus

$$B_{1/2}(n) \cap \mathbb{Z} = \{n\}.$$

So every integer is isolated.

Since every point of A is isolated, A has no limit points. Hence

$$A' = \varnothing.$$

(c) Let $A = \{x \in \mathbb{Q} : 0 < x < 1\}$. I claim that

$$\overline{A} = [0,1], \qquad A' = [0,1], \qquad \text{and } A \text{ has no isolated points.}$$

First, let $x \in [0,1]$. Fix $r > 0$. Since $\mathbb{Q}$ is dense in $\mathbb{R}$, there exists $q \in \mathbb{Q}$ with

$$\max\left\{0, x - \frac{r}{2}\right\} < q < \min\left\{1, x + \frac{r}{2}\right\}.$$

Then $0 < q < 1$, so $q \in A$, and also $|q - x| < r$. Hence

$$B_r(x) \cap A \neq \varnothing.$$

So every point of $[0,1]$ is adherent to A, and therefore

$$[0,1] \subseteq \overline{A}.$$

Now let $x \notin [0,1]$. If $x < 0$, choose $r = \frac{|x|}{2} > 0$; then $B_r(x) \subseteq (-\infty, 0)$, so $B_r(x) \cap A = \varnothing$. If $x > 1$, choose $r = \frac{x-1}{2} > 0$; then $B_r(x) \subseteq (1, \infty)$, so again $B_r(x) \cap A = \varnothing$. Thus no point outside $[0,1]$ is adherent to A. Therefore

$$\overline{A} = [0,1].$$

Next, let $x \in A$ and let $r > 0$. Since $\mathbb{Q}$ is dense in $\mathbb{R}$, there exists a rational number

$$y \in \left(x, \min\left\{x + \frac{r}{2}, 1\right\}\right).$$

Then $y \in A$, $y \neq x$, and $|y - x| < r$. So every ball around x contains another point of A, which shows that no point of A is isolated.

Finally, let $x \in [0,1]$ and let $r > 0$. By the same density argument used above, there exists $q \in A$ with $q \neq x$ and $|q - x| < r$. Thus every point of $[0,1]$ is a limit point of A. Conversely, every limit point is adherent, so it must lie in $\overline{A} = [0,1]$. Hence

$$A' = [0,1].$$

Therefore:

(a) $\overline{A} = [0,1], \quad A' = [0,1],$ no isolated points;

(b) $\overline{A} = \mathbb{Z}, \quad A' = \varnothing,$ all points isolated;

(c) $\overline{A} = [0,1], \quad A' = [0,1],$ no isolated points.

□

Problem 2

Let (X, d) be a metric space, let A be a nonempty subset of X, and let U be an open subset of X.

(a) Prove that $U \cap \overline{A} \subseteq \overline{U \cap A}$.

(b) Prove that $\overline{U} \cap \overline{A} = \overline{U \cap A}$. (Hint: Use part (a) and properties of closure from lecture.)

(c) Conclude that if $U \cap A = \varnothing$, then $U \cap \overline{A} = \varnothing$.

Discussion & Scratch Work

For this problem, I want to use the definition of closure directly and keep the inclusions separate. In part (a), if I start with a point in $U \cap \overline{A}$, then I know two things: the point lies in the open set U, and every ball around it meets A. Since U is open, I can first choose a small ball staying inside U. Then, because the point is in $\overline{A}$, that smaller ball must still meet A. That gives a point in $U \cap A$, which is exactly what I need to prove the point lies in $\overline{U \cap A}$.

For part (b), I use both part (a) and the closure facts from lecture. Part (a) already gives one inclusion once I replace U by $\overline{U}$, since $\overline{U}$ is closed and contains U. For the other inclusion, I use monotonicity of closure: because $U \cap A \subseteq U$ and $U \cap A \subseteq A$, its closure must lie inside both $\overline{U}$ and $\overline{A}$.

Part (c) is then a short consequence of part (b). If $U \cap A = \varnothing$, then its closure is also empty, so part (b) forces $\overline{U} \cap \overline{A} = \varnothing$. Since $U \subseteq \overline{U}$, I immediately get $U \cap \overline{A} = \varnothing$.

Proof.
(a) Let $x \in U \cap \overline{A}$. We show that $x \in \overline{U \cap A}$.

Since $x \in U$ and U is open, there exists $r_0 > 0$ such that

$$B_{r_0}(x) \subseteq U.$$

Now let $r > 0$ be arbitrary, and set

$$\rho := \min\{r, r_0\} > 0.$$

Because $x \in \overline{A}$, every open ball around x meets A, so

$$B_\rho(x) \cap A \neq \varnothing.$$

Choose $y \in B_\rho(x) \cap A$. Since $\rho \leq r_0$, we have

$$y \in B_{r_0}(x) \subseteq U,$$

and since $y \in A$, it follows that

$$y \in U \cap A.$$

Also, because $\rho \le r$, we have $y \in B_r(x)$. Thus

$$B_r(x) \cap (U \cap A) \neq \varnothing.$$

Since this holds for every $r > 0$, the point x is adherent to $U \cap A$. Therefore

$$x \in \overline{U \cap A}.$$

Hence

$$U \cap \overline{A} \subseteq \overline{U \cap A}.$$

(b) We prove that

$$\overline{U} \cap \overline{A} = \overline{U \cap A}.$$

First, since U is open, part (a) applied to the set A and the open set U gives

$$U \cap \overline{A} \subseteq \overline{U \cap A}.$$

Because $U \subseteq \overline{U}$, we certainly have

$$U \cap \overline{A} \subseteq \overline{U} \cap \overline{A}.$$

To prove the desired equality, it is cleaner to establish both inclusions directly.
Since

$$U \cap A \subseteq U \qquad \text{and} \qquad U \cap A \subseteq A,$$

monotonicity of closure gives

$$\overline{U \cap A} \subseteq \overline{U} \qquad \text{and} \qquad \overline{U \cap A} \subseteq \overline{A}.$$

Therefore

$$\overline{U \cap A} \subseteq \overline{U} \cap \overline{A}.$$

For the reverse inclusion, let $x \in \overline{U} \cap \overline{A}$. Since U is open and $x \in \overline{U}$, every ball around x meets U; indeed, if some ball $B_r(x)$ were disjoint from U, then x would not be adherent to U. Now apply part (a) with the set A replaced by A and the open set U: because $x \in \overline{A}$ and every sufficiently small ball around x meets U, the argument from part (a) shows that every ball around x meets $U \cap A$. Hence

$$x \in \overline{U \cap A}.$$

So

$$\overline{U} \cap \overline{A} \subseteq \overline{U \cap A}.$$

Combining the two inclusions gives

$$\overline{U} \cap \overline{A} = \overline{U \cap A}.$$

(c) Suppose that

$$U \cap A = \varnothing.$$

Then

$$\overline{U \cap A} = \overline{\varnothing} = \varnothing.$$

By part (b),

$$\overline{U} \cap \overline{A} = \varnothing.$$

Since $U \subseteq \overline{U}$, we obtain

$$U \cap \overline{A} \subseteq \overline{U} \cap \overline{A} = \varnothing.$$

Therefore

$$U \cap \overline{A} = \varnothing.$$

$\square$

Problem 3

Recall that d_2 is a metric on $\mathbb{R}^2$. Consider the subspace

$$Y = \{(x,0) : x \in \mathbb{R}\}$$

with the induced metric d_Y.

(a) Provide an example of a set $A \subseteq Y$ where A is open in (Y, d_Y) but not open in $(\mathbb{R}^2, d_2)$.

(b) Prove that for any $B \subseteq Y$, if B is closed in (Y, d_Y) then B is closed in $(\mathbb{R}^2, d_2)$.

Discussion & Scratch Work

For part (a), I just need one example of a subset of Y that is open in the subspace metric but not open in all of $\mathbb{R}^2$. Since

$$Y = \{(x,0) : x \in \mathbb{R}\}$$

is the x-axis, a natural choice is an open interval sitting inside that line, such as

$$A = \{(x,0) : 0 < x < 1\}.$$

It should be open in Y because, inside the line, every point has a small ball that stays on the line and inside the interval. But it should fail to be open in $\mathbb{R}^2$ because every ordinary open ball in the plane contains points off the x-axis, and those points are not in A.

For part (b), I want to prove a closed-set statement for a subspace. The key fact from lecture is that a set is closed if and only if it contains the limits of all convergent sequences in it, and also that convergence in the subspace metric agrees with convergence in the ambient space because the induced metric is just the same distance restricted to Y. So the cleanest proof is sequential: take a sequence in $B \subseteq Y$ that converges in $\mathbb{R}^2$, notice it is automatically a sequence in the subspace Y with the same limit, and then use that B is closed in Y to conclude the limit lies in B. That shows B is closed in $\mathbb{R}^2$.

Proof.
(a) Let

$$A := \{(x,0) : 0 < x < 1\} \subseteq Y.$$

We claim that A is open in (Y, d_Y) but not open in $(\mathbb{R}^2, d_2)$.

First, let $(x,0) \in A$, so $0 < x < 1$. Set

$$r := \frac{1}{2}\min\{x, 1-x\} > 0.$$

If $(y,0) \in Y$ and $d_Y((x,0),(y,0)) < r$, then since d_Y is induced by d_2,

$$|y - x| = d_Y((x,0),(y,0)) < r.$$

Hence
$$0 < x - r < y < x + r < 1,$$
so $(y, 0) \in A$. Therefore
$$B_r^Y((x,0)) \subseteq A,$$
and A is open in (Y, d_Y).

Now we show that A is not open in $(\mathbb{R}^2, d_2)$. Let $(x, 0) \in A$ and let $r > 0$. Then the point
$$(x, r/2)$$
satisfies
$$d_2((x,0),(x,r/2)) = r/2 < r,$$
so $(x, r/2) \in B_r^{\mathbb{R}^2}((x,0))$. But $(x, r/2) \notin A$ since its second coordinate is not 0. Thus no open ball in $\mathbb{R}^2$ centered at a point of A is contained in A, and therefore A is not open in $(\mathbb{R}^2, d_2)$.

(b) Let $B \subseteq Y$, and suppose that B is closed in (Y, d_Y). We prove that B is closed in $(\mathbb{R}^2, d_2)$.

Let $\{b_n\}_{n\geq 1}$ be a sequence in B that converges in $(\mathbb{R}^2, d_2)$ to some point $b \in \mathbb{R}^2$. Since every b_n lies in Y, we may write
$$b_n = (x_n, 0) \qquad \text{for some } x_n \in \mathbb{R}.$$
Because
$$d_2(b_n, b) \to 0,$$
the second coordinates of b_n converge to the second coordinate of b. But the second coordinate of every b_n is 0, so the second coordinate of b must also be 0. Hence
$$b \in Y.$$

Now, for points of Y, the induced metric is just the restriction of d_2, so
$$d_Y(b_n, b) = d_2(b_n, b) \to 0.$$

Thus the sequence $\{b_n\}$ converges to b in the subspace (Y, d_Y). Since B is closed in (Y, d_Y) and every term of the sequence lies in B, it follows that
$$b \in B.$$

Therefore every sequence in B that converges in $(\mathbb{R}^2, d_2)$ has its limit in B. Hence B is closed in $(\mathbb{R}^2, d_2)$. $\square$

Problem 4

Let (X, d) be a metric space, $\{x_n\}_{n\geq 1}$ a sequence in X, and $x \in X$. Prove that the sequence $\{x_n\}_{n\geq 1}$ converges to x if and only if for every open set U that contains x, we have $x_n \in U$ for all but finitely many $n \in \mathbb{N}$.

Discussion & Scratch Work

> This problem is asking me to connect the usual ε-definition of convergence with the open-set language from metric spaces. The statement "for all but finitely many n" means that eventually every term of the sequence lies in the set under discussion. So I want to show that saying
>
> $$d(x_n, x) < \varepsilon$$
>
> for all sufficiently large n is equivalent to saying that, for every open set U containing x, all sufficiently large terms x_n lie in U.
>
> So the proof should go in two directions. If I assume $x_n \to x$, then given an open set U containing x, openness gives me some ball $B_r(x) \subseteq U$, and convergence tells me that eventually $x_n \in B_r(x)$, hence eventually $x_n \in U$. Conversely, if I assume the open-set condition, then I test it on the particular open set $U = B_\varepsilon(x)$. Since every open ball is open and contains x, the hypothesis gives exactly the usual convergence condition.

Proof.
We prove that

$$x_n \to x \quad \Longleftrightarrow \quad \text{for every open set } U \ni x,\ x_n \in U \text{ for all but finitely many } n.$$

Suppose first that $x_n \to x$. Let U be an open set containing x. Since U is open and $x \in U$, there exists $r > 0$ such that

$$B_r(x) \subseteq U.$$

Because $x_n \to x$, there exists $N \in \mathbb{N}$ such that

$$d(x_n, x) < r \qquad \text{for all } n \geq N.$$

Equivalently,

$$x_n \in B_r(x) \qquad \text{for all } n \geq N.$$

Since $B_r(x) \subseteq U$, it follows that

$$x_n \in U \qquad \text{for all } n \geq N.$$

Hence $x_n \in U$ for all but finitely many n.

Conversely, suppose that for every open set U containing x, we have $x_n \in U$ for all but finitely many n. Let $\varepsilon > 0$. The open ball

$$B_\varepsilon(x)$$

is an open set containing x. Therefore there exists $N \in \mathbb{N}$ such that

$$x_n \in B_\varepsilon(x) \qquad \text{for all } n \geq N.$$

By definition of the ball, this means

$$d(x_n, x) < \varepsilon \qquad \text{for all } n \geq N.$$

Since $\varepsilon > 0$ was arbitrary, it follows that $x_n \to x$. $\square$

Problem 5

Recall that d_2 is a metric on $\mathbb{R}^2$. Suppose that $x^n = (x_1^n, x_2^n)$ is a sequence of points in $\mathbb{R}^2$.

(a) Prove that $|x_1^n - x_1| \le d_2(x^n, x)$ for any $x = (x_1, x_2) \in \mathbb{R}^2$.

(b) Prove that the sequence $\{x^n\}_{n\ge1}$ converges in $(\mathbb{R}^2, d_2)$ if and only if the sequences $\{x_1^n\}_{n\ge1}$ and $\{x_2^n\}_{n\ge1}$ of real numbers both converge.

Discussion & Scratch Work

For part (a), the key idea is that in $\mathbb{R}^2$ with the Euclidean metric, the distance between two points controls the distance between each individual coordinate. So if $x^n \to x$ in d_2, then the first coordinates should automatically converge as real numbers. The proof is just an inequality: I compare $|x_1^n - x_1|$ to the full Euclidean distance.

For part (b), I want to prove the usual coordinatewise characterization of convergence in $\mathbb{R}^2$. One direction should follow from part (a): if the vectors converge, then each coordinate converges. For the other direction, if both coordinate sequences converge, then I use the definition of d_2 and estimate

$$d_2(x^n, x) = \sqrt{(x_1^n - x_1)^2 + (x_2^n - x_2)^2}$$

by making each coordinate difference small. So the proof is just two short implications, both using the definition of the Euclidean metric.

Proof.
(a) Let $x^n = (x_1^n, x_2^n)$ and $x = (x_1, x_2)$ be points in $\mathbb{R}^2$. Then by definition of d_2,

$$d_2(x^n, x) = \sqrt{(x_1^n - x_1)^2 + (x_2^n - x_2)^2}.$$

Since $(x_2^n - x_2)^2 \ge 0$, we have

$$(x_1^n - x_1)^2 \le (x_1^n - x_1)^2 + (x_2^n - x_2)^2.$$

Taking square roots gives

$$|x_1^n - x_1| \le d_2(x^n, x).$$

(b) We prove that $\{x^n\}_{n\ge1}$ converges in $(\mathbb{R}^2, d_2)$ if and only if the real sequences $\{x_1^n\}_{n\ge1}$ and $\{x_2^n\}_{n\ge1}$ both converge.

Suppose first that $x^n \to x = (x_1, x_2)$ in $(\mathbb{R}^2, d_2)$. By part (a),

$$|x_1^n - x_1| \le d_2(x^n, x) \qquad \text{and} \qquad |x_2^n - x_2| \le d_2(x^n, x)$$

for all n. Since $d_2(x^n, x) \to 0$, it follows that

$$|x_1^n - x_1| \to 0 \qquad \text{and} \qquad |x_2^n - x_2| \to 0.$$

Hence

$$x_1^n \to x_1 \qquad \text{and} \qquad x_2^n \to x_2$$

as sequences of real numbers.

Conversely, suppose that

$$x_1^n \to x_1 \qquad \text{and} \qquad x_2^n \to x_2.$$

Let $x = (x_1, x_2)$. We show that $x^n \to x$ in $(\mathbb{R}^2, d_2)$. Let $\varepsilon > 0$. Since $x_1^n \to x_1$, there exists $N_1 \in \mathbb{N}$ such that

$$|x_1^n - x_1| < \frac{\varepsilon}{\sqrt{2}} \qquad \text{for all } n \geq N_1.$$

Since $x_2^n \to x_2$, there exists $N_2 \in \mathbb{N}$ such that

$$|x_2^n - x_2| < \frac{\varepsilon}{\sqrt{2}} \qquad \text{for all } n \geq N_2.$$

Set

$$N := \max\{N_1, N_2\}.$$

Then for all $n \geq N$,

$$d_2(x^n, x) = \sqrt{(x_1^n - x_1)^2 + (x_2^n - x_2)^2} < \sqrt{\frac{\varepsilon^2}{2} + \frac{\varepsilon^2}{2}} = \varepsilon.$$

Therefore $x^n \to x$ in $(\mathbb{R}^2, d_2)$. $\square$

Problem 6

Let (X, d) be a metric space and $\{x_n\}_{n\geq 1}$ a sequence in X. Prove that if $\{x_n\}_{n\geq 1}$ converges to $x \in X$, then for any $y \in X$, the sequence $\{d(x_n, y)\}_{n\geq 1}$ of real numbers converges to $d(x, y)$.

Discussion & Scratch Work

Here,

$$\{d(x_n, y)\}_{n\geq 1}$$

is a sequence of real numbers, so saying it converges to $d(x, y)$ means exactly that

$$\forall \varepsilon > 0 \ \exists N \in \mathbb{N} \text{ s.t. } \left|d(x_n, y) - d(x, y)\right| < \varepsilon \qquad \forall n \geq N.$$

So, to my understanding, I think the problem is asking me to show that if $x_n \to x$ in the metric space (X, d), then the distances from x_n to a fixed point y behave continuously and converge to the distance from x to y.

The key idea is to compare $d(x_n, y)$ and $d(x, y)$ using the triangle inequality. The standard trick is to write two inequalities,

$$d(x_n, y) \leq d(x_n, x) + d(x, y) \qquad \text{and} \qquad d(x, y) \leq d(x, x_n) + d(x_n, y),$$

and then combine them to get

$$\left|d(x_n, y) - d(x, y)\right| \leq d(x_n, x).$$

Once I have that estimate, the conclusion is immediate because $x_n \to x$ means $d(x_n, x) \to 0$.

Proof.
Assume that $x_n \to x$ in (X, d), and fix $y \in X$. We must show that the real sequence $\{d(x_n, y)\}_{n\geq 1}$ converges to $d(x, y)$.

Let $n \geq 1$. By the triangle inequality,

$$d(x_n, y) \leq d(x_n, x) + d(x, y).$$

Rearranging gives

$$d(x_n, y) - d(x, y) \leq d(x_n, x).$$

Also, by the triangle inequality,

$$d(x, y) \leq d(x, x_n) + d(x_n, y) = d(x_n, x) + d(x_n, y),$$

so

$$d(x, y) - d(x_n, y) \leq d(x_n, x).$$

Combining these two inequalities, we obtain

$$\left|d(x_n, y) - d(x, y)\right| \leq d(x_n, x) \qquad \text{for all } n \geq 1.$$

Now let $\varepsilon > 0$. Since $x_n \to x$, there exists $N \in \mathbb{N}$ such that

$$d(x_n, x) < \varepsilon \qquad \text{for all } n \geq N.$$

Therefore, for all $n \geq N$,

$$\left| d(x_n, y) - d(x, y) \right| \leq d(x_n, x) < \varepsilon.$$

This shows that

$$d(x_n, y) \to d(x, y)$$

as a sequence of real numbers. $\square$

Appendix: Toolbox

Lecture 22

Definition 1. *Let (X, d) be a metric space and $A \subseteq X$. We say:*

(1) *$a \in X$ is an interior point of A if $\exists\, r > 0$ s.t. $B_r(a) \subseteq A$.*

(2)

$$A^\circ = \{a \in X : \ a \text{ is an interior point of } A\}$$

is the interior of A.

(3) *A is open if $A^\circ = A$.*

Example 1. *$X = \mathbb{R}^2$ with the metric d_2, and $A = [0, 1] \times [0, 1]$.*

- *$x = \left(\frac{3}{4}, \frac{1}{2}\right)$ is an interior point of A: $B_{1/4}(x) \subseteq A$.*
- *$y = \left(\frac{1}{2}, 1\right)$ is not an interior point of A: $\forall r > 0,\ B_r(y) \nsubseteq A$.*
- *$A^\circ = (0, 1) \times (0, 1)$.*
- *A is not open in $(\mathbb{R}^2, d_2)$.*

Example 2. *Let (X, d) be a metric space.*

- *$\varnothing$ and X are open sets.*
- *$B_r(a)$ is an open set $\forall a \in X$ and $\forall r > 0$:*
  Claim: $B_r(a) \subseteq (B_r(a))^\circ$.
  Fix $x \in B_r(a)$, i.e. $d(x, a) < r$.
  Want: $x \in (B_r(a))^\circ$, i.e. $\exists\, s > 0$ s.t. $B_s(x) \subseteq B_r(a)$. Set

  $$s = r - d(x, a) > 0.$$

  Then

  $$y \in B_s(x) \Rightarrow d(y, x) < s \Rightarrow d(y, a) \leq d(y, x) + d(x, a) < s + d(x, a) = r.$$

  So $B_s(x) \subseteq B_r(a)$.

Remark. *Note that $A^\circ \subseteq A$ for any set A:*

$$x \in A^\circ \Rightarrow \exists\, r > 0 \text{ s.t. } B_r(x) \subseteq A \Rightarrow x \in B_r(x) \subseteq A$$

since $d(x, x) = 0 < r$.
So, to prove that a set is open, it suffices to show $A^\circ \supseteq A$.

Proposition 1. *Let (X, d) be a metric space and $A, B \subseteq X$.*

(1) *If $A \subseteq B$, then $A^\circ \subseteq B^\circ$.*

(2) *$A^\circ \cup B^\circ \subseteq (A \cup B)^\circ$.*

(3) *$A^\circ \cap B^\circ = (A \cap B)^\circ$.*

(4) *$(A^\circ)^\circ = A^\circ$. In particular, A° is an open set.*

(5) *A° is the largest open set contained in A.*

(6) *A finite intersection of open sets is open.*

(7) *An arbitrary union of open sets is open.*

Remark. *An arbitrary intersection of open sets need not be open. Eg. in $(\mathbb{R}, |\cdot|)$,*

$$\bigcap_{n\in\mathbb{N}} \left(-\frac{1}{n}, \frac{1}{n}\right) = \{0\}.$$

Lecture 23

Definition 2. *Let (X, d) be a metric space. A set $A \subseteq X$ is closed if A^c is open.*

Example 3. *Let (X, d) be a metric space.*

(1) *$\varnothing$ and X are closed: since $\varnothing^c = X$ and $X^c = \varnothing$ are open.*

(2)

$$(B_r(a))^c = \{x \in X : \ d(x, a) \geq r\}$$

is closed $\forall r > 0$ and $\forall a \in X$: since

$$[(B_r(a))^c]^c = B_r(a)$$

is open.

(3)

$$A = \{x \in X : \ d(x, a) \leq r\}$$

is closed $\forall r > 0$ and $\forall a \in X$.

Want: $A^c = \{x \in X : \ d(x, a) > r\}$ is open. Fix $x \in A^c$. Then $d(x, a) > r$. Set $s = d(x, a) - r > 0$. Then $B_s(x) \subseteq A^c$ since

$$y \in B_s(x) \Rightarrow d(y, x) < s \Rightarrow d(a, x) \leq d(a, y) + d(y, x) < d(a, y) + s \Rightarrow d(a, y) > d(a, x) - s = r.$$

Remark. *"closed" $\neq$ "not open". Eg., in $(\mathbb{R}, |\cdot|)$,*

- *$(0, 1)$ is open*
- *$[0, 1]$ is closed*
- *$\varnothing$ and $\mathbb{R}$ are open and closed*

- $(0,1]$ *is not open and not closed*

Definition 3. *Let (X,d) be a metric space and $A \subseteq X$. We say:*

(1) *$x \in X$ is an adherent point of A if:*

$$\forall r > 0,\ B_r(x) \cap A \neq \varnothing.$$

(2)

$$\overline{A} = \{x \in X : \ x \text{ is an adherent point of } A\}$$

is the closure of A.

(3) *An adherent point x of A is called:*

- *isolated if $\exists\, r > 0$ s.t. $B_r(x) \cap A = \{x\}$*
- *a limit point of A otherwise*

(4)

$$A' = \{x \in X : \ x \text{ is a limit point of } A\}.$$

Remark. *$A \subseteq \overline{A}$ for any set A:*

$$x \in A \Rightarrow x \in B_r(x) \cap A \qquad \forall r > 0.$$

$$x \in A' \Leftrightarrow \text{ the intersection of } B_r(x) \text{ and } A \setminus \{x\} \text{ is nonempty } \forall r > 0$$

$$\overline{A} = A \cup A'.$$

Example 4. $(\mathbb{R}, |\cdot|)$, $\qquad A = \left\{\frac{1}{n} : n \in \mathbb{N}\right\}$.

(1) *$\frac{1}{n}$ is an isolated point of A $\ \forall n \in \mathbb{N}$:*

Fix n. Set

$$r = \frac{1}{n} - \frac{1}{n+1} > 0.$$

Then

$$B_r\left(\frac{1}{n}\right) \cap A = \left\{\frac{1}{n}\right\}.$$

(2) *0 is a limit point for A:*

Fix $r > 0$. Pick $n \in \mathbb{N}$ s.t. $\frac{1}{n} < r$. Then

$$\frac{1}{n} \in (-r, r) \cap A,$$

so $(-r,r) \cap A \neq \varnothing$.

(3)

$$\overline{A} = A \cup \{0\}.$$

Lemma. *Let (X, d) be a metric space and $A \subseteq X$.*

(1) $(\overline{A})^c = (A^c)^\circ$

(2) $(A^\circ)^c = \overline{A^c}$

(3) $A = \overline{A} \iff A$ *is a closed set.*

Exercise 1. *Let (X, d) be a metric space and $A, B \subseteq X$.*

(1) *If $A \subseteq B$ then $\overline{A} \subseteq \overline{B}$.*

(2) $\overline{A \cap B} \subseteq \overline{A} \cap \overline{B}$.

(3) $\overline{A \cup B} = \overline{A} \cup \overline{B}$.

(4) $\overline{\overline{A}} = \overline{A}$. *In particular, $\overline{A}$ is a closed set.*

(5) *$\overline{A}$ is the smallest closed set containing A.*

(6) *A finite union of closed sets is a closed set.*

(7) *An arbitrary intersection of closed sets is a closed set.*

Use the proposition from last lecture and take the complement of each statement.
*Eg., for **(1)**,*

$$A \subseteq B \Rightarrow A^c \supseteq B^c \Rightarrow (A^c)^\circ \supseteq (B^c)^\circ \Rightarrow \overline{A}^c \supseteq \overline{B}^c \Rightarrow \overline{A} \subseteq \overline{B}.$$

Definition 4. *Let (X, d_X) be a metric space and let $\varnothing \neq Y \subseteq X$. Then $d_Y : Y \times Y \to \mathbb{R}$,*

$$d_Y(x, y) = d_X(x, y)$$

is a metric on Y, called the induced metric on Y. The metric space (Y, d_Y) is called a subspace of (X, d_X).

Example 5. *$\mathbb{R}$ with $d_{\mathbb{R}}(x, y) = |x - y|$ is a metric space.*
$[0, 1]$ with $d_{[0,1]}(x, y) = |x - y|$ is a subspace.

Lecture 24

Proposition 2. *Let (X, d_X) be a metric space and (Y, d_Y) a subspace (with the induced metric).*

(1) *A set $V \subseteq Y$ is open in (Y, d_Y) $\Leftrightarrow$ $\exists U \subseteq X$ open in (X, d_X) s.t. $V = U \cap Y$.*

(2) *A set $G \subseteq Y$ is closed in (Y, d_Y) $\Leftrightarrow$ $\exists F \subseteq X$ closed in (X, d_X) s.t. $G = F \cap Y$.*

Example 6. ***(1)*** *$[0, 1)$ is not open in $(\mathbb{R}, |\cdot|)$.*

$[0, 1)$ is open in $([0, 2], |\cdot|)$, since

$$[0, 1) = (-1, 1) \cap [0, 2].$$

(2) $[0,1)$ *is not closed in* $(\mathbb{R}, |\cdot|)$.

$[0,1)$ *is closed in* $((-1,1), |\cdot|)$, *since*

$$[0,1) = (-1,1) \cap [0,2].$$

Corollary 1. *Let* (X, d_X) *be a metric space and* (Y, d_Y) *a subspace. The following are equivalent:*

(1) Y *is open (closed) in* X.

(2) *Any* $A \subseteq Y$ *that is open (closed) in* Y *is also open (closed) in* X.

Proposition 3. *Let* (X, d_X) *be a metric space and* (Y, d_Y) *a subspace. For any* $A \subseteq Y$,

$$\overline{A}^Y = \overline{A}^X \cap Y.$$

Definition 5. *Let* (X, d) *be a metric space and let* $\{x_n\}_{n\geq 1}$ *be a sequence in* X. *We say* $\{x_n\}_{n\geq 1}$ *converges to* $x \in X$ *if:*

$$\forall \varepsilon > 0 \ \exists N \in \mathbb{N} \ \ s.t. \ \ d(x_n, x) < \varepsilon \qquad \forall n \geq N.$$

When this is true, we write "$\lim_{n\to\infty} x_n = x$*" or "*$x_n \to x$ *in* (X, d) *as* $n \to \infty$*".*

Example 7. ***(1)*** *For* $(\mathbb{R}, |\cdot|)$, *we have*

$$d(x_n, x) = |x_n - x|.$$

This agrees with the definition from lecture 7.

(2) *For* $(\mathbb{R}^2, d_p)$, *we have*

$$d(x_n, x) = \left(|x_{n,1} - x_1|^p + |x_{n,2} - x_2|^p\right)^{1/p}.$$

Remark. $x_n \to x$ *in* (X, d) *as* $n \to \infty$ $\Leftrightarrow$ *the sequence of real numbers* $d(x_n, x)$ *converges to* 0 *as* $n \to \infty$.

Lemma. *The limit of a sequence is unique, if it exists.*

Theorem 1. *A sequence* $\{x_n\}_{n\geq 1}$ *converges to* x $\Leftrightarrow$ *every subsequence* $\{x_{n_k}\}_{k\geq 1}$ *converges to* x.

Definition 6. *Let* (X, d) *be a metric space. We say that a sequence* $\{x_n\}_{n\geq 1}$ *is Cauchy if:*

$$\forall \varepsilon > 0 \ \exists N \in \mathbb{N} \ \ s.t. \ \ d(x_n, x_m) < \varepsilon \qquad \forall n, m \geq N.$$

Proposition 4. *If* $\{x_n\}_{n\geq 1}$ *converges, then it is Cauchy.*

Remark. *Not every Cauchy sequence converges! Eg.,*

(1) $((0,1), |\cdot|)$ *is a metric space.*

$\left\{\frac{1}{n}\right\}_{n\geq 1}$ *is Cauchy, but does not converge in* $(0,1)$.

(2) $(\mathbb{Q}, |\cdot|)$ *is a metric space.*

$$x_1 = 3, \qquad x_{n+1} = \frac{x_n}{2} + \frac{3}{2x_n} \qquad \forall n \geq 1.$$

Then $\{x_n\}_{n\geq 1}$ *is Cauchy, but doesn't converge in* $\mathbb{Q}$.

By lecture 12, this sequence converges to $\sqrt{3}$.

Lecture 25

Definition 7. *A metric space (X, d) is complete if every Cauchy sequence in X converges in X.*

Example 8. ***(1)*** $(\mathbb{R}, |\cdot|)$ *is complete (by lecture 12)*

(2) $(\mathbb{Q}, |\cdot|)$ *is not complete (by the previous example)*

Proposition 5. *Let $1 \leq p \leq \infty$. $\forall n \in \mathbb{N}$, $(\mathbb{R}^n, d_p)$ is complete.*

Proposition 6. *Let (X, d) be a metric space, $A \subseteq X$, and $x \in X$. Then:*

$$x \in \overline{A} \;\Leftrightarrow\; \exists \text{ a sequence } \{a_n\}_{n\geq 1} \text{ in } A \text{ that converges to } x.$$

Remark. *Recall:*

$$\overline{A} = A \cup A'$$

$$x \in A' \Leftrightarrow \exists \text{ a sequence in } A \setminus \{x\} \text{ that converges to } x.$$

Definition 8. *Let (X, d) be a metric space and $A \subseteq X$. We say:*

(1) *a collection of sets $\{U_i\}_{i\in I}$ is an open cover of A, if $I \neq \varnothing$ is a set, U_i is open $\forall i \in I$, and*

$$A \subseteq \bigcup_{i\in I} U_i.$$

(2) *the set A is compact if: $\forall$ open cover $\{U_i\}_{i\in I}$ of A, $\exists$ a finite set $J \subseteq I$ s.t. $\{U_j\}_{j\in J}$ is an open cover of A. We call $\{U_j\}_{j\in J}$ a finite subcover.*

Example 9. *Any finite set $\{x_1, \ldots, x_n\} \subseteq X$ is compact.*
Let $\{U_i\}_{i\in I}$ be an open cover of $\{x_1, \ldots, x_n\}$.
Then

$$x_1 \in \bigcup_{i\in I} U_i \Rightarrow \exists\, i_1 \in I \text{ s.t. } x_1 \in U_{i_1}$$

$$x_2 \in \bigcup_{i\in I} U_i \Rightarrow \exists\, i_2 \in I \text{ s.t. } x_2 \in U_{i_2}$$

$$\vdots$$

So $\{U_{i_1}, \ldots, U_{i_n}\}$ is a finite subcover.

Math 521, Section 2
Homework 10

Problem 1. Let (X, d) be a metric space and let $Y \subseteq X$. We call the set Y complete if the subspace Y with the induced metric is complete. Prove that if Y is complete, then Y is closed in X. (Hint: It suffices to show $\overline{Y} \subseteq Y$. Use sequences.)

Problem 2. Let (X, d) be a metric space. Prove that if $A, B \subseteq X$ are both compact then $A \cup B$ is compact.

Problem 3. Let (X, d) be a metric space.

(a) Prove that if $A \subseteq B \subseteq X$, A is closed in X, and B is compact, then A is compact. (Hint: Given an open cover of A, we can add A^c to get an open cover of B.)

(b) Prove that if $C, D \subseteq X$ are both compact, then $C \cap D$ is compact. (Hint: Use (a).)

Problem 4. Let (X, d) be a metric space and let $Y \subseteq X$. Prove that if Y is compact, then Y is complete.

Homework 10

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Contents

Problem 1

Let (X, d) be a metric space and let $Y \subseteq X$. We call the set Y complete if the subspace Y with the induced metric is complete. Prove that if Y is complete, then Y is closed in X. (Hint: It suffices to show $\overline{Y} \subseteq Y$. Use sequences.)

Discussion & Scratch Work

> The key definitions here come straight from lecture. First, saying that $Y \subseteq X$ is *complete* means that every Cauchy sequence in the subspace (Y, d_Y) converges to some point of Y, where d_Y is the induced metric. Since the induced metric is just the restriction of d to pairs of points in Y, a sequence in Y is Cauchy in (Y, d_Y) exactly when it is Cauchy in (X, d).
>
> The most important theorem from lecture for this problem is the sequential characterization of closure: for a subset $A \subseteq X$,
>
> $$x \in \overline{A} \iff \text{there exists a sequence } \{a_n\} \subseteq A \text{ such that } a_n \to x.$$
>
> We also proved that every convergent sequence in a metric space is Cauchy, and that limits in a metric space are unique.
>
> So the strategy is very direct. Since the hint says it suffices to prove $\overline{Y} \subseteq Y$, I start with an arbitrary point $x \in \overline{Y}$. By the closure theorem, there is a sequence $\{y_n\} \subseteq Y$ with $y_n \to x$ in X. Because convergent sequences are Cauchy, $\{y_n\}$ is Cauchy. Since Y is complete, that same sequence must converge to some point $y \in Y$. But now the sequence $\{y_n\}$ converges to both x and y in the metric space X, so uniqueness of limits forces $x = y$. Hence $x \in Y$. Since every point of $\overline{Y}$ lies in Y, we get $\overline{Y} \subseteq Y$, and therefore Y is closed.

Proof.
We will show that

$$\overline{Y} \subseteq Y.$$

Since always $Y \subseteq \overline{Y}$, this will imply $Y = \overline{Y}$, and hence Y is closed in X.

So let $x \in \overline{Y}$. By the sequential characterization of closure proved in lecture, there exists a sequence $\{y_n\}_{n \geq 1}$ in Y such that

$$y_n \to x \qquad \text{in } (X, d).$$

Since every convergent sequence in a metric space is Cauchy, the sequence $\{y_n\}$ is Cauchy in (X, d). Because each y_n lies in Y and the induced metric on Y is just the restriction of d, it follows that $\{y_n\}$ is also a Cauchy sequence in the subspace (Y, d_Y).

Now Y is complete by assumption, so every Cauchy sequence in (Y, d_Y) converges to a point of Y. Therefore there exists some $y \in Y$ such that

$$y_n \to y \qquad \text{in } (Y, d_Y).$$

But convergence in the subspace metric means exactly the same distance condition as convergence in (X, d), so in fact

$$y_n \to y \qquad \text{in } (X, d).$$

Thus the sequence $\{y_n\}$ converges in (X, d) to both x and y. By uniqueness of limits in metric spaces, we must have

$$x = y.$$

Since $y \in Y$, it follows that $x \in Y$.
Because $x \in \overline{Y}$ was arbitrary, we conclude that

$$\overline{Y} \subseteq Y.$$

Therefore Y is closed in X. $\square$

Problem 2

Let (X, d) be a metric space. Prove that if $A, B \subseteq X$ are both compact then $A \cup B$ is compact.

Discussion & Scratch Work

> The main definition from lecture that matters here is the definition of compactness: a subset $K \subseteq X$ is compact if every open cover of K has a finite subcover. So for this problem, I think I should start with an arbitrary open cover of $A \cup B$ and try to show that only finitely many of those open sets are needed.
>
> The big thing is that any open cover of $A \cup B$ automatically gives an open cover of A and also an open cover of B, since both sets are contained in $A \cup B$. Because A is compact, I can extract finitely many sets from the cover that already cover A. Likewise, because B is compact, I can extract finitely many sets from the same cover that cover B. Then I simply combine those two finite collections. Their union is still finite, and together they cover both A and B, hence they cover $A \cup B$.
>
> So the proof template is: begin with an arbitrary open cover of $A \cup B$, apply compactness once to A, apply compactness again to B, and then merge the two finite subcovers into one finite subcover of the union.

Proof.
Let $\{U_i\}_{i \in I}$ be an open cover of $A \cup B$. Thus each U_i is open in X and

$$A \cup B \subseteq \bigcup_{i \in I} U_i.$$

Since $A \subseteq A \cup B$, it follows that

$$A \subseteq \bigcup_{i \in I} U_i.$$

So $\{U_i\}_{i \in I}$ is an open cover of A. Because A is compact, there exists a finite set $J_A \subseteq I$ such that

$$A \subseteq \bigcup_{i \in J_A} U_i.$$

Similarly, since $B \subseteq A \cup B$, we also have

$$B \subseteq \bigcup_{i \in I} U_i.$$

So $\{U_i\}_{i \in I}$ is an open cover of B. Because B is compact, there exists a finite set $J_B \subseteq I$ such that

$$B \subseteq \bigcup_{i \in J_B} U_i.$$

Now set

$$J := J_A \cup J_B.$$

Since J_A and J_B are finite, the set J is finite. Moreover,

$$A \cup B \subseteq \left(\bigcup_{i \in J_A} U_i \right) \cup \left(\bigcup_{i \in J_B} U_i \right) = \bigcup_{i \in J} U_i.$$

Thus $\{U_i\}_{i \in J}$ is a finite subcover of $A \cup B$.

Since the open cover $\{U_i\}_{i \in I}$ was arbitrary, we conclude that $A \cup B$ is compact. $\square$

Problem 3

Let (X, d) be a metric space.

(a) Prove that if $A \subseteq B \subseteq X$, A is closed in X, and B is compact, then A is compact. (Hint: Given an open cover of A, we can add A^c to get an open cover of B.)

(b) Prove that if $C, D \subseteq X$ are both compact, then $C \cap D$ is compact. (Hint: Use (a).)

Discussion & Scratch Work

Both parts of this problem seem like applications of the open-cover definition of compactness, together with one of the closure results from lecture. For part (a), we are given that $A \subseteq B \subseteq X$, that A is closed in X, and that B is compact. Since compactness is about open covers, I should begin with an arbitrary open cover of A and try to turn it into an open cover of B. The hint tells me exactly how: because A is closed, its complement A^c is open, so if I add A^c to my cover of A, then every point of B is covered either because it lies in A or because it lies in $B \setminus A \subseteq A^c$. Then compactness of B gives a finite subcover, and when I restrict back to A, the set A^c becomes unnecessary.

For part (b), I want to prove that the intersection of two compact sets is compact. The cleanest way to use part (a) is to observe that if C and D are compact, then by the theorem from lecture each compact set is closed. In particular, C is closed in X, and of course $C \cap D \subseteq D$. Since D is compact, part (a) applies with $A = C \cap D$ and $B = D$. That immediately gives compactness of $C \cap D$.

Proof.
(a) Let $\{U_i\}_{i \in I}$ be an arbitrary open cover of A, where each U_i is open in X and

$$A \subseteq \bigcup_{i \in I} U_i.$$

Since A is closed in X, the complement A^c is open in X. I claim that

$$\{U_i\}_{i \in I} \cup \{A^c\}$$

is an open cover of B.

Indeed, let $x \in B$. If $x \in A$, then since $\{U_i\}_{i \in I}$ covers A, we have $x \in U_i$ for some $i \in I$. If $x \notin A$, then $x \in A^c$. In either case, x lies in one of the sets from $\{U_i\}_{i \in I} \cup \{A^c\}$. Hence

$$B \subseteq \left(\bigcup_{i \in I} U_i \right) \cup A^c.$$

So this is an open cover of B.

Because B is compact, there exist finitely many indices $i_1, \ldots, i_n \in I$ such that

$$B \subseteq U_{i_1} \cup \cdots \cup U_{i_n} \cup A^c.$$

Now intersect both sides with A. Since $A \subseteq B$, we obtain

$$A \subseteq A \cap (U_{i_1} \cup \cdots \cup U_{i_n} \cup A^c)\,.$$

Distributing the intersection gives

$$A \subseteq (A \cap U_{i_1}) \cup \cdots \cup (A \cap U_{i_n}) \cup (A \cap A^c).$$

But $A \cap A^c = \varnothing$, so

$$A \subseteq (A \cap U_{i_1}) \cup \cdots \cup (A \cap U_{i_n}).$$

Since each $A \cap U_{i_k} \subseteq U_{i_k}$, it follows that

$$A \subseteq U_{i_1} \cup \cdots \cup U_{i_n}.$$

Thus $\{U_{i_1}, \ldots, U_{i_n}\}$ is a finite subcover of A. Since the original open cover of A was arbitrary, we conclude that A is compact.

(b) Suppose that $C, D \subseteq X$ are both compact. By the theorem from lecture, every compact subset of a metric space is closed. Therefore C is closed in X.

Now note that

$$C \cap D \subseteq D \subseteq X.$$

Since C is closed in X, the set $C \cap D$ is also closed in D; equivalently, we may simply apply part (a) with

$$A = C \cap D, \qquad B = D.$$

Indeed, $A \subseteq B$, the set $A = C \cap D$ is closed in X as the intersection of two closed sets, and $B = D$ is compact. Therefore, by part (a), the set $C \cap D$ is compact. □

Problem 4

Let (X, d) be a metric space and let $Y \subseteq X$. Prove that if Y is compact, then Y is complete.

Discussion & Scratch Work

> The main definition here is the definition of completeness: a metric space is complete if every Cauchy sequence converges in that space. Since the problem assumes that $Y \subseteq X$ is compact, the idea is to start with an arbitrary Cauchy sequence in Y and show that it must converge to a point of Y.
>
> The theorem from lecture that does the heavy lifting is the sequential compactness result: if a set is compact, then every sequence in that set has a subsequence that converges in the set. We also know from lecture that every convergent sequence is Cauchy, and in any metric space a Cauchy sequence can have at most one possible limit.
>
> So the proof template is very standard. I take a Cauchy sequence $\{y_n\}$ in Y. Because Y is compact, the sequence has a subsequence $\{y_{n_k}\}$ that converges to some point $y \in Y$. Then I use the fact that the whole sequence is Cauchy to show that actually the entire sequence must converge to that same point y. That will prove that every Cauchy sequence in Y converges in Y, which is exactly what it means for Y to be complete.

Proof.
To prove that Y is complete, let $\{y_n\}_{n\geq 1}$ be a Cauchy sequence in the subspace (Y, d_Y). Since the induced metric d_Y is just the restriction of d to Y, we may regard $\{y_n\}$ simply as a Cauchy sequence of points in Y.

Because Y is compact, every sequence in Y has a subsequence that converges to a point of Y. Therefore there exists a subsequence $\{y_{n_k}\}_{k\geq 1}$ and a point $y \in Y$ such that

$$y_{n_k} \to y.$$

We claim that in fact the whole sequence $\{y_n\}$ converges to y.

Let $\varepsilon > 0$. Since $\{y_n\}$ is Cauchy, there exists $N_1 \in \mathbb{N}$ such that

$$d(y_n, y_m) < \frac{\varepsilon}{2} \qquad \text{for all } n, m \geq N_1.$$

Since $y_{n_k} \to y$, there exists $N_2 \in \mathbb{N}$ such that

$$d(y_{n_k}, y) < \frac{\varepsilon}{2} \qquad \text{for all } k \geq N_2.$$

Choose $k \geq N_2$ large enough so that also $n_k \geq N_1$. Now let $n \geq N_1$. By the triangle inequality,

$$d(y_n, y) \leq d(y_n, y_{n_k}) + d(y_{n_k}, y).$$

Since $n \geq N_1$ and $n_k \geq N_1$, the Cauchy condition gives

$$d(y_n, y_{n_k}) < \frac{\varepsilon}{2},$$

and since $k \geq N_2$, the convergence of the subsequence gives

$$d(y_{n_k}, y) < \frac{\varepsilon}{2}.$$

Therefore

$$d(y_n, y) < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Since this holds for all $n \geq N_1$, we conclude that

$$y_n \to y.$$

Because $y \in Y$, the sequence $\{y_n\}$ converges in Y.

Thus every Cauchy sequence in Y converges to a point of Y, so Y is complete. $\square$

Math 521, Section 2
Homework 11

Problem 1. Consider the metric space $(\mathbb{R}, d)$ with the usual metric $d(x, y) = |x - y|$. Note that the set $\mathbb{Q}$ is not an interval, and so from lecture we know that it must be disconnected.

(a) Find two sets $A, B \subseteq \mathbb{R}$ that are nonempty and separated so that $\mathbb{Q} = A \cup B$, and prove your answer.
(b) Find a set $\emptyset \neq C \subsetneq \mathbb{Q}$ that is both open and closed in $\mathbb{Q}$, and prove your answer.

Problem 2. Let (X, d) be a metric space.

(a) Prove that if $F \subseteq X$ is disconnected and closed in X, then there exist $A, B \subseteq X$ that are nonempty, separated, and closed in X so that $F = A \cup B$.
(b) Prove that if $U \subseteq X$ is disconnected and open in X, then there exist $A, B \subseteq X$ that are nonempty, separated, and open in X so that $U = A \cup B$.

Problem 3. Suppose that (X, d) is a metric space such that for any pair of points $a, b \in X$, there exists a connected set $C \subseteq X$ so that $a, b \in C$. Prove that X is connected.

Problem 4. Let (X, d) be a metric space and $A \subseteq X$. Prove that if A is connected, then $\overline{A}$ is connected.

Problem 5. Let (X, d) be a metric space and $A, B \subseteq X$. Prove that if A and B are both connected and $A \cap B \neq \emptyset$, then $A \cup B$ is connected.

Problem 6. Let (X, d) be a metric space.

(a) Let $A, B, C \subseteq X$. Prove that if the sets A and B are separated and the sets A and C are separated, then the sets A and $B \cup C$ are separated.
(b) Prove that if $F, G \subseteq X$ are closed and $F \cup G$ and $F \cap G$ are connected, then F is connected.

Homework 11

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Contents

Problem 1

Consider the metric space $(\mathbb{R}, d)$ with the usual metric $d(x, y) = |x - y|$. Note that the set $\mathbb{Q}$ is not an interval, and so from lecture we know that it must be disconnected.

(a) Find two sets $A, B \subseteq \mathbb{R}$ that are nonempty and separated so that $\mathbb{Q} = A \cup B$, and prove your answer.

(b) Find a set $\varnothing \neq C \subsetneq \mathbb{Q}$ that is both open and closed in $\mathbb{Q}$, and prove your answer.

Discussion & Scratch Work

The problem already gives an important fact from lecture: in $(\mathbb{R}, |\cdot|)$, every interval is connected, so if a subset of $\mathbb{R}$ is not an interval, then it cannot be connected. Lecture 29 proves that every interval in $\mathbb{R}$ is connected, and the contrapositive is exactly the idea being invoked here: since $\mathbb{Q}$ is not an interval, it must be disconnected. In Lectures 28 and 29, disconnected means that the set can be written as the union of two nonempty separated sets, where "separated" means

$$\overline{A} \cap B = \varnothing \qquad \text{and} \qquad A \cap \overline{B} = \varnothing.$$

This tells me what kind of sets I should try to build.

The easiest way to split $\mathbb{Q}$ is to cut at an irrational number, because that creates a genuine gap inside $\mathbb{Q}$. The most natural choice is $\sqrt{2}$. So I define

$$A := (-\infty, \sqrt{2}) \cap \mathbb{Q}, \qquad B := (\sqrt{2}, \infty) \cap \mathbb{Q}.$$

These are clearly nonempty, disjoint, and their union is all of $\mathbb{Q}$. The real work is to prove they are separated. The key observation is that, as subsets of $\mathbb{R}$, their closures are

$$\overline{A} = (-\infty, \sqrt{2}], \qquad \overline{B} = [\sqrt{2}, \infty),$$

so the only possible boundary point where they might "touch" is $\sqrt{2}$, and that point is not rational. Once that is checked, part (a) is done.

For part (b), the problem is asking for a proper nonempty subset of $\mathbb{Q}$ that is both open and closed in the subspace metric on $\mathbb{Q}$. In plain English, that means a subset of $\mathbb{Q}$ that looks like an open set when viewed from inside $\mathbb{Q}$, but whose complement in $\mathbb{Q}$ also looks open. The same irrational cut works perfectly. If I take

$$C := (-\infty, \sqrt{2}) \cap \mathbb{Q},$$

then it should be open in $\mathbb{Q}$ because it is the intersection of the open set $(-\infty, \sqrt{2})$ in $\mathbb{R}$ with $\mathbb{Q}$, and it should be closed in $\mathbb{Q}$ because it is also the intersection of the closed set $(-\infty, \sqrt{2}]$ in $\mathbb{R}$ with $\mathbb{Q}$. That uses exactly the subspace results from lecture: a set is open or closed in a subspace precisely when it is the intersection of the subspace with a set that is open or closed in the ambient space.

Proof.
(a) Define
$$A := (-\infty, \sqrt{2}) \cap \mathbb{Q}, \qquad B := (\sqrt{2}, \infty) \cap \mathbb{Q}.$$
Then $A, B \subseteq \mathbb{R}$, and both are nonempty because, for example,
$$1 \in A, \qquad 2 \in B.$$
Also,
$$A \cup B = \mathbb{Q},$$
since every rational number is either less than $\sqrt{2}$ or greater than $\sqrt{2}$, and no rational number equals $\sqrt{2}$ because $\sqrt{2} \notin \mathbb{Q}$.

It remains to show that A and B are separated. First note that
$$\overline{A} = (-\infty, \sqrt{2}] \qquad \text{and} \qquad \overline{B} = [\sqrt{2}, \infty)$$
in $\mathbb{R}$. Hence
$$\overline{A} \cap B = (-\infty, \sqrt{2}] \cap \big((\sqrt{2}, \infty) \cap \mathbb{Q}\big) = \varnothing,$$
because no number can be both at most $\sqrt{2}$ and strictly greater than $\sqrt{2}$. Similarly,
$$A \cap \overline{B} = \big((-\infty, \sqrt{2}) \cap \mathbb{Q}\big) \cap [\sqrt{2}, \infty) = \varnothing.$$
Therefore A and B are separated.

Since A and B are nonempty, separated, and satisfy $\mathbb{Q} = A \cup B$, this gives a separation of $\mathbb{Q}$.

(b) Let
$$C := (-\infty, \sqrt{2}) \cap \mathbb{Q}.$$
Then $C \neq \varnothing$ since $1 \in C$, and $C \subsetneq \mathbb{Q}$ since, for example, $2 \in \mathbb{Q} \setminus C$.

We show that C is open in $\mathbb{Q}$. The set $(-\infty, \sqrt{2})$ is open in $\mathbb{R}$, so by the subspace topology result from lecture,
$$C = (-\infty, \sqrt{2}) \cap \mathbb{Q}$$
is open in $\mathbb{Q}$.

We next show that C is closed in $\mathbb{Q}$. The set $(-\infty, \sqrt{2}]$ is closed in $\mathbb{R}$, and since $\sqrt{2} \notin \mathbb{Q}$ we have
$$(-\infty, \sqrt{2}] \cap \mathbb{Q} = (-\infty, \sqrt{2}) \cap \mathbb{Q} = C.$$
Again by the subspace result from lecture, this implies that C is closed in $\mathbb{Q}$.

Thus C is a nonempty proper subset of $\mathbb{Q}$ that is both open and closed in $\mathbb{Q}$. □

Problem 2

Let (X, d) be a metric space.

(a) Prove that if $F \subseteq X$ is disconnected and closed in X, then there exist $A, B \subseteq X$ that are nonempty, separated, and closed in X so that $F = A \cup B$.

(b) Prove that if $U \subseteq X$ is disconnected and open in X, then there exist $A, B \subseteq X$ that are nonempty, separated, and open in X so that $U = A \cup B$.

Discussion & Scratch Work

The starting point for both parts is the definition of disconnectedness from Lectures 28 and 29: a set is disconnected if it can be written as the union of two nonempty separated sets. And "separated" means

$$\overline{A} \cap B = \varnothing \qquad \text{and} \qquad A \cap \overline{B} = \varnothing.$$

So in both parts, I do not need to invent the sets from scratch. The definition of disconnectedness already gives me two nonempty separated pieces whose union is the set I started with. The actual work is to prove that those pieces inherit the extra property I want: closedness in part (a), and openness in part (b).

For part (a), I start with $F \subseteq X$ disconnected and closed in X. By disconnectedness, there exist nonempty separated sets $A, B \subseteq F$ with $F = A \cup B$. Since F is closed, I want to use it together with the separation conditions to show that A and B are closed in X. The key observation is that separation forces

$$F \cap \overline{A} = A \qquad \text{and} \qquad F \cap \overline{B} = B.$$

Once I have that, each set is an intersection of closed sets, so it is closed.

For part (b), I start with $U \subseteq X$ disconnected and open in X. Again, disconnectedness gives nonempty separated sets $A, B \subseteq U$ with $U = A \cup B$. Now I want to prove that A and B are open in X. The right way to use separation is to write each piece as an intersection of U with the complement of the closure of the other piece:

$$A = U \cap (\overline{B})^c, \qquad B = U \cap (\overline{A})^c.$$

Since U is open and closures are closed, these are open sets. So the proof in both parts is: use disconnectedness first, and then use either closedness or openness of the ambient set together with the separation conditions to upgrade the two pieces.

Proof.

(a) Suppose that $F \subseteq X$ is disconnected and closed in X. Since F is disconnected, by definition there exist sets $A, B \subseteq X$ such that

$$A \neq \varnothing, \qquad B \neq \varnothing, \qquad F = A \cup B,$$

and A and B are separated; that is,

$$\overline{A} \cap B = \varnothing \qquad \text{and} \qquad A \cap \overline{B} = \varnothing.$$

It remains to show that A and B are closed in X.

We claim that

$$F \cap \overline{A} = A.$$

Indeed, since $A \subseteq F$ and $A \subseteq \overline{A}$, we have $A \subseteq F \cap \overline{A}$. Conversely, let $x \in F \cap \overline{A}$. Since $F = A \cup B$, either $x \in A$ or $x \in B$. But $x \in B$ is impossible, because then $x \in \overline{A} \cap B$, contradicting that A and B are separated. Hence $x \in A$. This proves the claim.

Now both F and $\overline{A}$ are closed in X, so $F \cap \overline{A}$ is closed in X. Since $F \cap \overline{A} = A$, it follows that A is closed in X.

Similarly,

$$F \cap \overline{B} = B,$$

and therefore B is closed in X.

Thus there exist $A, B \subseteq X$ that are nonempty, separated, and closed in X such that

$$F = A \cup B.$$

(b) Suppose that $U \subseteq X$ is disconnected and open in X. Since U is disconnected, by definition there exist sets $A, B \subseteq X$ such that

$$A \neq \varnothing, \qquad B \neq \varnothing, \qquad U = A \cup B,$$

and A and B are separated; that is,

$$\overline{A} \cap B = \varnothing \qquad \text{and} \qquad A \cap \overline{B} = \varnothing.$$

It remains to show that A and B are open in X.

We claim that

$$A = U \cap (\overline{B})^c.$$

First, let $x \in A$. Then $x \in U$ since $A \subseteq U$, and also $x \notin \overline{B}$ because $A \cap \overline{B} = \varnothing$. Hence

$$x \in U \cap (\overline{B})^c.$$

So $A \subseteq U \cap (\overline{B})^c$.

Conversely, let $x \in U \cap (\overline{B})^c$. Since $x \in U = A \cup B$, either $x \in A$ or $x \in B$. But $x \in B$ is impossible, because every set is contained in its closure, so $x \in B$ would imply $x \in \overline{B}$, contradicting $x \in (\overline{B})^c$. Therefore $x \in A$. This proves the claim.

Since U is open and $(\overline{B})^c$ is open, the set

$$A = U \cap (\overline{B})^c$$

is open in X.

By the same argument,

$$B = U \cap (\overline{A})^c,$$

and therefore B is open in X.

Thus there exist $A, B \subseteq X$ that are nonempty, separated, and open in X such that

$$U = A \cup B.$$

$\square$

Problem 3

Suppose that (X, d) is a metric space such that for any pair of points $a, b \in X$, there exists a connected set $C \subseteq X$ so that $a, b \in C$. Prove that X is connected.

Discussion & Scratch Work

> The statement is saying that any two points of X can be placed inside some connected subset of X. Intuitively, that means the space cannot break into two separated pieces, because if it did, then a connected set containing one point from each side would be forced to lie entirely in one side or entirely in the other, which is impossible. So the result I want to prove is really a global connectedness statement coming from the existence of enough connected subsets joining pairs of points.
>
> From Lecture 29, the main result I want is the proposition that if $C \subseteq X$ is connected and $A, B \subseteq X$ are separated with $C \subseteq A \cup B$, then either $C \subseteq A$ or $C \subseteq B$. That is exactly the theorem that prevents a connected set from straddling a separation. From Lectures 28 and 29, I also need the definition of disconnectedness: X is disconnected if there exist nonempty separated sets $A, B \subseteq X$ such that $X = A \cup B$. And of course, X is connected if it is not disconnected.
>
> So the proof template is contradiction. I assume X is disconnected. Then there exist nonempty separated sets A and B whose union is X. I pick one point $a \in A$ and one point $b \in B$. By hypothesis, there is a connected set $C \subseteq X$ containing both a and b. But since $C \subseteq X = A \cup B$, the lecture proposition says that either $C \subseteq A$ or $C \subseteq B$. That cannot happen because C contains both a and b. This contradiction shows that X must be connected.

Proof.
Suppose, toward a contradiction, that X is disconnected. Then by definition there exist sets $A, B \subseteq X$ such that

$$A \neq \varnothing, \qquad B \neq \varnothing, \qquad X = A \cup B,$$

and A and B are separated; that is,

$$\overline{A} \cap B = \varnothing \qquad \text{and} \qquad A \cap \overline{B} = \varnothing.$$

Choose points

$$a \in A, \qquad b \in B.$$

By the hypothesis of the problem, there exists a connected set $C \subseteq X$ such that

$$a, b \in C.$$

Since $X = A \cup B$, we have

$$C \subseteq A \cup B.$$

Now apply the proposition from lecture: if a connected set is contained in the union of two separated sets, then it must lie entirely in one of them. Therefore either

$$C \subseteq A \qquad \text{or} \qquad C \subseteq B.$$

But this is impossible, because $a \in C \cap A$ and $b \in C \cap B$. If $C \subseteq A$, then $b \in A$, contradicting that $b \in B$ and A, B are separated (hence disjoint). Similarly, if $C \subseteq B$, then $a \in B$, again a contradiction.

Thus our assumption that X is disconnected is false. Therefore X is connected. $\square$

Problem 4

Let (X, d) be a metric space and $A \subseteq X$. Prove that if A is connected, then $\overline{A}$ is connected.

Discussion & Scratch Work

> This problem is really asking me to show that connectedness is preserved when I pass from a set to its closure. The relevant definition from Lectures 28 and 29 is that a set is disconnected if it can be written as the union of two nonempty separated sets; otherwise it is connected. So to prove that $\overline{A}$ is connected, the natural strategy is contradiction: assume that $\overline{A}$ is disconnected, and then try to use that separation to force a separation of A itself.
>
> The key lecture result I want is the proposition from Lecture 29: if C is connected and E, F are separated with $C \subseteq E \cup F$, then either $C \subseteq E$ or $C \subseteq F$. Since A is connected and $A \subseteq \overline{A}$, this proposition says that if $\overline{A}$ were split into two separated pieces, then all of A would have to lie in just one of them. The remaining idea is then to use the fact that points of the closure of A are limits of sequences in A from Lecture 25. If all of A lies in one side of the separation, then every limit point of A must also lie on that same side, which rules out the other side entirely. That contradiction shows $\overline{A}$ cannot be disconnected.

Proof.
Suppose, toward a contradiction, that $\overline{A}$ is disconnected. Then by definition there exist nonempty separated sets $E, F \subseteq X$ such that

$$\overline{A} = E \cup F.$$

Since $A \subseteq \overline{A} = E \cup F$ and A is connected, the proposition from lecture implies that either

$$A \subseteq E \qquad \text{or} \qquad A \subseteq F.$$

Without loss of generality, assume that

$$A \subseteq E.$$

Now let $x \in F$. Since $F \subseteq \overline{A}$, we have $x \in \overline{A}$. By the sequential characterization of closure from lecture, there exists a sequence $\{a_n\} \subseteq A$ such that

$$a_n \to x.$$

But every $a_n \in A \subseteq E$, so $\{a_n\}$ is a sequence in E converging to x. Hence

$$x \in \overline{E}.$$

Since also $x \in F$, this shows that

$$x \in \overline{E} \cap F,$$

which contradicts the fact that E and F are separated.

Therefore $F = \varnothing$, contradicting that E and F were both nonempty. This contradiction shows that $\overline{A}$ is not disconnected. Hence $\overline{A}$ is connected. $\square$

Problem 5

Let (X, d) be a metric space and $A, B \subseteq X$. Prove that if A and B are both connected and $A \cap B \neq \varnothing$, then $A \cup B$ is connected.

Discussion & Scratch Work

> This is one of the standard closure properties of connected sets. The main definition from Lectures 28 and 29 is that a set is disconnected if it can be written as the union of two nonempty separated sets; otherwise it is connected. So the natural proof strategy is contradiction: assume that $A \cup B$ is disconnected, and then analyze what that separation would force on the connected sets A and B individually.
>
> The key lecture result I need is the proposition from Lecture 29 that says: if C is connected and E, F are separated with $C \subseteq E \cup F$, then either $C \subseteq E$ or $C \subseteq F$. That theorem is exactly designed for this problem. If $A \cup B$ were disconnected, then I could write
>
> $$A \cup B = E \cup F$$
>
> with E and F nonempty and separated. Since A is connected and lies in $E \cup F$, the proposition forces all of A to lie in one side. Likewise, all of B must lie in one side. Because $A \cap B \neq \varnothing$, the sets A and B cannot land in different sides. So they must both lie in the same side, which would force the other side to be empty. That contradiction shows that $A \cup B$ is connected.

Proof.
Suppose, toward a contradiction, that $A \cup B$ is disconnected. Then by definition there exist nonempty separated sets $E, F \subseteq X$ such that

$$A \cup B = E \cup F.$$

Since $A \subseteq A \cup B = E \cup F$ and A is connected, the proposition from lecture implies that either

$$A \subseteq E \qquad \text{or} \qquad A \subseteq F.$$

Similarly, since $B \subseteq E \cup F$ and B is connected, either

$$B \subseteq E \qquad \text{or} \qquad B \subseteq F.$$

We claim that A and B must lie in the same side. Indeed, suppose not. Then, after relabeling if necessary, we would have

$$A \subseteq E \qquad \text{and} \qquad B \subseteq F.$$

But $A \cap B \neq \varnothing$, so choose some point $x \in A \cap B$. Then $x \in E$ and $x \in F$, which implies

$$x \in E \cap F.$$

This is impossible because separated sets are disjoint. Hence A and B must both be contained in E, or both be contained in F.

If $A \subseteq E$ and $B \subseteq E$, then

$$A \cup B \subseteq E,$$

so from $A \cup B = E \cup F$ we get $F = \varnothing$, contradicting that F is nonempty. The case $A \subseteq F$ and $B \subseteq F$ similarly implies $E = \varnothing$, also a contradiction.

Thus our assumption that $A \cup B$ is disconnected is false. Therefore $A \cup B$ is connected. $\square$

Problem 6

Let (X, d) be a metric space.

(a) Let $A, B, C \subseteq X$. Prove that if the sets A and B are separated and the sets A and C are separated, then the sets A and $B \cup C$ are separated.

(b) Prove that if $F, G \subseteq X$ are closed and $F \cup G$ and $F \cap G$ are connected, then F is connected.

Discussion & Scratch Work

For part (a), the whole problem is about carefully unpacking the definition of separated sets. From lecture, E and F are separated if

$$\overline{E} \cap F = \varnothing \qquad \text{and} \qquad E \cap \overline{F} = \varnothing.$$

So if I want to prove that A and $B \cup C$ are separated, I need to check exactly those two conditions with $B \cup C$ in place of one of the sets. The first condition is easy, because

$$\overline{A} \cap (B \cup C) = (\overline{A} \cap B) \cup (\overline{A} \cap C),$$

and both pieces are empty by hypothesis. For the second condition, the key topological fact is

$$\overline{B \cup C} \subseteq \overline{B} \cup \overline{C}.$$

Then

$$A \cap \overline{B \cup C} \subseteq (A \cap \overline{B}) \cup (A \cap \overline{C}),$$

and again both pieces are empty. So part (a) is really just a direct closure computation.

For part (b), the assumptions are that F and G are closed, that $F \cup G$ is connected, and that $F \cap G$ is connected. I want to prove that F itself is connected. The natural strategy is contradiction: assume F is disconnected. Then by Problem 2(a), because F is also closed, I can write

$$F = A \cup B$$

where A and B are nonempty, separated, and closed in X. Since $F \cap G$ is connected and sits inside $A \cup B$, the connected-set proposition from Lecture 29 says that $F \cap G$ must lie entirely in one side. Without loss of generality, that means

$$F \cap G \subseteq A.$$

Then $G \cap B = \varnothing$. Since both G and B are closed, that actually makes G and B separated. We already know A and B are separated, so by part (a), B is separated from $A \cup G$. But then

$$F \cup G = (A \cup G) \cup B$$

is a separation of $F \cup G$, contradicting connectedness. So F must be connected.

Proof.
(a) Suppose that A and B are separated and that A and C are separated. We show that A and $B \cup C$ are separated.

First,
$$\overline{A} \cap (B \cup C) = (\overline{A} \cap B) \cup (\overline{A} \cap C).$$
Since A and B are separated, we have $\overline{A} \cap B = \varnothing$, and since A and C are separated, we have $\overline{A} \cap C = \varnothing$. Therefore
$$\overline{A} \cap (B \cup C) = \varnothing.$$

Next, because $B \subseteq B \cup C$ and $C \subseteq B \cup C$, we have
$$\overline{B \cup C} \subseteq \overline{B} \cup \overline{C}.$$

Hence
$$A \cap \overline{B \cup C} \subseteq A \cap (\overline{B} \cup \overline{C}) = (A \cap \overline{B}) \cup (A \cap \overline{C}).$$
Again, since A is separated from both B and C, we have
$$A \cap \overline{B} = \varnothing \qquad \text{and} \qquad A \cap \overline{C} = \varnothing.$$

Therefore
$$A \cap \overline{B \cup C} = \varnothing.$$

Thus
$$\overline{A} \cap (B \cup C) = \varnothing \qquad \text{and} \qquad A \cap \overline{B \cup C} = \varnothing,$$
so A and $B \cup C$ are separated.

(b) Suppose, toward a contradiction, that F is disconnected. Since F is also closed in X, Problem 2(a) implies that there exist sets $A, B \subseteq X$ such that
$$A \neq \varnothing, \qquad B \neq \varnothing, \qquad F = A \cup B,$$
and A and B are separated and closed in X.

Now consider the set $F \cap G$. Since $F \cap G \subseteq F = A \cup B$ and $F \cap G$ is connected by assumption, the proposition from Lecture 29 implies that either
$$F \cap G \subseteq A \qquad \text{or} \qquad F \cap G \subseteq B.$$

Without loss of generality, assume that
$$F \cap G \subseteq A.$$

We claim that G and B are separated. Indeed, since $B \subseteq F$, we have
$$G \cap B \subseteq G \cap F = F \cap G \subseteq A.$$

But A and B are separated, hence disjoint, so $A \cap B = \varnothing$. Therefore
$$G \cap B = \varnothing.$$

Because G and B are both closed in X, we have $\overline{G} = G$ and $\overline{B} = B$, and so

$$\overline{G} \cap B = G \cap B = \varnothing, \qquad G \cap \overline{B} = G \cap B = \varnothing.$$

Thus G and B are separated.

We already know that A and B are separated. Applying part (a) with the roles of the sets relabeled, we conclude that B and $A \cup G$ are separated.

Now

$$F \cup G = (A \cup B) \cup G = (A \cup G) \cup B.$$

Also, $B \neq \varnothing$ by construction, and $A \cup G \neq \varnothing$ because $A \neq \varnothing$. Since B and $A \cup G$ are separated, this shows that $F \cup G$ is disconnected, contradicting the assumption that $F \cup G$ is connected.

Therefore F must be connected. $\square$

Appendix: Toolbox

Math 521, Section 2
Homework 12

Problem 1. Let (X, d_X) and (Y, d_Y) be metric spaces, $f : X \to Y$, and $x_0 \in X$. Prove that f is continuous at x_0 if and only if for any open set $V \subseteq Y$ containing $f(x_0)$, there is an open set $U \subseteq X$ containing x_0 so that $x \in U$ implies $f(x) \in V$.

Problem 2. Let (X, d_X) and (Y, d_Y) be metric spaces and $f : X \to Y$ a function. Prove that f is continuous if and only if $f(\overline{A}) \subseteq \overline{f(A)}$ for any $A \subseteq X$.

Problem 3. Let $f : \mathbb{R} \to \mathbb{R}$ be the function $f(x) = \frac{1}{1+x^2}$.

(a) Prove that f is continuous.
(b) Find an open set $U \subseteq \mathbb{R}$ so that $f(U)$ is not open.
(c) Find a closed set $F \subseteq \mathbb{R}$ so that $f(F)$ is not closed.
(d) Find a set $A \subseteq \mathbb{R}$ so that $f(\overline{A}) \neq \overline{f(A)}$.

Problem 4. Consider the metric spaces $(\mathbb{R}, |\cdot|)$ and $(\mathbb{R}^2, d_2)$.

(a) Prove that the functions $f, g : \mathbb{R}^2 \to \mathbb{R}$ given by $f(x, y) = x$ and $g(x, y) = y$ are continuous. (Hint: Use sequences.)
(b) Prove that if $h : [0, 1] \to \mathbb{R}^2$ is a continuous function such that $h(0) = (x_0, y_0)$, $h(1) = (x_1, y_1)$, and $x_0 < 0 < x_1$, then there exists a point $t \in (0, 1)$ so that $h(t)$ lies on the y-axis.

Problem 5. Going forward, you may assume (no proof necessary) any fact about $\sin x$ and $\cos x$ from calculus, such as $\sin, \cos : \mathbb{R} \to [-1, 1]$ are continuous functions.

Prove that the function $f : \mathbb{R}^2 \to \mathbb{R}$ given by

$$f(x, y) = x^2 + y^2 + \sin(x - y)$$

has a global minimum. (Hint: Notice that we cannot directly apply the extreme value theorem here. However, we can apply it to any compact subset of $\mathbb{R}^2$.)

Problem 6. Let X and Y be metric spaces and $f : X \to Y$ a function. Prove that if X is compact and f is continuous and bijective, then the inverse function $f^{-1} : Y \to X$ is also continuous. (Hint: It suffices to show that the preimage of any closed subset of X under f^{-1} is closed in Y.)

Homework 12

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Contents

Problem 1

Let (X, d_X) and (Y, d_Y) be metric spaces, $f : X \to Y$, and $x_0 \in X$. Prove that f is continuous at x_0 if and only if for any open set $V \subseteq Y$ containing $f(x_0)$, there is an open set $U \subseteq X$ containing x_0 so that $x \in U$ implies $f(x) \in V$.

Discussion & Scratch Work

This problem is asking me to prove that two definitions of continuity at a point are equivalent. From lecture, I know the standard ε–δ definition: f is continuous at x if for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$d_X(x', x) < \delta \implies d_Y(f(x'), f(x)) < \varepsilon.$$

I also know the open-set formulation: f is continuous at x if for every open set $V \subseteq Y$ with $f(x) \in V$, there exists an open set $U \subseteq X$ such that

$$x \in U \qquad \text{and} \qquad f(U) \subseteq V.$$

So the goal is to show these two definitions are equivalent. Since this is an "if and only if," I prove both directions.

For the forward direction, I assume the ε–δ definition. If V is open and contains $f(x)$, then there exists an open ball $B_\varepsilon(f(x)) \subseteq V$. Then continuity gives a $\delta > 0$ such that $f(B_\delta(x)) \subseteq B_\varepsilon(f(x)) \subseteq V$. So I take $U = B_\delta(x)$.

For the reverse direction, I assume the open-set condition. Given $\varepsilon > 0$, I consider the open ball $B_\varepsilon(f(x))$. By hypothesis, there exists an open set U containing x such that $f(U) \subseteq B_\varepsilon(f(x))$. Since U is open, there exists $\delta > 0$ such that $B_\delta(x) \subseteq U$. This gives the ε–δ condition.

Proof.
($\Rightarrow$) Assume that f is continuous at $x \in X$. Let $V \subseteq Y$ be open with $f(x) \in V$. Then there exists $\varepsilon > 0$ such that

$$B_\varepsilon(f(x)) \subseteq V.$$

By continuity at x, there exists $\delta > 0$ such that

$$d_X(x', x) < \delta \implies d_Y(f(x'), f(x)) < \varepsilon.$$

Let $U := B_\delta(x)$. Then U is open, $x \in U$, and if $x' \in U$, then $f(x') \in B_\varepsilon(f(x)) \subseteq V$. Hence

$$f(U) \subseteq V.$$

($\Leftarrow$) Assume that for every open set $V \subseteq Y$ with $f(x) \in V$, there exists an open set $U \subseteq X$ such that $x \in U$ and $f(U) \subseteq V$.

Let $\varepsilon > 0$ and set

$$V := B_\varepsilon(f(x)).$$

Then V is open and contains $f(x)$, so there exists an open set $U \subseteq X$ such that

$$x \in U \qquad \text{and} \qquad f(U) \subseteq B_\varepsilon(f(x)).$$

Since U is open and $x \in U$, there exists $\delta > 0$ such that

$$B_\delta(x) \subseteq U.$$

Now if $d_X(x', x) < \delta$, then $x' \in U$, so

$$f(x') \in B_\varepsilon(f(x)),$$

which implies

$$d_Y(f(x'), f(x)) < \varepsilon.$$

Therefore f is continuous at x. $\square$

Problem 2

Let (X, d_X) and (Y, d_Y) be metric spaces and $f : X \to Y$ a function. Prove that f is continuous if and only if $f(\overline{A}) \subseteq \overline{f(A)}$ for any $A \subseteq X$.

Discussion & Scratch Work

> This problem is asking me to prove an equivalent characterization of continuity using closures. I am given a function $f : X \to Y$ between metric spaces, and I need to show that
>
> $$f \text{ is continuous} \iff f(\overline{A}) \subseteq \overline{f(A)} \quad \text{for all } A \subseteq X.$$
>
> So I need to prove both directions.
>
> First, I unpack what this statement means. The set $\overline{A}$ consists of all points x that are either in A or are limits of sequences from A. Similarly, $\overline{f(A)}$ consists of all points in Y that are limits of sequences from $f(A)$.
>
> So the statement $f(\overline{A}) \subseteq \overline{f(A)}$ is saying: if x is either in A or is a limit of points in A, then $f(x)$ should be either in $f(A)$ or a limit of points in $f(A)$.
>
> This strongly suggests using the sequential characterization of closure and the fact from lecture that:
>
> $$f \text{ continuous} \iff x_n \to x \Rightarrow f(x_n) \to f(x).$$
>
> For the forward direction, I assume f is continuous. Then if $x \in \overline{A}$, I can find a sequence $a_n \in A$ with $a_n \to x$. Continuity will give $f(a_n) \to f(x)$, which means $f(x) \in \overline{f(A)}$.
>
> For the reverse direction, I should not use subsequences. Instead, I will use the open-set characterization from Problem 1. So I fix an open set $V \subseteq Y$ and a point $x \in X$ such that $f(x) \in V$, and I define
>
> $$A := X \setminus f^{-1}(V).$$
>
> If x were in $\overline{A}$, then the hypothesis would imply
>
> $$f(x) \in f(\overline{A}) \subseteq \overline{f(A)}.$$
>
> But every point of $f(A)$ lies in $Y \setminus V$, so $\overline{f(A)} \subseteq Y \setminus V$ because $Y \setminus V$ is closed. That would force $f(x) \notin V$, a contradiction. Hence $x \notin \overline{A}$.
>
> Since $x \notin \overline{A}$, there exists an open set U containing x with $U \cap A = \varnothing$. Therefore $U \subseteq X \setminus A = f^{-1}(V)$, so $f(U) \subseteq V$. This is exactly the open-set characterization of continuity.

Proof.

($\Rightarrow$) Assume that f is continuous. Let $A \subseteq X$, and let $x \in \overline{A}$. By the sequential characterization of closure, there exists a sequence $\{a_n\} \subseteq A$ such that

$$a_n \to x.$$

Since f is continuous, we have

$$f(a_n) \to f(x).$$

Each $f(a_n) \in f(A)$, so $f(x)$ is a limit of points in $f(A)$. Hence

$$f(x) \in \overline{f(A)}.$$

Therefore

$$f(\overline{A}) \subseteq \overline{f(A)}.$$

($\Leftarrow$) Assume that for all $A \subseteq X$, we have

$$f(\overline{A}) \subseteq \overline{f(A)}.$$

We show that f is continuous.

Let $V \subseteq Y$ be open, and let $x \in X$ satisfy

$$f(x) \in V.$$

Set

$$A := X \setminus f^{-1}(V).$$

We claim that $x \notin \overline{A}$. Suppose for contradiction that

$$x \in \overline{A}.$$

Then, since $x \in \overline{A}$, we have $f(x) \in f(\overline{A})$. By the hypothesis,

$$f(\overline{A}) \subseteq \overline{f(A)},$$

so

$$f(x) \in \overline{f(A)}.$$

Now if $y \in f(A)$, then $y = f(a)$ for some $a \in A$. Since $a \notin f^{-1}(V)$, we must have

$$f(a) \notin V.$$

Thus

$$f(A) \subseteq Y \setminus V.$$

Because V is open, the set $Y \setminus V$ is closed. Therefore

$$\overline{f(A)} \subseteq Y \setminus V.$$

It follows that

$$f(x) \in Y \setminus V,$$

which contradicts $f(x) \in V$.

Hence $x \notin \overline{A}$. Since $\overline{A}$ is closed, its complement is open, so there exists an open set $U \subseteq X$ such that

$$x \in U \qquad \text{and} \qquad U \cap A = \varnothing.$$

Therefore

$$U \subseteq X \setminus A = f^{-1}(V).$$

So if $u \in U$, then $u \in f^{-1}(V)$, which means $f(u) \in V$. Hence

$$f(U) \subseteq V.$$

We have shown that for every open set $V \subseteq Y$ containing $f(x)$, there exists an open set $U \subseteq X$ containing x such that $f(U) \subseteq V$. By Problem 1, this means that f is continuous at x. Since $x \in X$ was arbitrary, f is continuous on X. $\square$

Problem 3

Let $f : \mathbb{R} \to \mathbb{R}$ be the function $f(x) = \frac{1}{1+x^2}$.

(a) Prove that f is continuous.

(b) Find an open set $U \subseteq \mathbb{R}$ so that $f(U)$ is not open.

(c) Find a closed set $F \subseteq \mathbb{R}$ so that $f(F)$ is not closed.

(d) Find a set $A \subseteq \mathbb{R}$ so that $f(\overline{A}) \neq \overline{f(A)}$.

Discussion & Scratch Work

This problem is asking me to analyze a specific function

$$f(x) = \frac{1}{1+x^2}$$

from several perspectives: continuity, behavior of images of open/closed sets, and how closure interacts with the function.

For part (a), I need to prove f is continuous. From Lecture 30, I know I can use the sequential definition: f is continuous if $x_n \to x$ implies $f(x_n) \to f(x)$. This is the easiest approach here since rational functions behave well under limits.

For part (b), I need to find an open set U such that $f(U)$ is not open. So I need to recall that a set is not open if it fails to contain an open ball around some point. Intuitively, f maps $\mathbb{R}$ into $(0, 1]$, and the maximum value 1 is only achieved at $x = 0$. That suggests that images of open sets containing 0 will include 1 but not values larger than 1, which creates a boundary point—so the image won't be open.

For part (c), I need a closed set F such that $f(F)$ is not closed. From Lecture 31, continuous functions do not necessarily preserve closed sets unless the domain is compact. So I should look for a closed set whose image "misses" a limit point.

For part (d), I need to find a set A where

$$f(\overline{A}) \neq \overline{f(A)}.$$

From Problem 2, I know that continuity gives only one inclusion:

$$f(\overline{A}) \subseteq \overline{f(A)}.$$

So I need an example where this inclusion is strict. This typically happens when limit points are "collapsed" under the function, meaning some points in $\overline{f(A)}$ are not hit by $f(\overline{A})$.

Proof.

(a) Let $\{x_n\}$ be a sequence in $\mathbb{R}$ such that $x_n \to x$. Then We use the standard limit laws from earlier lectures: polynomials and reciprocal functions are continuous where defined. Then

$$x_n^2 \to x^2,$$

so

$$1 + x_n^2 \to 1 + x^2.$$

Since the function $t \mapsto \frac{1}{t}$ is continuous for $t \neq 0$, we have

$$\frac{1}{1 + x_n^2} \to \frac{1}{1 + x^2}.$$

Thus $f(x_n) \to f(x)$, so f is continuous.

(b) Let $U = \mathbb{R}$, which is open. Then

$$f(U) = (0, 1] .$$

This set is not open in $\mathbb{R}$ because $1 \in f(U)$, but there is no open interval around 1 contained in $(0, 1]$. Hence $f(U)$ is not open.

(c) Let $F = [0, \infty)$, which is closed in $\mathbb{R}$. Then

$$f(F) = (0, 1] .$$

As above, $(0, 1]$ is not closed because it does not contain 0, which is a limit point. Hence $f(F)$ is not closed.

(d) Let

$$A = (0, \infty).$$

Then

$$\overline{A} = [0, \infty).$$

So

$$f(\overline{A}) = f([0, \infty)) = (0, 1].$$

On the other hand,

$$f(A) = f((0, \infty)) = (0, 1),$$

so

$$\overline{f(A)} = [0, 1].$$

Thus

$$f(\overline{A}) = (0, 1] \neq [0, 1] = \overline{f(A)}.$$

□

Problem 4

Consider the metric spaces $(\mathbb{R}, |\cdot|)$ and $(\mathbb{R}^2, d_2)$.

(a) Prove that the functions $f, g : \mathbb{R}^2 \to \mathbb{R}$ given by $f(x, y) = x$ and $g(x, y) = y$ are continuous. (Hint: Use sequences.)

(b) Prove that if $h : [0, 1] \to \mathbb{R}^2$ is a continuous function such that $h(0) = (x_0, y_0)$, $h(1) = (x_1, y_1)$, and $x_0 < 0 < x_1$, then there exists a point $t \in (0, 1)$ so that $h(t)$ lies on the y-axis.

Discussion & Scratch Work

For part (a), the problem explicitly tells me to use sequences, so I should use the sequential characterization of continuity from lecture: a function is continuous if whenever a sequence converges in the domain, its image converges in the codomain. Here the functions are just the coordinate projections

$$f(x, y) = x \qquad \text{and} \qquad g(x, y) = y.$$

So if $(x_n, y_n) \to (x, y)$ in $(\mathbb{R}^2, d_2)$, then by the definition of the Euclidean metric,

$$d_2((x_n, y_n), (x, y)) = \sqrt{(x_n - x)^2 + (y_n - y)^2} \to 0.$$

That immediately forces both $x_n \to x$ and $y_n \to y$, so the coordinate functions should be continuous.

For part (b), the key idea is to compose the given path $h : [0, 1] \to \mathbb{R}^2$ with the projection onto the first coordinate. If I define

$$\varphi(t) := f(h(t)),$$

then $\varphi : [0, 1] \to \mathbb{R}$ is continuous because h is continuous and part (a) says f is continuous. Also,

$$\varphi(0) = x_0 < 0 \qquad \text{and} \qquad \varphi(1) = x_1 > 0.$$

So now I can use the Intermediate Value Theorem from calculus on φ. That gives some $t \in (0, 1)$ with $\varphi(t) = 0$, which means the first coordinate of $h(t)$ is 0. In other words, $h(t)$ lies on the y-axis.

Proof.

(a) We first prove that $f : \mathbb{R}^2 \to \mathbb{R}$ given by $f(x, y) = x$ is continuous. Let $\{(x_n, y_n)\}$ be a sequence in $\mathbb{R}^2$ such that

$$(x_n, y_n) \to (x, y)$$

in $(\mathbb{R}^2, d_2)$. Then

$$d_2((x_n, y_n), (x, y)) = \sqrt{(x_n - x)^2 + (y_n - y)^2} \to 0.$$

Since
$$|x_n - x| \le \sqrt{(x_n - x)^2 + (y_n - y)^2} = d_2((x_n, y_n), (x, y)),$$
it follows by the squeeze theorem that
$$x_n \to x.$$
Hence
$$f(x_n, y_n) = x_n \to x = f(x, y).$$
Therefore f is continuous.

The proof for $g : \mathbb{R}^2 \to \mathbb{R}$ given by $g(x, y) = y$ is exactly the same. If $(x_n, y_n) \to (x, y)$, then
$$|y_n - y| \le \sqrt{(x_n - x)^2 + (y_n - y)^2} = d_2((x_n, y_n), (x, y)) \to 0,$$
so
$$y_n \to y.$$
Thus
$$g(x_n, y_n) = y_n \to y = g(x, y),$$
and therefore g is continuous.

(b) Let $h : [0, 1] \to \mathbb{R}^2$ be continuous, with
$$h(0) = (x_0, y_0), \qquad h(1) = (x_1, y_1), \qquad x_0 < 0 < x_1.$$
Define
$$\varphi : [0, 1] \to \mathbb{R}, \qquad \varphi(t) := f(h(t)),$$
where $f : \mathbb{R}^2 \to \mathbb{R}$ is the first-coordinate projection from part (a), namely $f(x, y) = x$.

Since h is continuous and f is continuous, the composition $\varphi = f \circ h$ is continuous on $[0, 1]$. Moreover,
$$\varphi(0) = f(h(0)) = f(x_0, y_0) = x_0 < 0,$$
and
$$\varphi(1) = f(h(1)) = f(x_1, y_1) = x_1 > 0.$$
By the Intermediate Value Theorem from calculus, which we may assume, there exists some $t \in (0, 1)$ such that
$$\varphi(t) = 0.$$
But $\varphi(t) = f(h(t))$ is exactly the first coordinate of $h(t)$. So the first coordinate of $h(t)$ is 0, which means
$$h(t) \in \{(0, y) : y \in \mathbb{R}\},$$
that is, $h(t)$ lies on the y-axis. $\square$

Problem 5

Going forward, you may assume (no proof necessary) any fact about $\sin x$ and $\cos x$ from calculus, such as $\sin, \cos : \mathbb{R} \to [-1, 1]$ are continuous functions. Prove that the function $f : \mathbb{R}^2 \to \mathbb{R}$ given by

$$f(x, y) = x^2 + y^2 + \sin(x - y)$$

has a global minimum. (Hint: Notice that we cannot directly apply the extreme value theorem here. However, we can apply it to any compact subset of $\mathbb{R}^2$.)

Discussion & Scratch Work

The goal is to prove that

$$f(x, y) = x^2 + y^2 + \sin(x - y)$$

has a global minimum on $\mathbb{R}^2$. The hint is important: I cannot apply the Extreme Value Theorem directly on all of $\mathbb{R}^2$, because $\mathbb{R}^2$ is not compact. So the idea is to first show that I only need to look on some compact subset of $\mathbb{R}^2$, and then apply the Extreme Value Theorem there.

First, I note that f is continuous. The functions $(x, y) \mapsto x^2 + y^2$ and $(x, y) \mapsto x - y$ are continuous, and by the problem statement I may assume that sin is continuous. Therefore $(x, y) \mapsto \sin(x - y)$ is continuous, and so f is continuous as a sum of continuous functions.

Next, I want a lower bound that tells me what happens far away from the origin. Since

$$-1 \leq \sin(x - y) \leq 1,$$

I get

$$f(x, y) \geq x^2 + y^2 - 1.$$

So if (x, y) is far from the origin, then $x^2 + y^2$ is large, and therefore $f(x, y)$ is large as well. This is the key coercive-type estimate.

To use this effectively, I want to compare that lower bound with the value of f at some specific point. A convenient choice is

$$\left(-\frac{1}{2}, \frac{1}{2}\right),$$

because then

$$x - y = -1,$$

so

$$f\left(-\frac{1}{2}, \frac{1}{2}\right) = \frac{1}{4} + \frac{1}{4} + \sin(-1) = \frac{1}{2} - \sin(1) < 0.$$

On the other hand, if $x^2 + y^2 \geq 1$, then

$$f(x, y) \geq x^2 + y^2 - 1 \geq 0.$$

So every point outside the closed unit disk has $f(x,y) \geq 0$, while the point $\left(-\frac{1}{2}, \frac{1}{2}\right)$ inside the closed unit disk gives a negative value. That means a global minimizer cannot occur outside the closed unit disk.

So I restrict to
$$K = \{(x,y) \in \mathbb{R}^2 : x^2 + y^2 \leq 1\}.$$
This set is closed and bounded in $\mathbb{R}^2$, hence compact. Since f is continuous, the Extreme Value Theorem implies that f attains a minimum on K. Because every point outside K has value at least 0, while some point in K has value < 0, that minimum on K is automatically a global minimum on all of $\mathbb{R}^2$.

Proof.
Define
$$f : \mathbb{R}^2 \to \mathbb{R}, \qquad f(x,y) = x^2 + y^2 + \sin(x-y).$$
First, f is continuous on $\mathbb{R}^2$. Indeed, $(x,y) \mapsto x^2 + y^2$ is continuous, $(x,y) \mapsto x - y$ is continuous, and $\sin : \mathbb{R} \to \mathbb{R}$ is continuous, so $(x,y) \mapsto \sin(x-y)$ is continuous. Therefore f is continuous as a sum of continuous functions.

Next, since
$$\sin(x-y) \geq -1$$
for all $(x,y) \in \mathbb{R}^2$, we have
$$f(x,y) = x^2 + y^2 + \sin(x-y) \geq x^2 + y^2 - 1.$$
In particular, whenever $x^2 + y^2 \geq 1$, we get
$$f(x,y) \geq 0.$$
Now consider the point
$$p = \left(-\frac{1}{2}, \frac{1}{2}\right).$$
Then
$$f(p) = f\left(-\frac{1}{2}, \frac{1}{2}\right) = \frac{1}{4} + \frac{1}{4} + \sin(-1) = \frac{1}{2} - \sin(1).$$
Since $\sin(1) > \frac{1}{2}$, it follows that
$$f(p) < 0.$$
Also, p lies in the closed unit disk
$$K := \{(x,y) \in \mathbb{R}^2 : x^2 + y^2 \leq 1\}.$$
The set K is closed and bounded in $\mathbb{R}^2$, hence compact. Since f is continuous, the Extreme Value Theorem implies that there exists some $q \in K$ such that
$$f(q) \leq f(z) \qquad \text{for all } z \in K.$$

We claim that q is a global minimizer of f on $\mathbb{R}^2$. Let $z = (x, y) \in \mathbb{R}^2$.
If $z \in K$, then by definition of q,

$$f(q) \le f(z).$$

If $z \notin K$, then $x^2 + y^2 > 1$, so from the estimate above,

$$f(z) \ge 0.$$

But

$$f(q) \le f(p) < 0,$$

so again

$$f(q) < 0 \le f(z).$$

Thus in every case,

$$f(q) \le f(z) \qquad \text{for all } z \in \mathbb{R}^2.$$

Therefore f has a global minimum on $\mathbb{R}^2$. $\square$

Problem 6

Let X and Y be metric spaces and $f : X \to Y$ a function. Prove that if X is compact and f is continuous and bijective, then the inverse function $f^{-1} : Y \to X$ is also continuous. (Hint: It suffices to show that the preimage of any closed subset of X under f^{-1} is closed in Y.)

Discussion & Scratch Work

The goal is to prove that $f^{-1} : Y \to X$ is continuous. The hint is telling me exactly which continuity criterion to use: it is enough to show that for every closed set $C \subseteq X$, the preimage of C under f^{-1} is closed in Y.

So I should start with an arbitrary closed set $C \subseteq X$ and understand what

$$(f^{-1})^{-1}(C)$$

actually is. Since f is bijective, this set is just

$$f(C).$$

So the real task is to prove that $f(C)$ is closed in Y whenever C is closed in X.

Now the compactness assumption is what makes this work. Since X is compact and C is closed in X, I know from lecture that C is compact. Since f is continuous, the image of a compact set under f is compact. And since Y is a metric space, compact subsets of Y are closed. So the chain is:

$$C \text{ closed in } X \implies C \text{ compact} \implies f(C) \text{ compact} \implies f(C) \text{ closed in } Y.$$

That is exactly what I need to show that f^{-1} is continuous.

Proof.
We prove that $f^{-1} : Y \to X$ is continuous by using the closed-set criterion.

Let $C \subseteq X$ be closed. Since X is compact and C is closed in X, it follows that C is compact.

Because $f : X \to Y$ is continuous, the image of a compact set under f is compact. Hence

$$f(C)$$

is compact in Y.

Now Y is a metric space, and in a metric space every compact set is closed. Therefore $f(C)$ is closed in Y.

Since f is bijective, we have

$$(f^{-1})^{-1}(C) = f(C).$$

Thus $(f^{-1})^{-1}(C)$ is closed in Y for every closed set $C \subseteq X$.

Therefore, by the closed-set criterion for continuity, f^{-1} is continuous. $\square$

Appendix: Toolbox

Math 521, Section 2
Homework 13

Going forward, you may assume (no proof necessary) the familiar differentiation properties of e^x, $\log x$, $\sin x$, and $\cos x$ from calculus.

Problem 1. Consider the sequence of functions $f_n : [0,1] \to \mathbb{R}$ given by

$$f_n(x) = \begin{cases} nx & \text{if } x \in [0, \frac{1}{n}], \\ 2 - nx & \text{if } x \in (\frac{1}{n}, \frac{2}{n}], \\ 0 & \text{if } x \in (\frac{2}{n}, 1]. \end{cases}$$

(a) Prove that f_n converges pointwise to the function $f : [0,1] \to \mathbb{R}$, $f(x) = 0$.
(b) Prove that f_n does not converge uniformly to any function $g : [0,1] \to \mathbb{R}$.

Problem 2. Consider the series of functions $\displaystyle\sum_{n=0}^{\infty} \frac{x^n}{1+x^n}$.

(a) Prove that for any $r \in (0,1)$, the series converges uniformly on $[0,r]$.
(b) Prove that the series does not converge uniformly on $[0,1)$.

Problem 3. Consider the power series $\displaystyle\sum_{n=1}^{\infty} \frac{x^n}{\sqrt{n}}$.

(a) Find its radius of convergence R.
(b) Determine if the series converges at each of the endpoints $x = R$ and $x = -R$.

Problem 4. For each of the following power series, determine the radius of convergence.

$$\text{(a)} \quad \sum_{n=0}^{\infty} n^2 x^n \qquad\qquad \text{(b)} \quad \sum_{n=1}^{\infty} \left(\frac{x}{n}\right)^n \qquad\qquad \text{(c)} \quad \sum_{n=0}^{\infty} n!\, x^n$$

Problem 5. The *hyperbolic sine* function is defined by $\sinh x = \frac{1}{2}(e^x - e^{-x})$.

(a) Find its Taylor series centered at $x = 0$.
(b) Prove that this series converges to $\sinh x$ for any $x \in \mathbb{R}$.

Problem 6. Consider the function $f : (-1, \infty) \to \mathbb{R}$, $f(x) = \log(1+x)$.

(a) Using induction, prove that $f^{(n)}(x) = (-1)^{n+1}(n-1)!\,(1+x)^{-n}$ for all $n \geq 1$.
(b) Using part (a), find the Taylor series of f centered at $x = 0$.
(c) Suppose we want to approximate the number $f(1) = \log 2$. Prove that for any $n \geq 2$,

$$\left| f(1) - \sum_{k=1}^{n-1} \frac{(-1)^{k+1}}{k} \right| \leq \frac{1}{n}.$$

This gives us a way to approximate $\log 2$ and to guarantee that the approximation is as accurate as we want, without having to know the value of $\log 2$ beforehand.